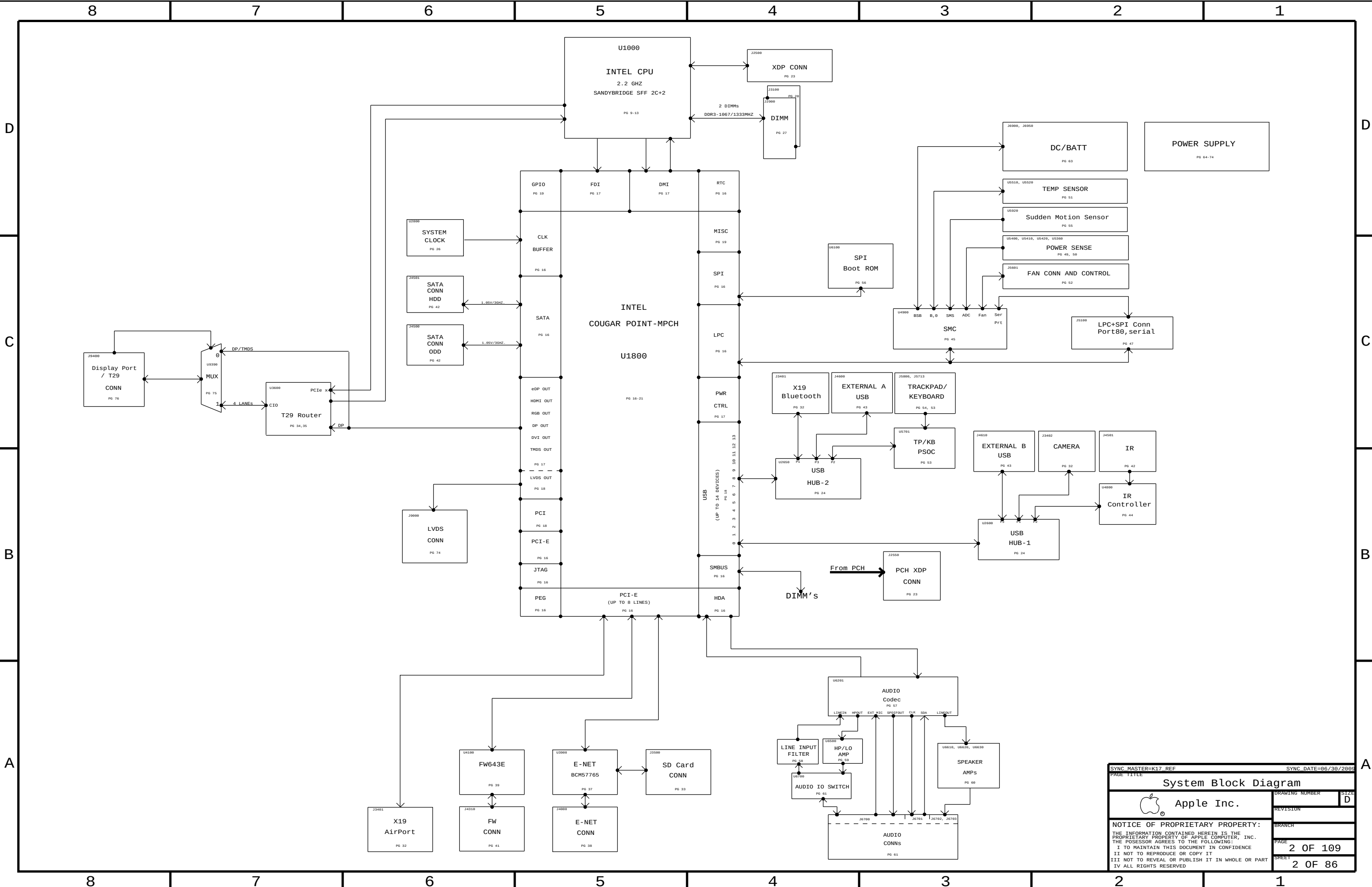
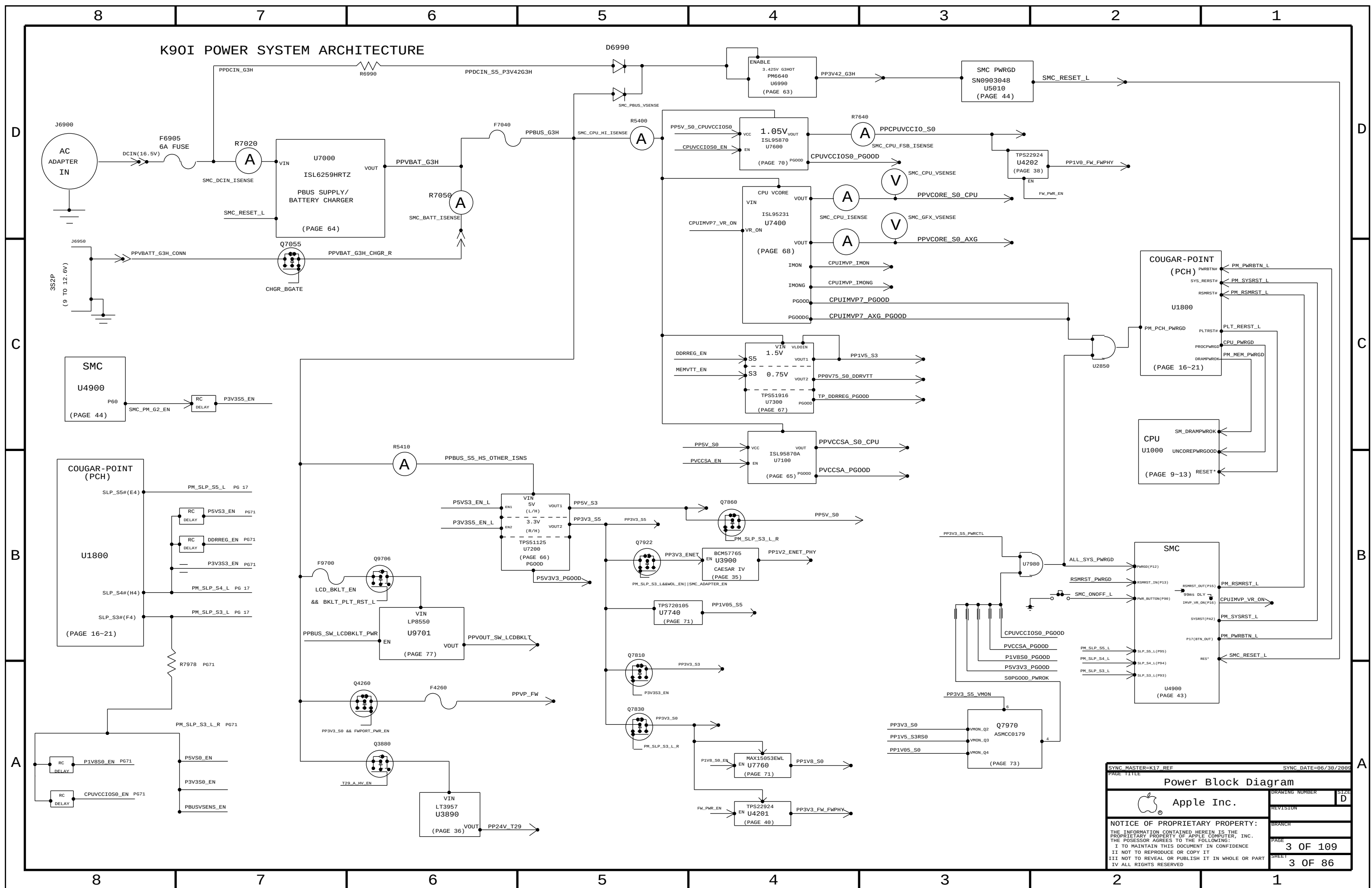
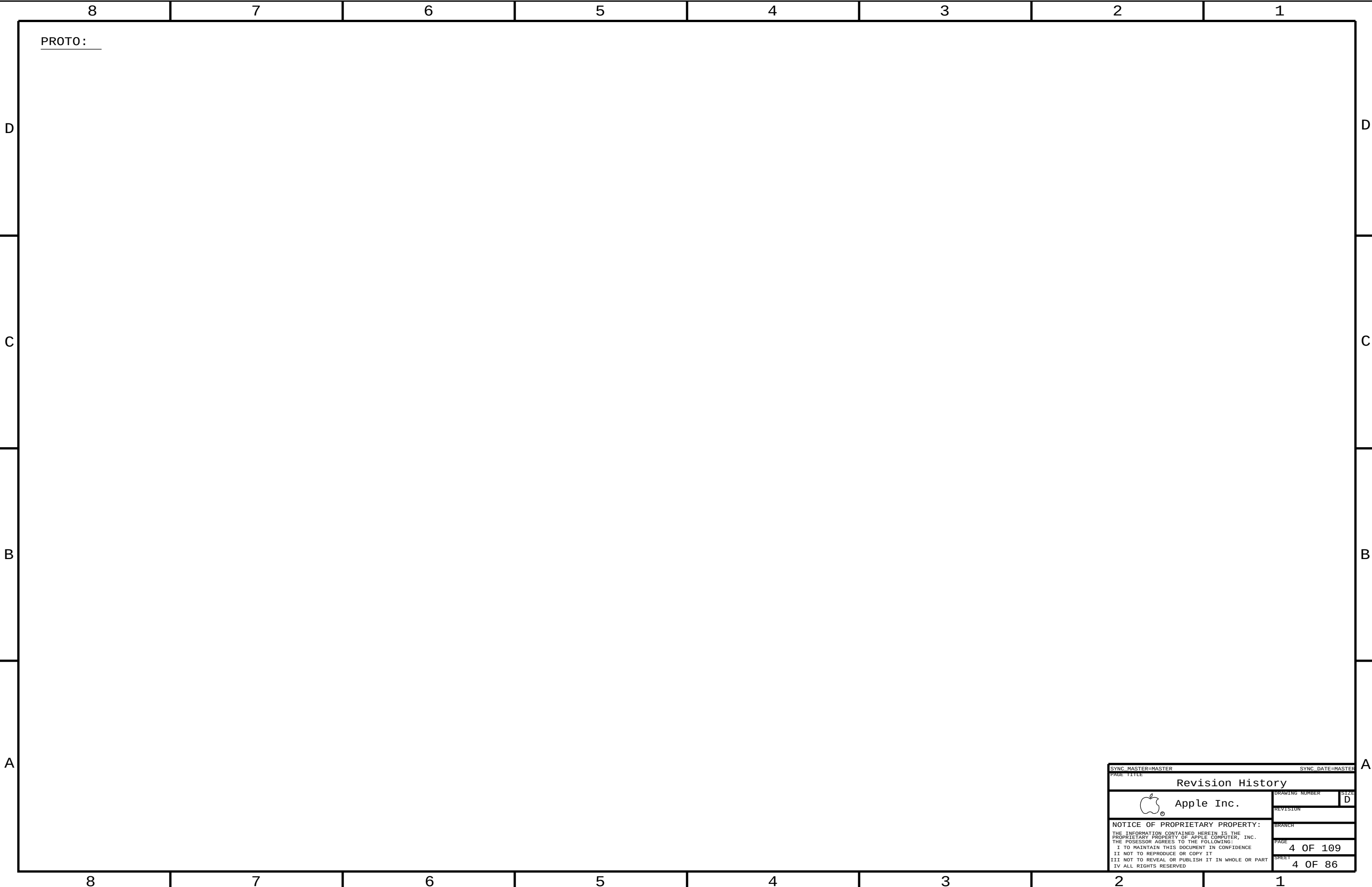








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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-1294	PCBA, 2.5G, K901	K901_COMMON, CPU_2.5GHZ, EEEE_DDRQ
639-1581	PCBA, 2.7G, K901	K901_COMMON, CPU_2.7GHZ, EEEE_DH78
639-1698	PCBA, 2.6G, K901	K901_COMMON, CPU_2.6GHZ, EEEE_DHB8
639-1699	PCBA, 2.3G, K901	K901_COMMON, CPU_2.3GHZ, EEEE_DHB8
085-1998	K901 MLB DEVELOPMENT BOM	K901_DEVEL : ENG

K90i BOM GROUPS

BOM GROUP	BOM OPTIONS
K90i__COMMON	ALTERNATE,COMMON,K90i__COMMON1,K90i__COMMON2,K90i__DEBUG:ENG,K90i__PROGPARTS,USBHUB_2513B,T29BST:Y
K90i__COMMON1	BATT_3S,CPUMEM_S0,SMC_DEBUG_YES,HUB1_2NONREM,T29:YES,DP_SDRV:A2,SDRV_PD,SDRVI2C:MCU
K90i__COMMON2	MIKEY,KB_BL
K90i__PROGPARTS	BOOTROM_PROG,SMC_PROG,TPAD_PROG,ENET_PROG,T29ROM:PROG,T29MCU:PROG
K90i__DEVEL:ENG	BKLT:ENG,BMON:ENG,XDP_CONN,XDP_CPU:BPM,XDP_PCH,LPCPLUS,VREFMRGN,S9PGOOD_ISL,IMVPISNS_ENG
K90i__DEVEL:PVT	LPCPLUS,XDP_CONN,XDP_PCH
K90i__DEBUG:ENG	DEVEL_BOM,SMC_DEBUG_YES,XDP
K90i__DEBUG:PVT	DEVEL_BOM,BKLT:PROD,BMON:PROD,SMC_DEBUG_YES,XDP,VREFMRGN_NOT
K90i__DEBUG:PROD	BKLT:PROD,BMON:PROD,SMC_DEBUG_YES,XDP,VREFMRGN_NOT,LPCPLUS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3934	1	SNB, 2C, QXXX, ES1, 2, 2, 35W, B2, 3M, GT1, BGA	U1000	CRITICAL	CPU_2_2GHZ
337S4058	1	SNB, Q1RA, QS, J1, 2, 5, 35W, 2+2, 1:30, 3M, BGA	U1000	CRITICAL	CPU_2_5GHZ
337S4057	1	SNB, Q1R3, QS, J1, 2, 7, 35W, 2+2, 1:30, 4M, BGA	U1000	CRITICAL	CPU_2_7GHZ
337S4024	1	SNB, Q1R9, QS, J1, 2, 3, 35W, 2+2, 1:30, 3M, BGA	U1000	CRITICAL	CPU_2_3GHZ
337S4064	1	SNB, Q1R7, QS, J1, 2, 6, 35W, 2+2, 1:30, 3M, BGA	U1000	CRITICAL	CPU_2_6GHZ
337S4029	1	IC, PCH, COUGARPOINT, SLH9D, PRQ, BD82HM65	U1800	CRITICAL	
343S0534	1	IC, BCM57765B0, ENET&SD, 8X8	U3900	CRITICAL	
338S0753	1	IC, PM643-E2, 3394B PHY/ONC1 LINK/PCI-E, 32	U4100	CRITICAL	
338S0921	1	IC, T29-C0, 229 FCBGA, 15x15MM	U3600	CRITICAL	T29: YES
353S3055	1	IC, P13VEDP212, X2 DISPLAYPORT 2:1 MUX, QFN	U9390	CRITICAL	

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
33550663	1	IC, FLASH, SERIAL, SPI, 1MBIT, ZV7, BP, SOIC	U3990	CRITICAL	ENET_BLANK
341S3026	1	IC ENET, 11MBITFLAH, CIV REV01, K60/K62	U3990	CRITICAL	ENET_PROG
335S0777	1	IC, EEPROM, SERIAL, SPI, 1Kx8, 1.8V, MLP8, LF	U3690	CRITICAL	T29ROM:BLANK
341T0317	1	IC, T29 ASSY	U3690	CRITICAL	T29ROM:PROG
337S3997	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:BLANK
341S2939	1	IC, PROGRMD, LPC1112A, T29 PORT MCU, HVQFN25	U9330	CRITICAL	T29MCU:PROG
338S0895	1	IC, SMC, HS8/2117/0MMx9MM, TLP	U4900	CRITICAL	SMC_BLANK
341T0300	1	IC, SMC, K901	U4900	CRITICAL	SMC_PROG
335S0770	1	64 MBIT SPI SERIAL DUAL 1/0 FLASH	U6100	CRITICAL	BOOTROM_BLANK
335S0769	1	64 MBIT SPI SERIAL DUAL 1/0 FLASH	U6100	CRITICAL	BOOTROM_BLANK
341T0299	1	IC, EFI ROM, K901	U6100	CRITICAL	BOOTROM_PROG
341S2384	1	IR, ENCORE II, CY7C53803-LQC	U4800	CRITICAL	
341S3024	1	IC, TP PSOC, K90, K901, K91, K91F, K92	U5701	CRITICAL	TPAD_PROG

Development BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM_OPTION
085-1998	1	K901 MLB DEVELOPMENT	DEVEL	CRITICAL	DEVEL_BOM

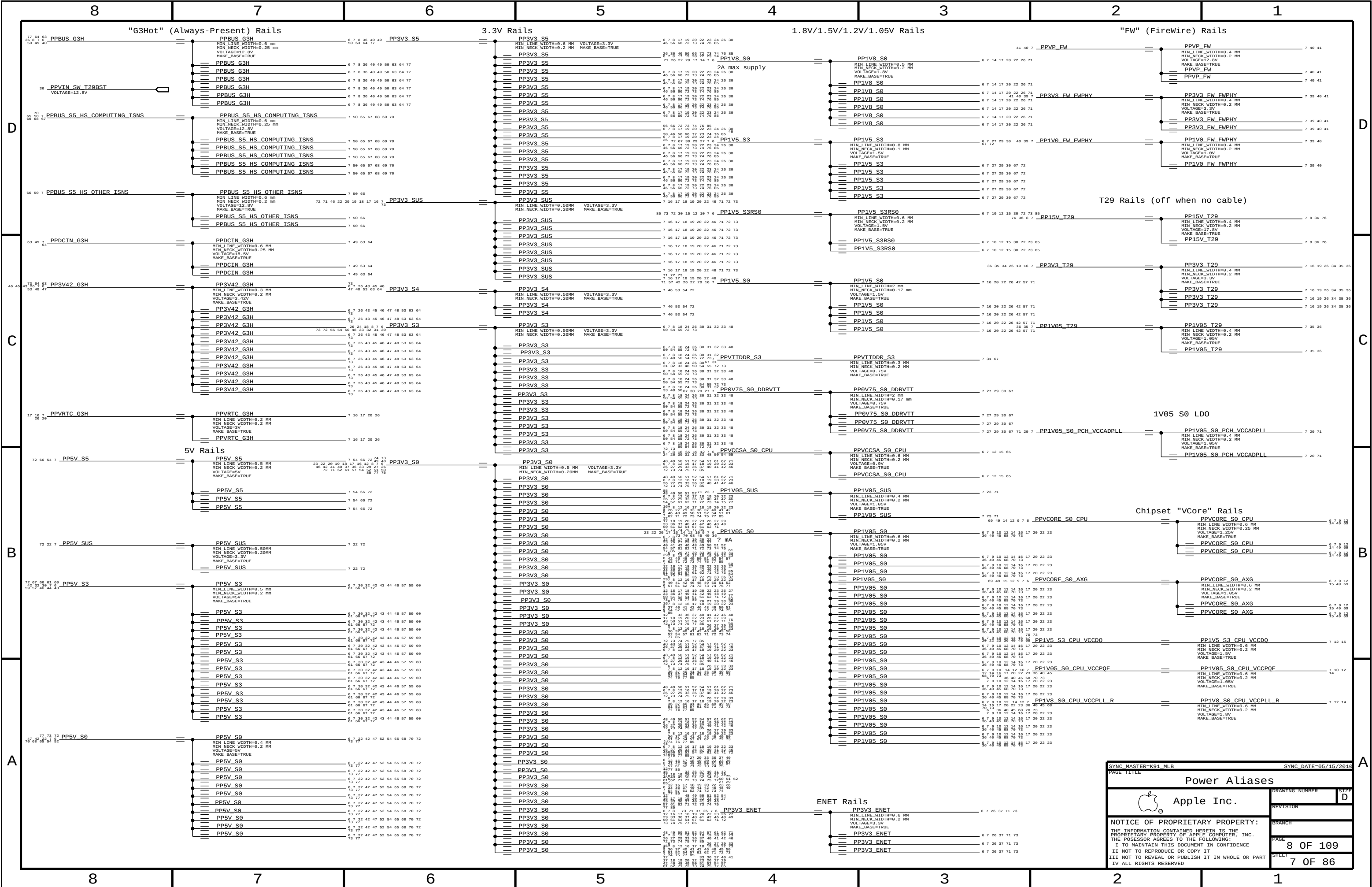
Bar Code Labels / EEEE #'s

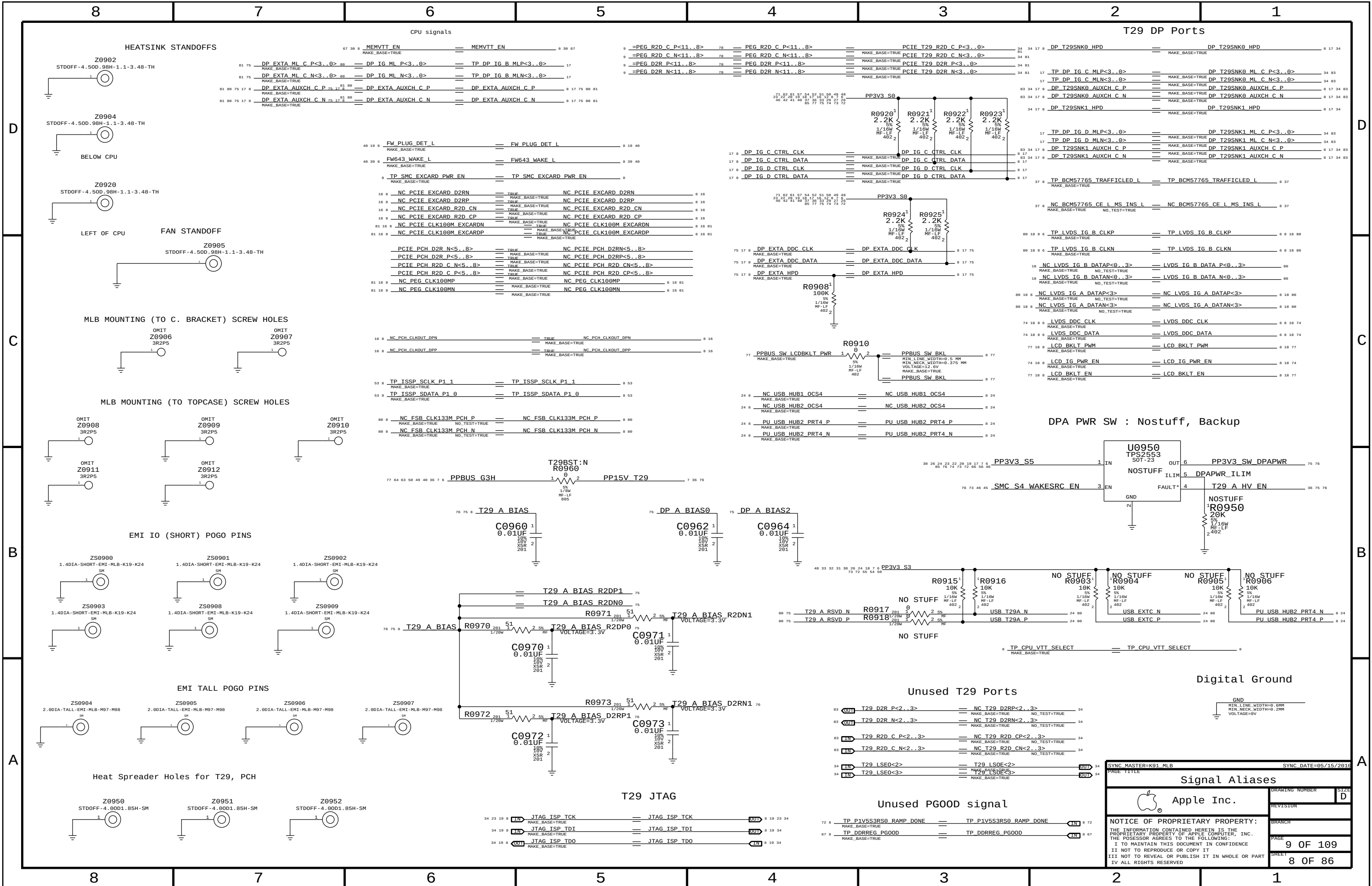
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DDRQ]	CRITICAL	EEEE_DDRQ
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DH78]	CRITICAL	EEEE_DH78
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DH8F]	CRITICAL	EEEE_DH8F
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DH8]	CRITICAL	EEEE_DH8G

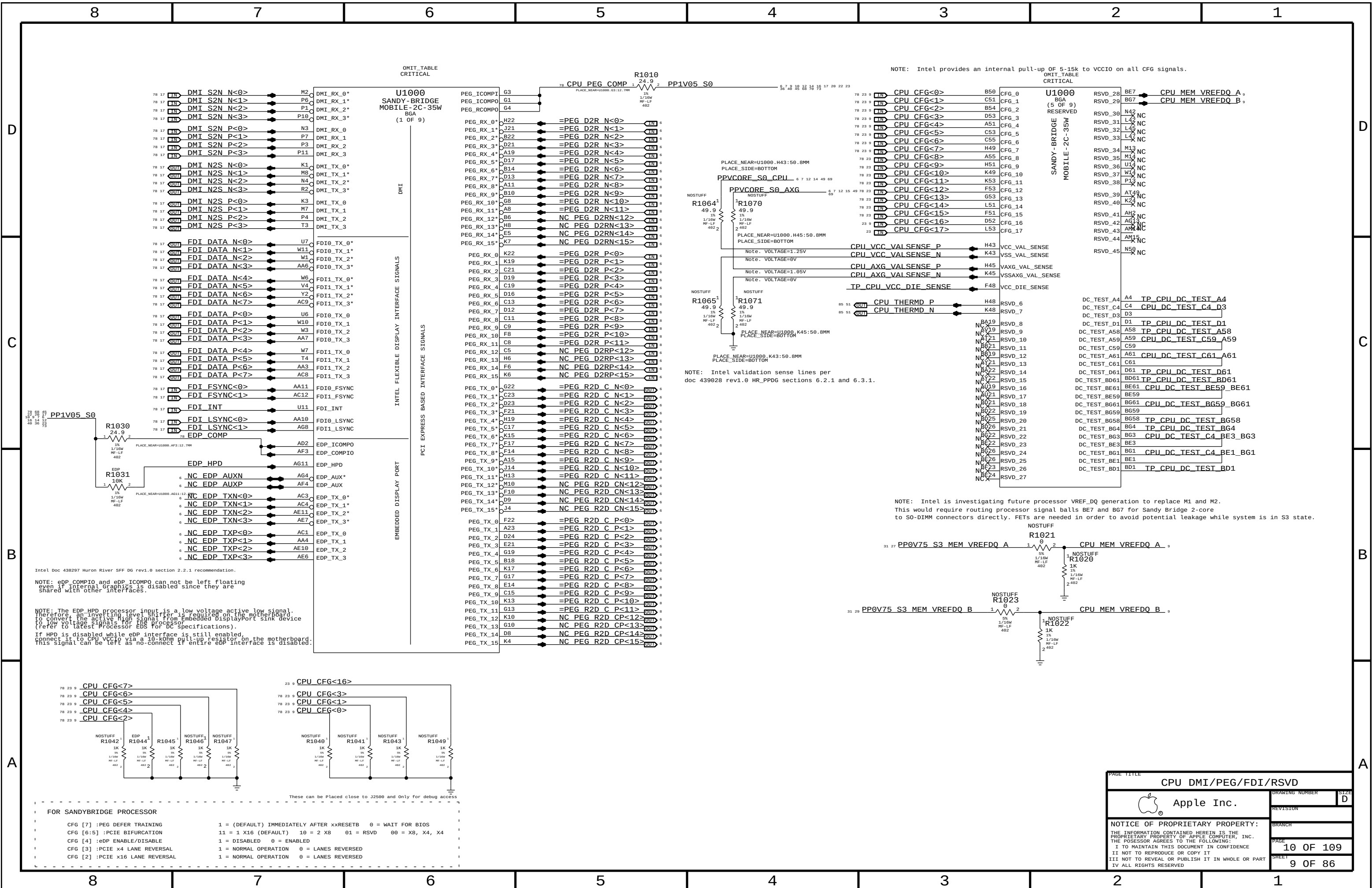
Alternate Parts

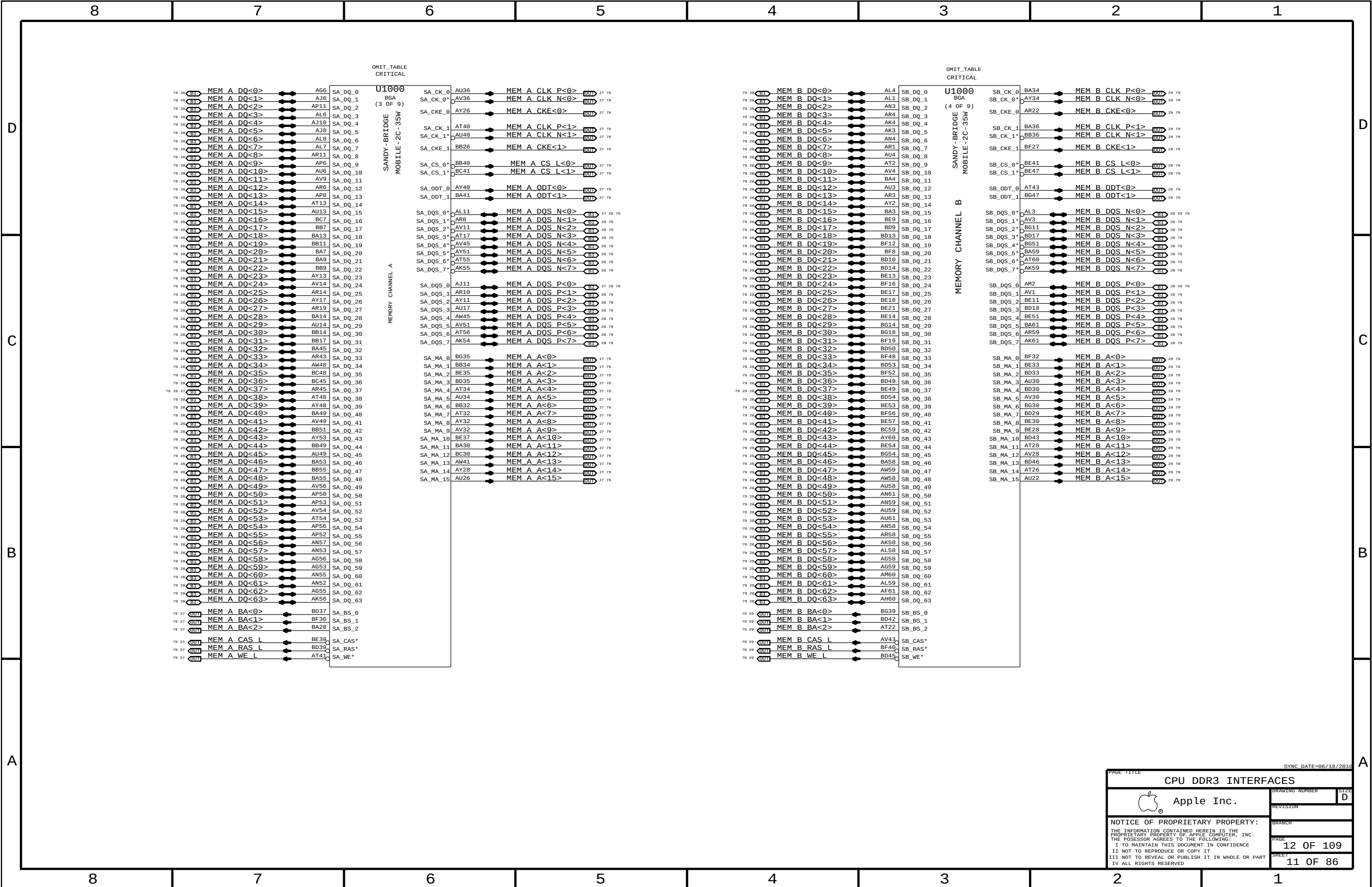
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0603	138S0602		ALL	Murata alt to Samsung
157S0058	157S0055		ALL	Delta alt to TDK Magnetics
516S0805	516S0806		ALL	Wolox alt to Foxconn
128S0303	128S0282		ALL	Panasonic alt to Sanyo
138S0676	138S0691		ALL	Murata alt to Samsung
152S0778	152S0693		ALL	Cyntec alt to Vishay
376S0855	376S0613		ALL	Diodes alt to Toshiba
376S0977	376S0859		ALL	Diodes alt to Toshiba
376S0972	376S0612		ALL	Rohe alt to Toshiba
376S0927	376S0966		ALL	Fairchild alt to Renesas
376S0927	376S0790		ALL	Fairchild alt to CIGLOW
376S0960	376S0801		ALL	Renesas alt to Renesas
376S0790	376S0928		ALL	CIGLOW alt to Fairchild
376S0928	376S0895		ALL	Fairchild alt to Renesas
376S0937	376S0845		ALL	Fairchild alt to Renesas
376S0777	376S0761		ALL	ADN alt to Siliconix
376S0957	376S0958		ALL	Fairchild alt to Fairchild
376S0953	376S0958		ALL	Fairchild alt to Renesas
353S3085	353S1058		ALL	STMicro alt to LT

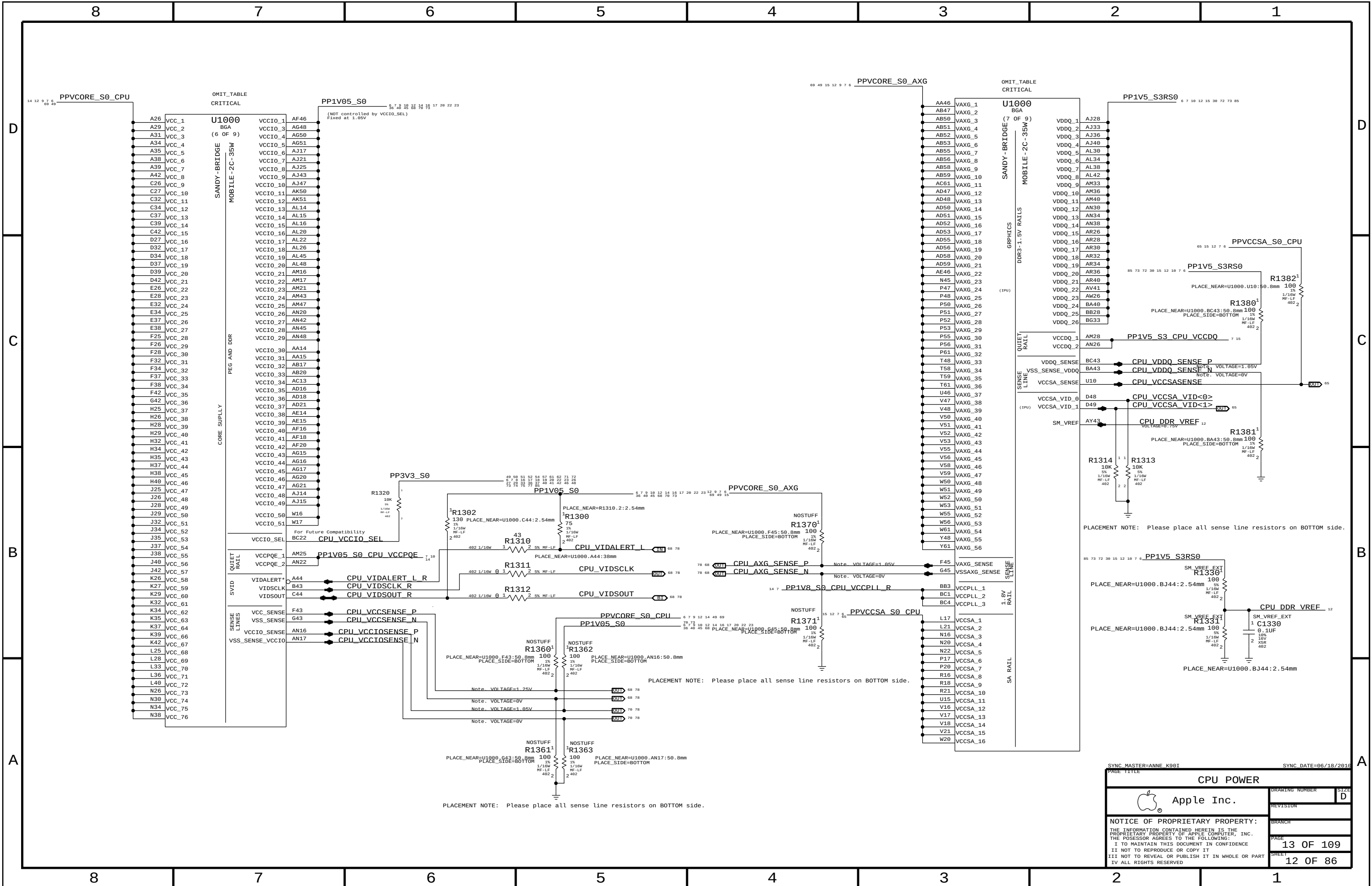
SYNC MASTER=K17 REF		SYNC DATE=95/28/2695	
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BOM Configuration			
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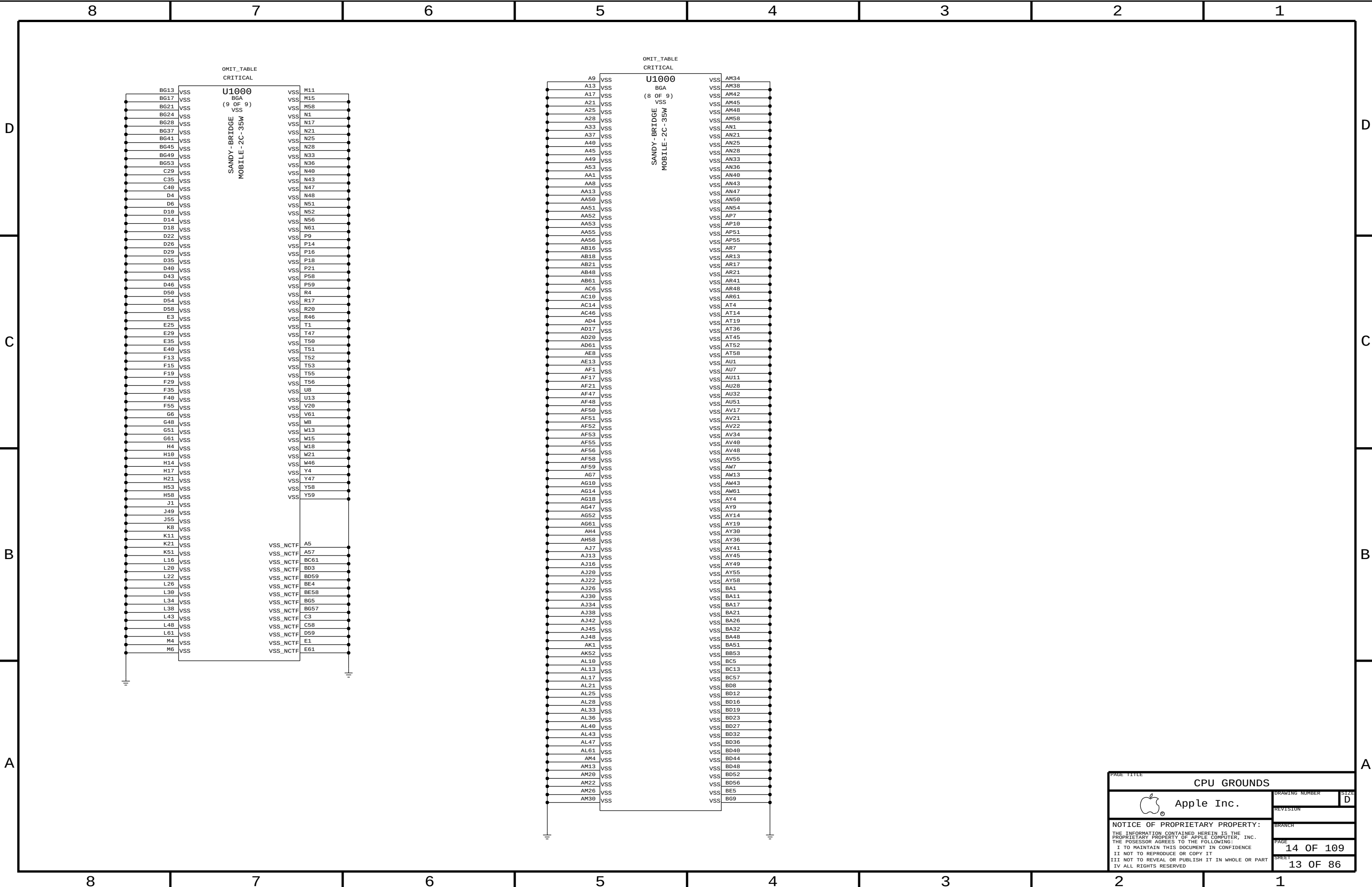












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69 49 12 9 7 6 PPVCORE S0 CPU



C1655, C1660, C1661, C1662, C1663, C1664, C1665, C1666, C1667, C1670, C1671, C1674, C1676, C1677, C1678, C1679

C

45-4836-23 DD1V05 SA

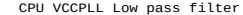
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B

PLACEMENT_NOTE (C1646-C1671):

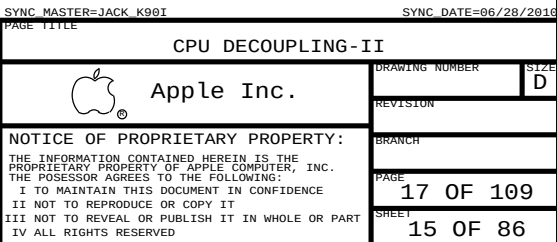
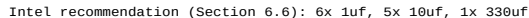
Place near U1000 on top side

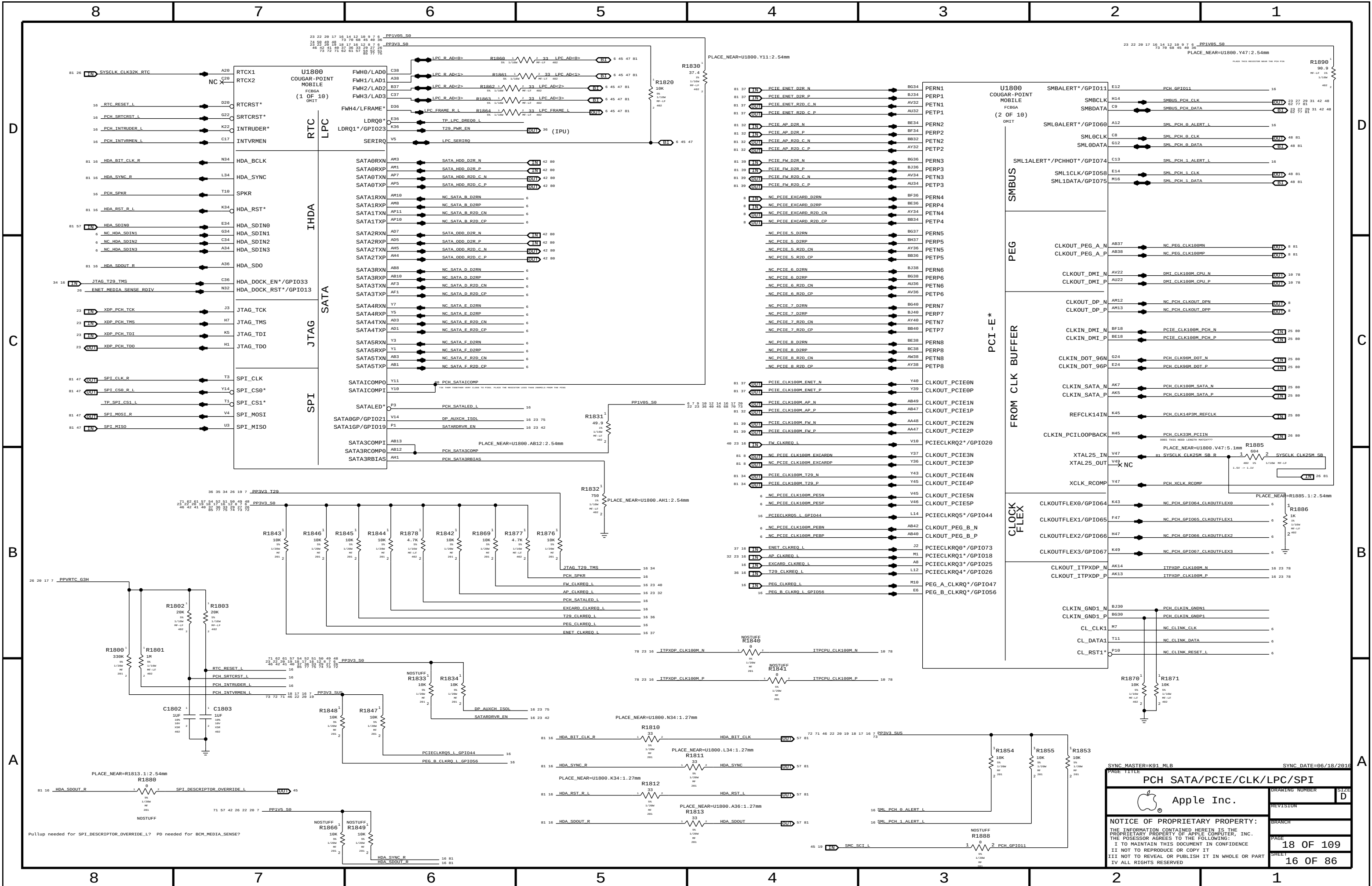
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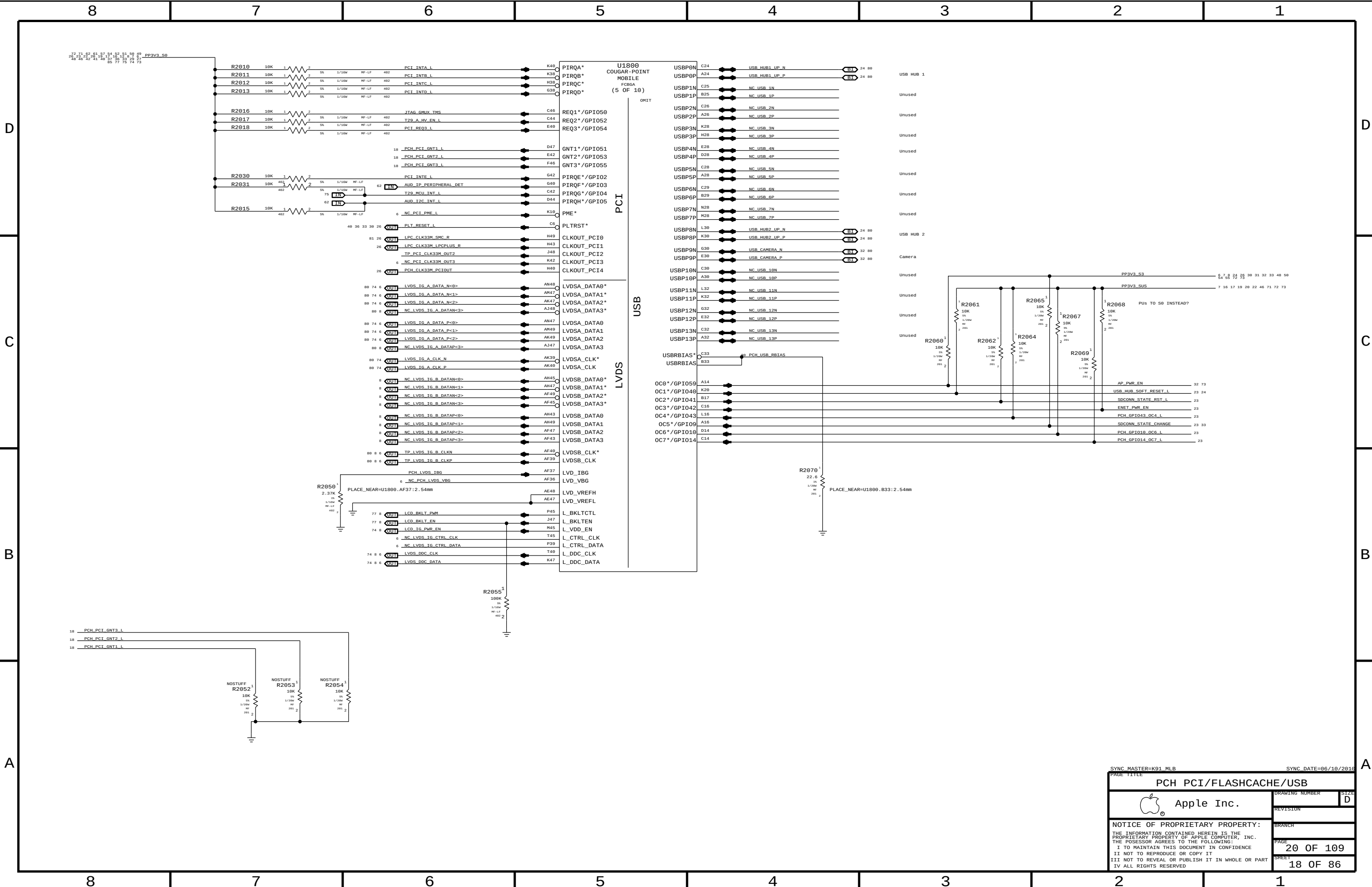
Intel recommendation (section 6.3): 21x 1uF, 6x 10uF, 6x 22uF, 2x 470uF

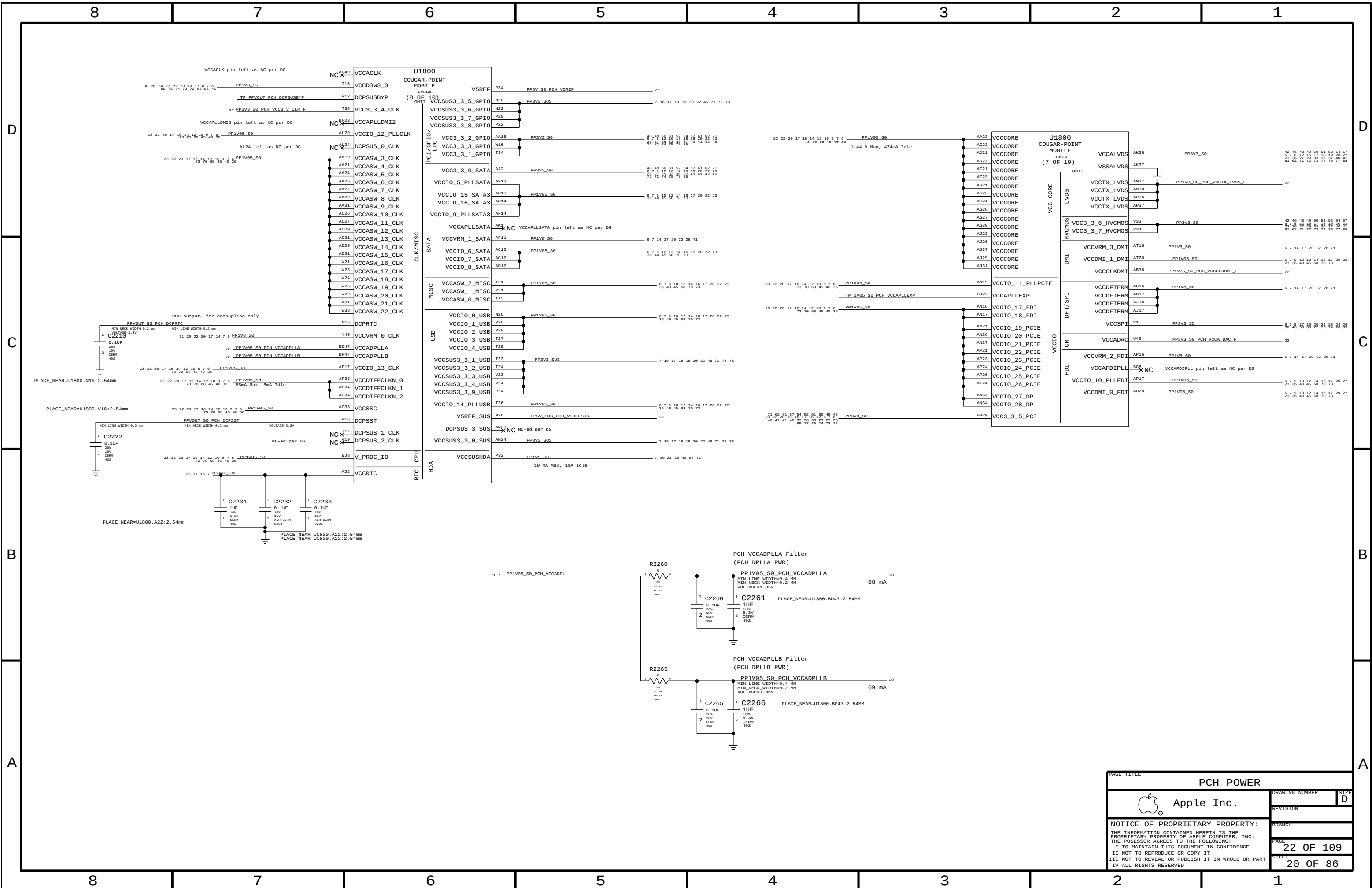


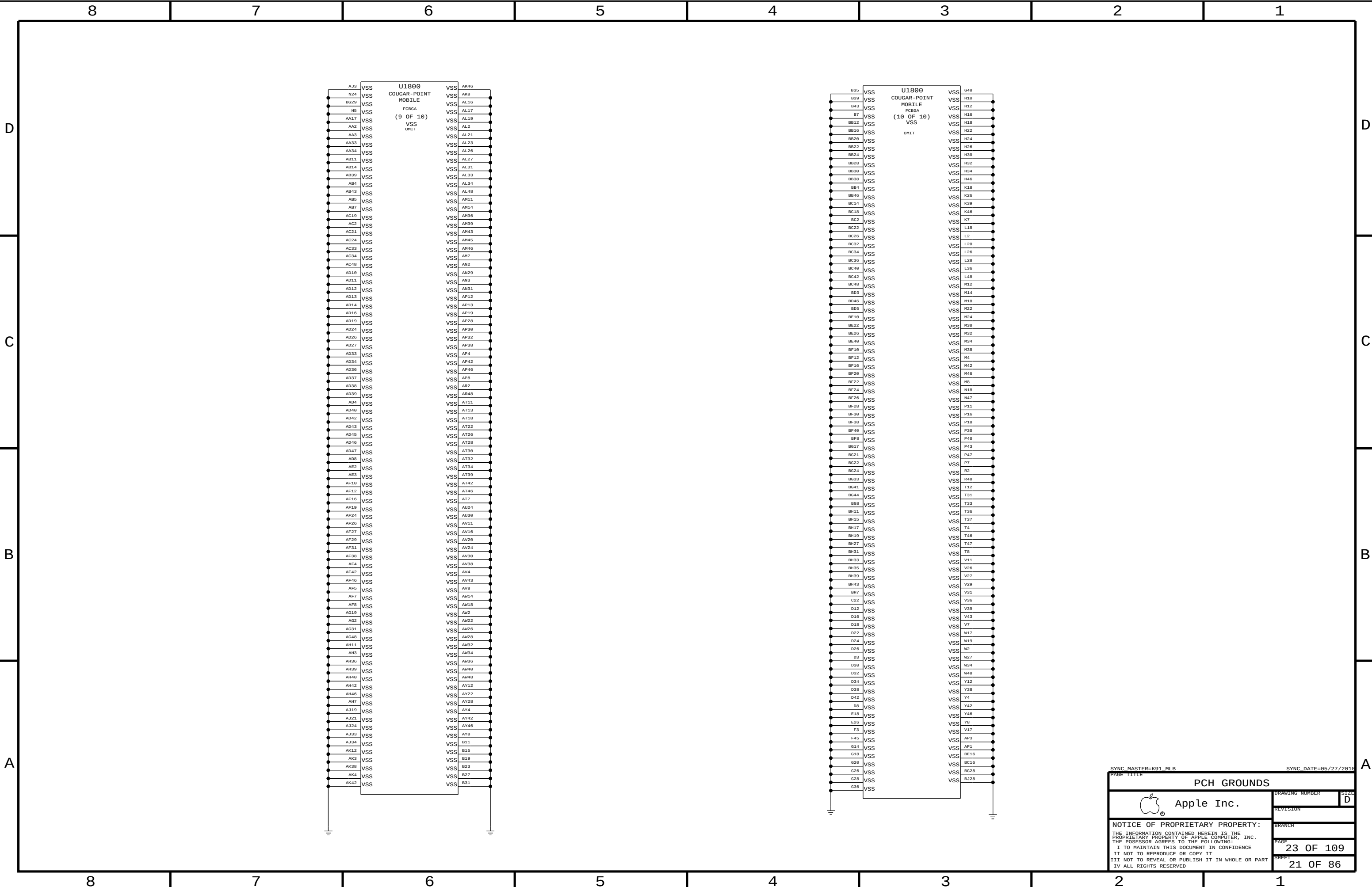
Intel recommendation (Section 6.5): 10x 1uF, 8x 10uF, 1x 330uF

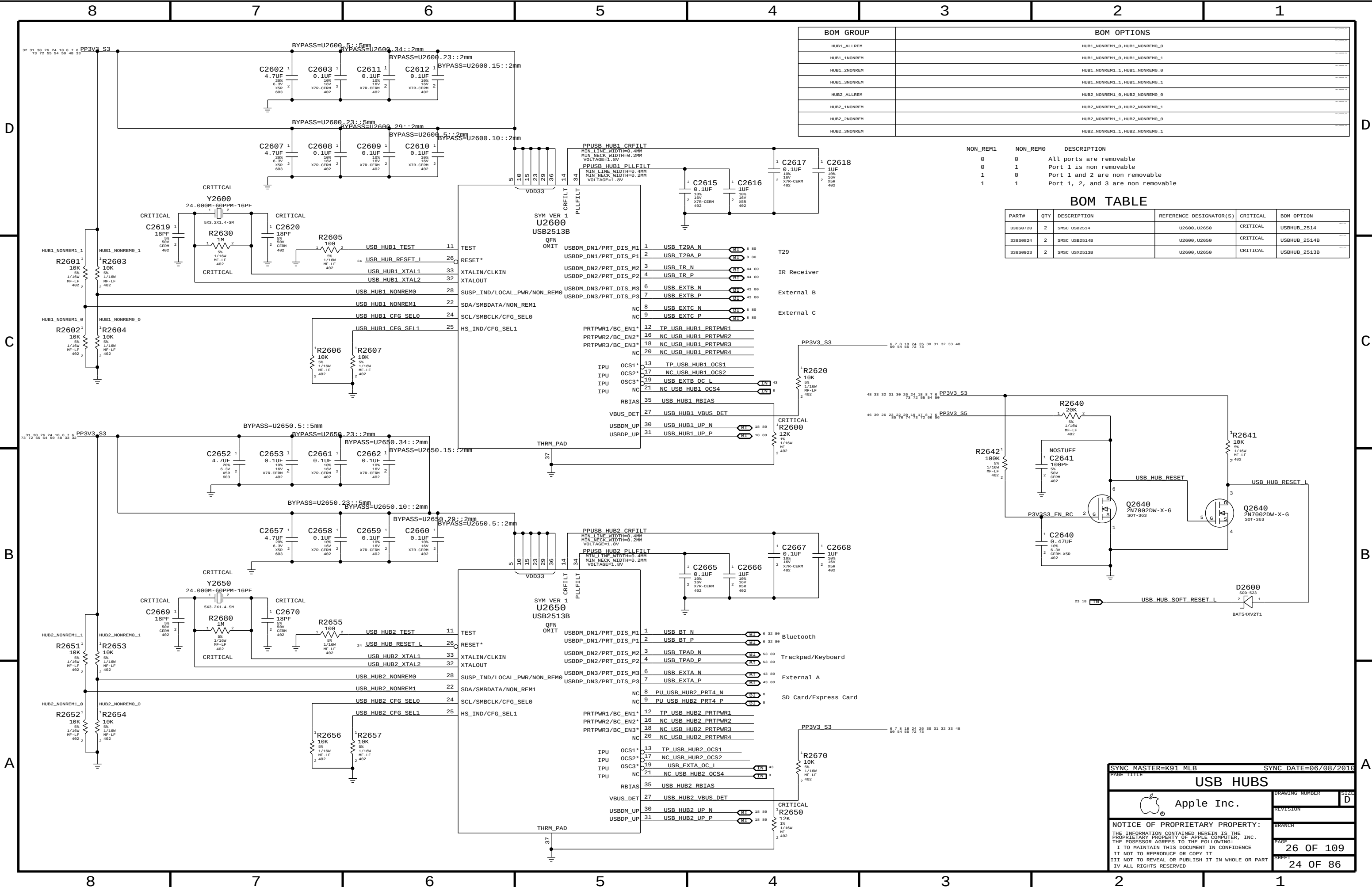












BOM GROUP	BOM OPTIONS
HUB1_ALLREM	HUB1_NONREM1_0, HUB1_NONREM0_0
HUB1_1NONREM	HUB1_NONREM1_0, HUB1_NONREM0_1
HUB1_2NONREM	HUB1_NONREM1_1, HUB1_NONREM0_0
HUB1_3NONREM	HUB1_NONREM1_1, HUB1_NONREM0_1
HUB2_ALLREM	HUB2_NONREM1_0, HUB2_NONREM0_0
HUB2_1NONREM	HUB2_NONREM1_0, HUB2_NONREM0_1
HUB2_2NONREM	HUB2_NONREM1_1, HUB2_NONREM0_0
HUB2_3NONREM	HUB2_NONREM1_1, HUB2_NONREM0_1

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE	DESIGNATOR(S)	CRITICAL	BOM OPTION
33850729	2	SMSC USB2514		U2600, U2650	CRITICAL	USBHUB_2514
33850824	2	SMSC USB2514B		U2600, U2650	CRITICAL	USBHUB_2514B
33850923	2	SMSC USX2513B		U2600, U2650	CRITICAL	USBHUB_2513B

SYNC MASTER=K91 MLB

SYNC DATE=06/08/2016

USB HUBS

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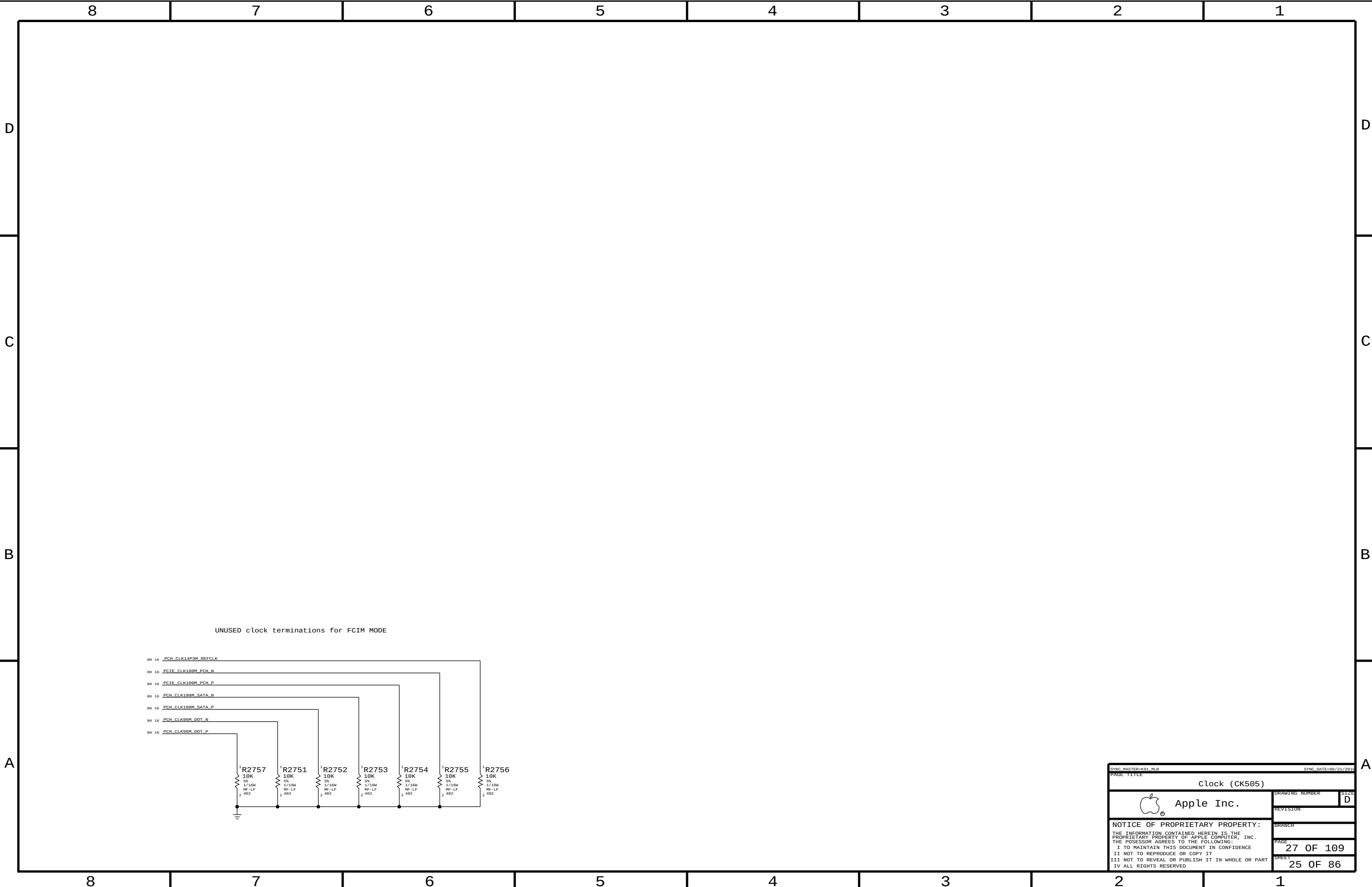
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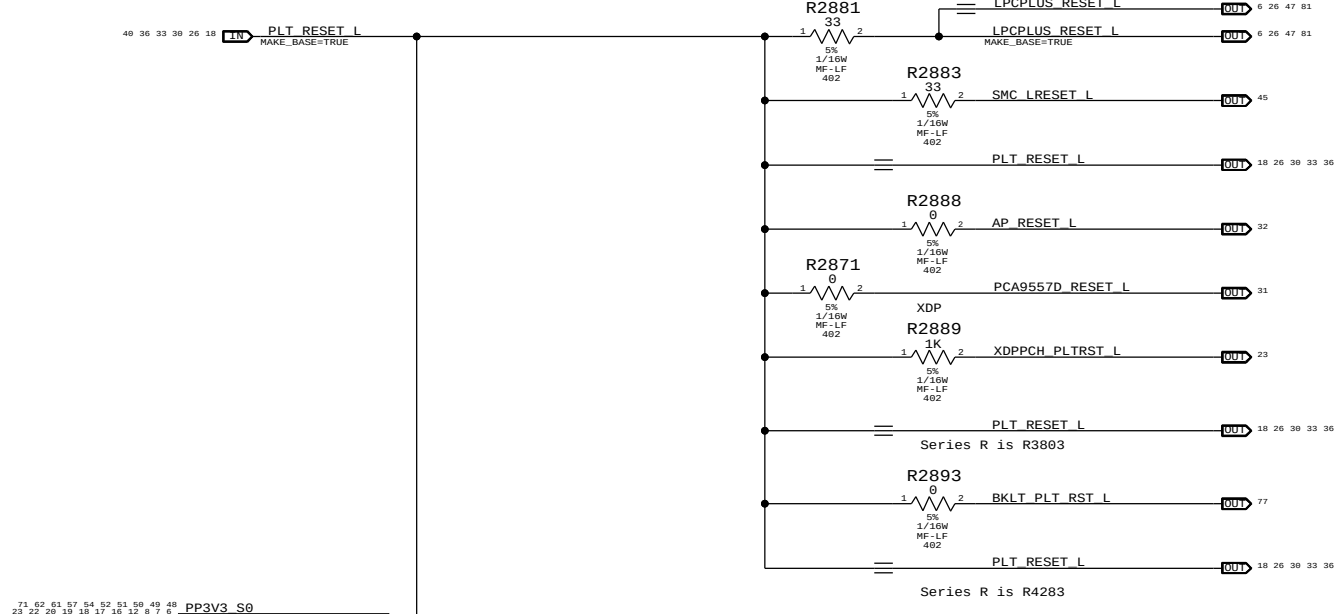
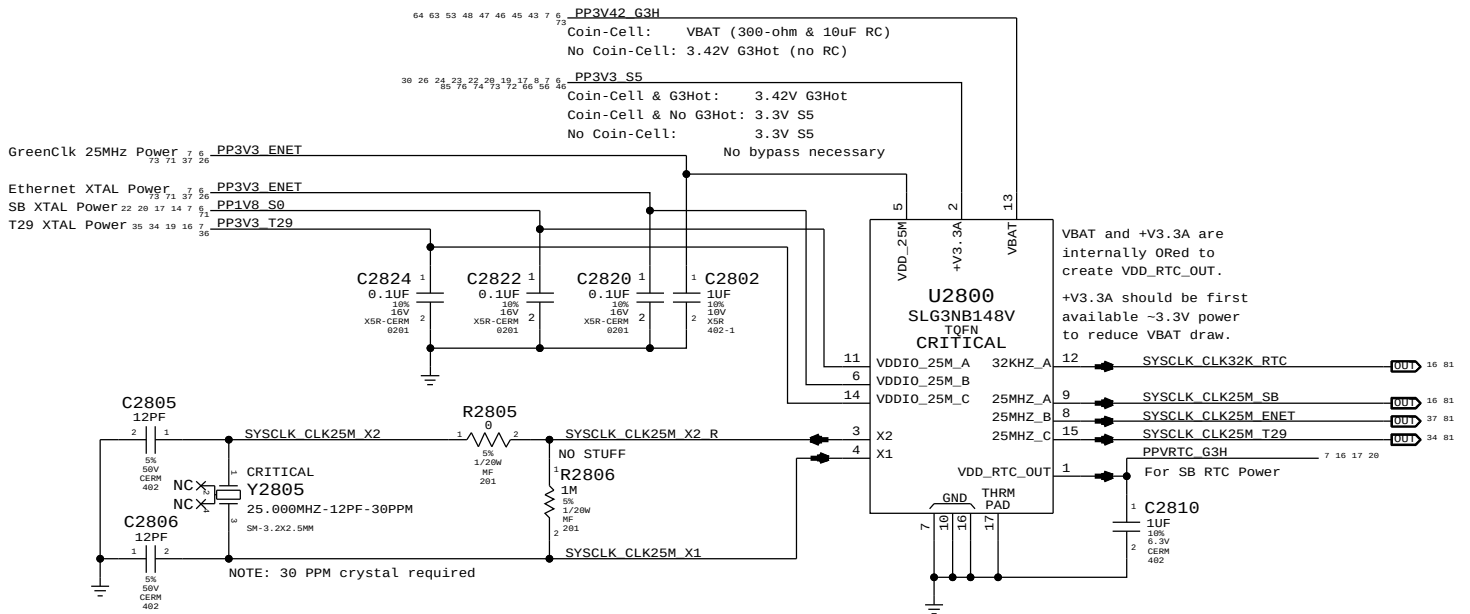
SYNC_DATE=00/21/2011

PAGE TITLE		Clock (CK505)	
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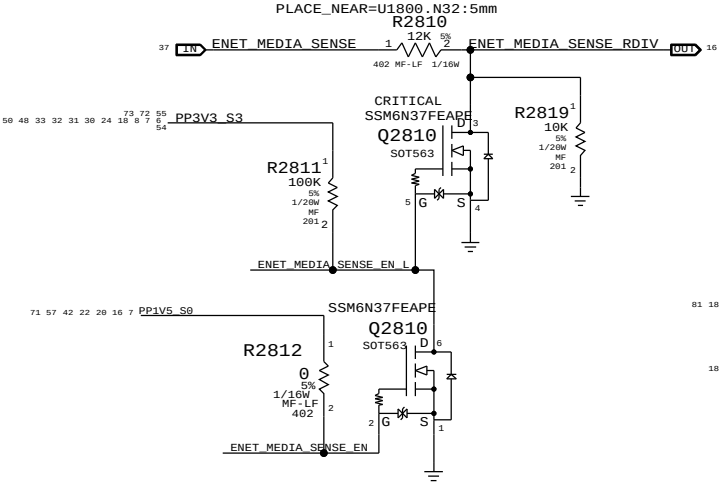
System RTC Power Source & 32kHz / 25MHz Clock Generator

Platform Reset Connections

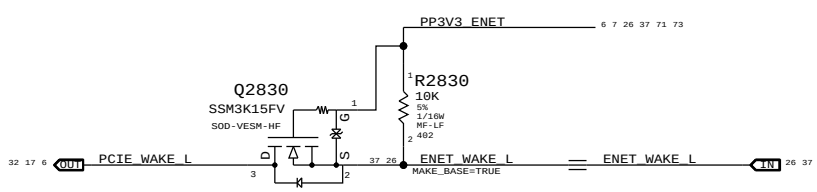
Unbuffered



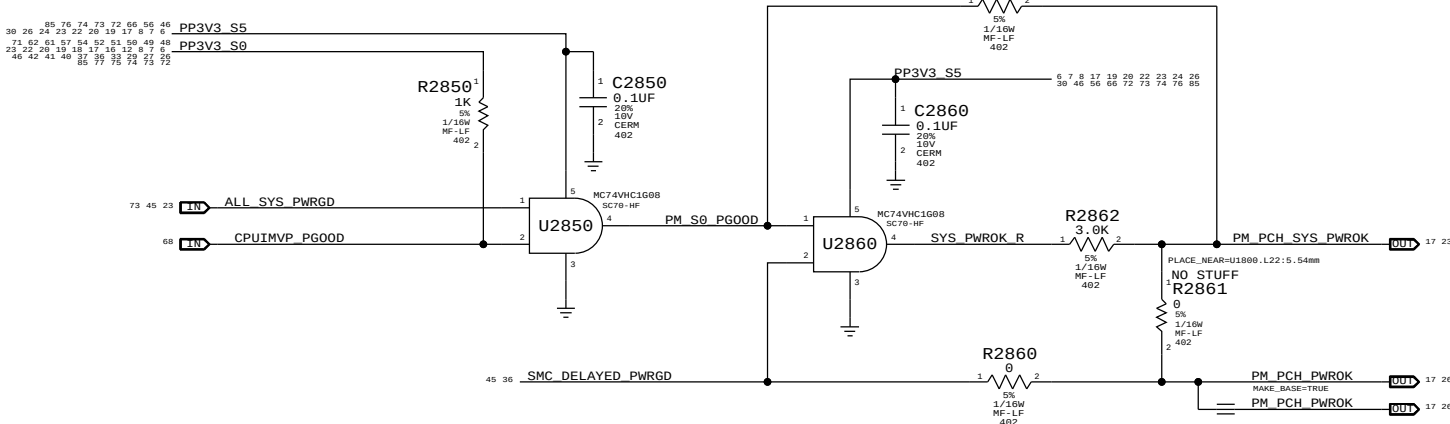
ENET_MEDIA_SENSE ISOLATION CIRCUIT



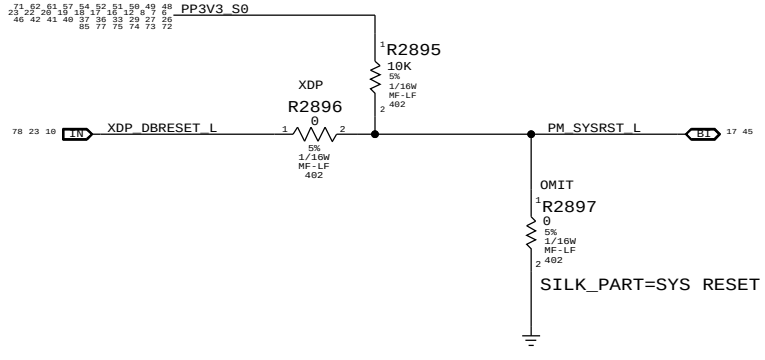
Ethernet WAKE# Isolation



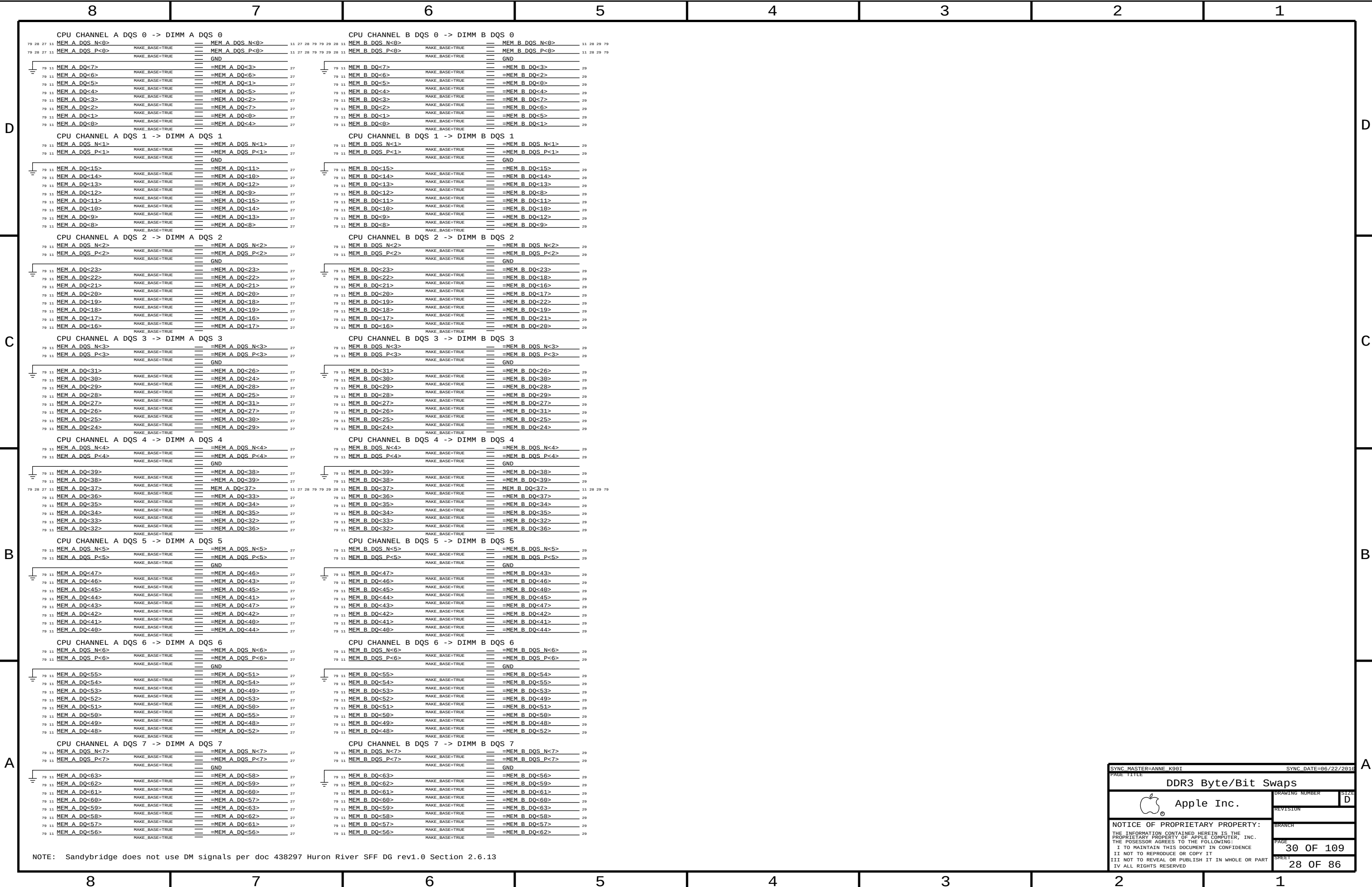
PCH S0 PWRGD



PCH Reset Button



SYNC MASTER=LINDA K901		SYNC DATE=07/08/2010	
PAGE TITLE		Chipset Support	
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NOTE: Sandybridge does not use DM signals per doc 438297 Huron River SFF DG rev1.0 Section 2.6.13

SYNC MASTER=ANNE_K901

SYNC DATE=06/22/2010

PAGE TITLE

DDR3 Byte/Bit Swaps

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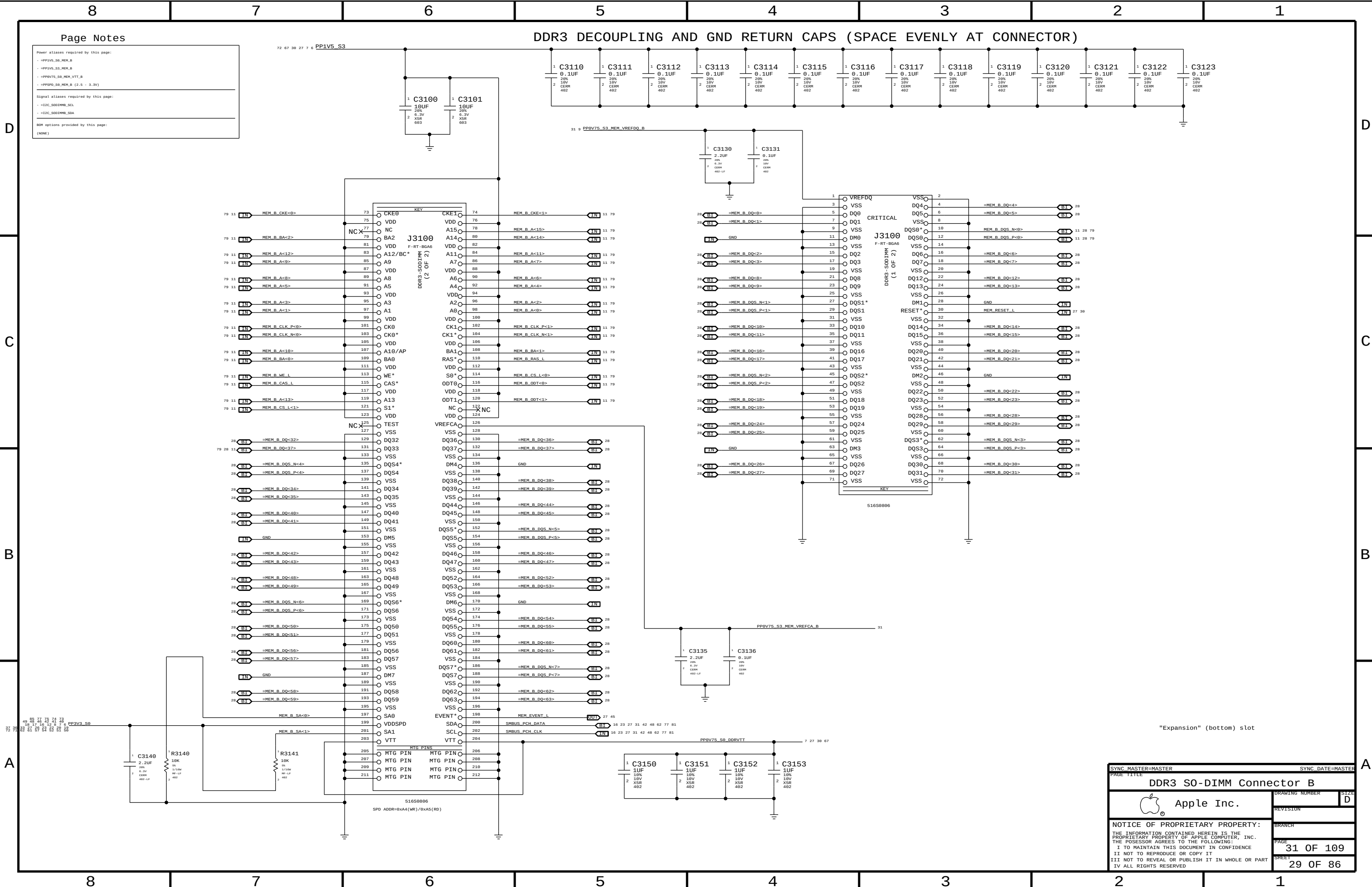
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Page Notes

Power aliases required by this page:
- PP1V5_S3_MEM_B
- PP1V5_S3_MEM_B
- PP0V75_S0_MEM_VTT_B
- PPSPD_S0_MEM_B (2.5 - 3.3V)
Signal aliases required by this page:
- I2C_S0D1MMB_SCL
- I2C_S0D1MMB_SDA
BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

"Expansion" (bottom) slot

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

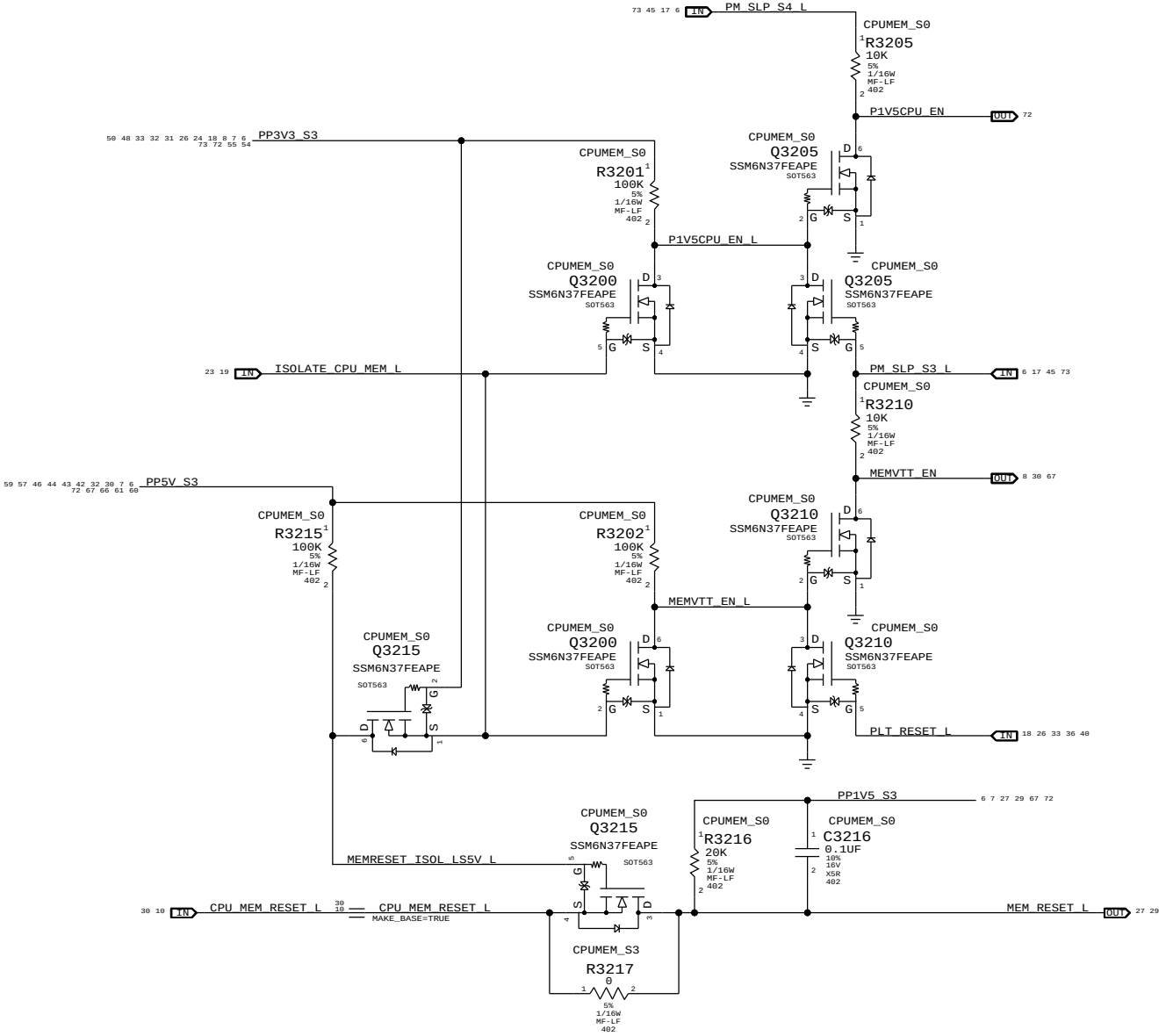
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

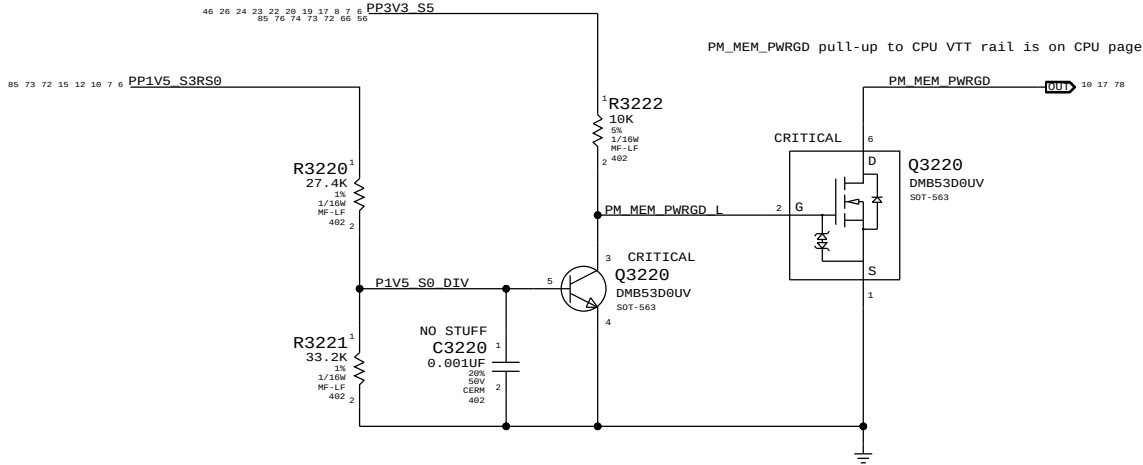
$P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L$

$MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L$

$MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L$

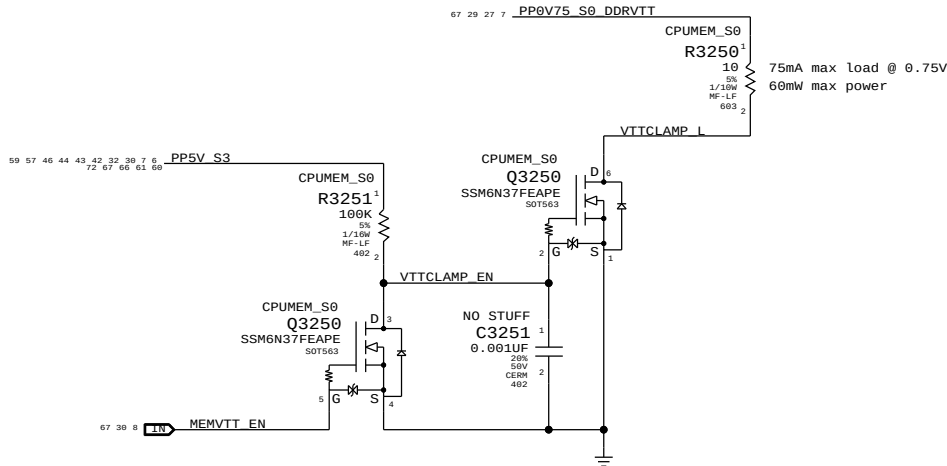


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

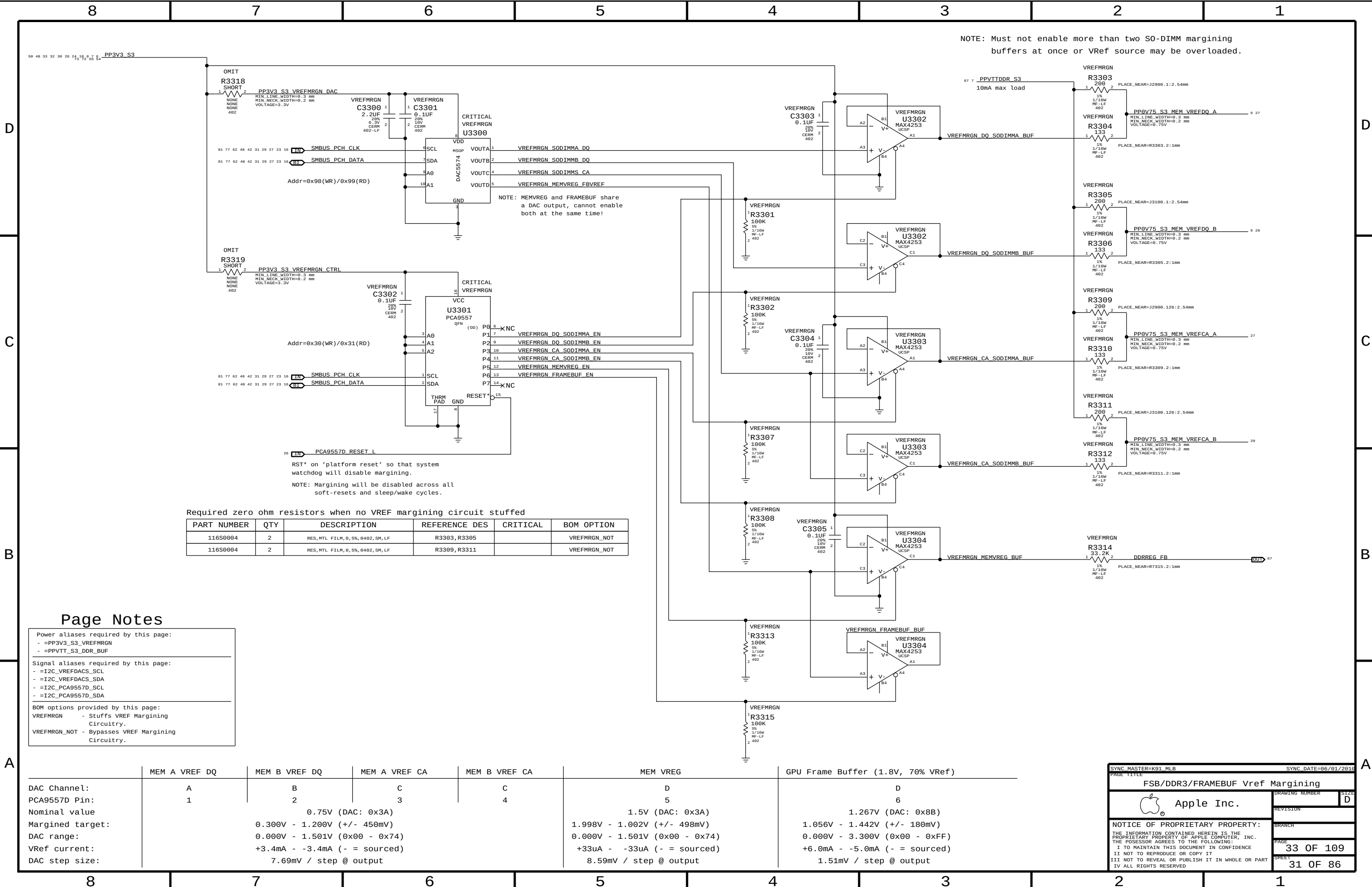


Step	SOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=ANNE_K901		SYNC DATE=06/22/2010	
PAGE TITLE			
CPU Memory S3 Support		DRAWING NUMBER	SIZE
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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMGRN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

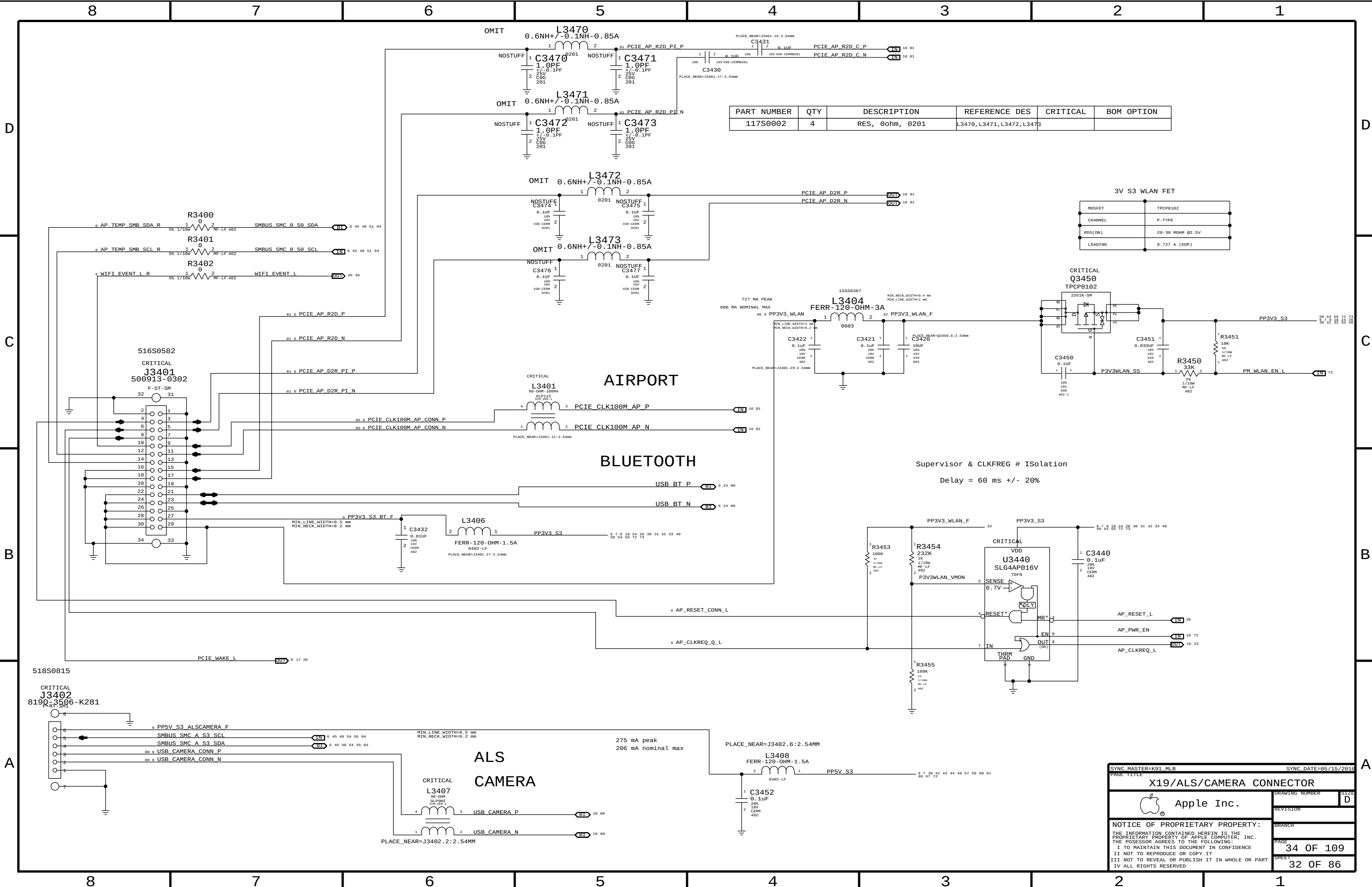
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- VREFMGRN - Stuffs VREF Margining Circuitry.
- VREFMGRN_NOT - Bypasses VREF Margining Circuitry.

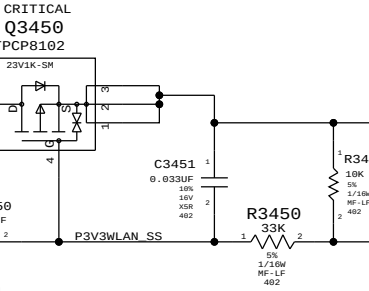
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)					1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)					1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)					0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)					+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output					1.51mV / step @ output

PAGE TITLE		DRAWING NUMBER	
FSB/DDR3/FRAMEBUF Vref Margining		D	
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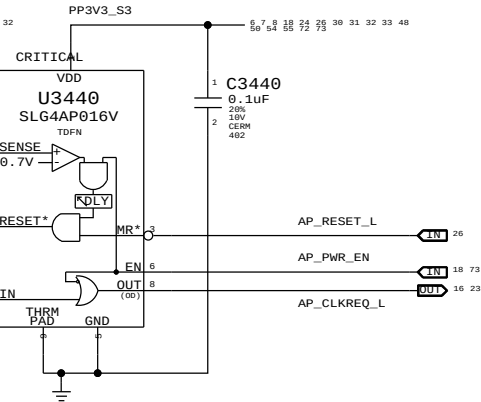


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0002	4	RES, 90hm, 0201	L3470, L3471, L3472, L3473		

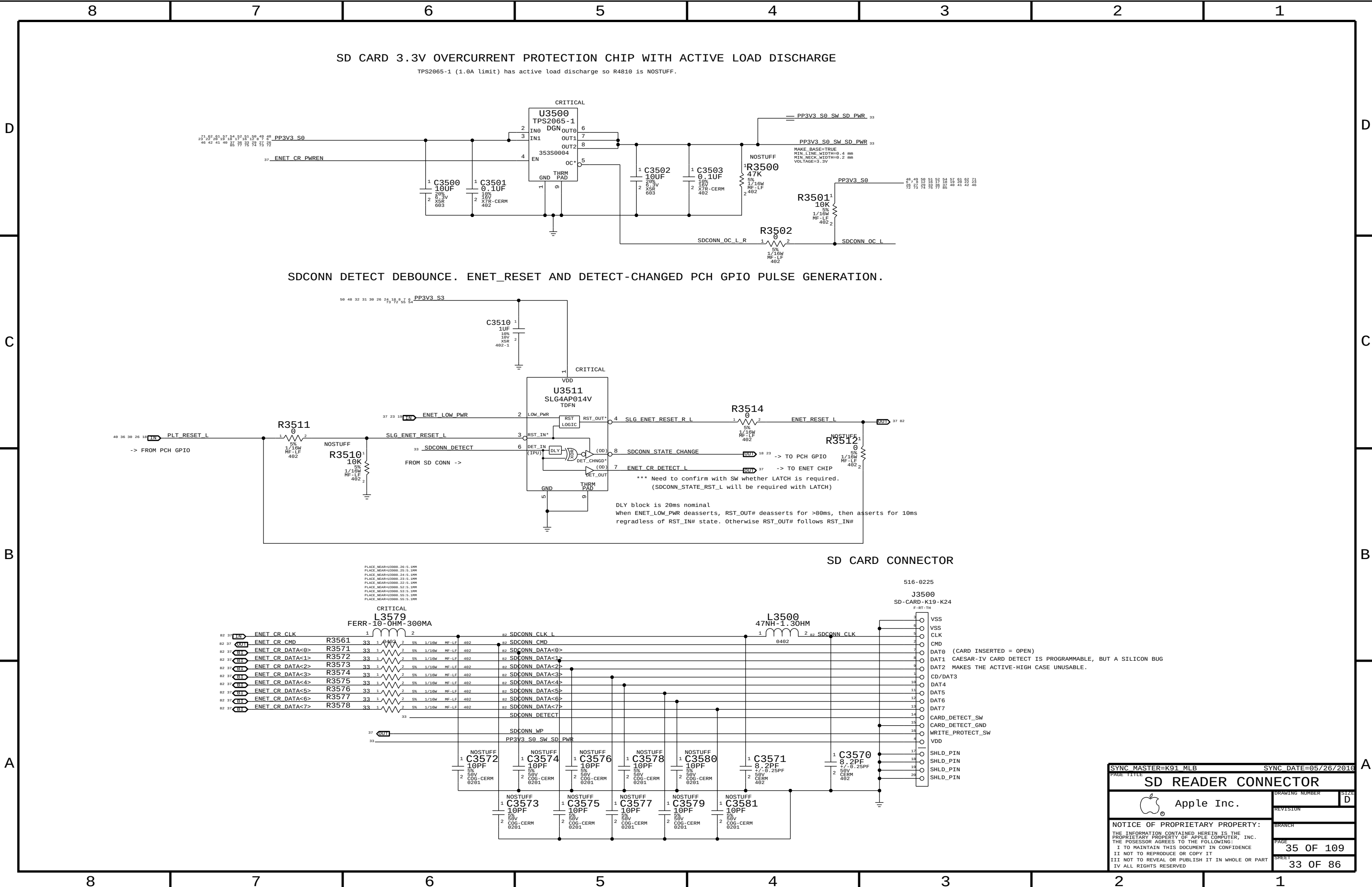
3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	26-30 MOHM @2.5V
LOADING	0.727 A (EDP)



Supervisor & CLKFREQ # Isolation
Delay = 60 ms +/- 20%



SYNC MASTER=K91 MLB		SYNC DATE=05/15/2010	
PAGE TITLE		DRAWING NUMBER	SIZE
X19/ALS/CAMERA CONNECTOR			D
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D

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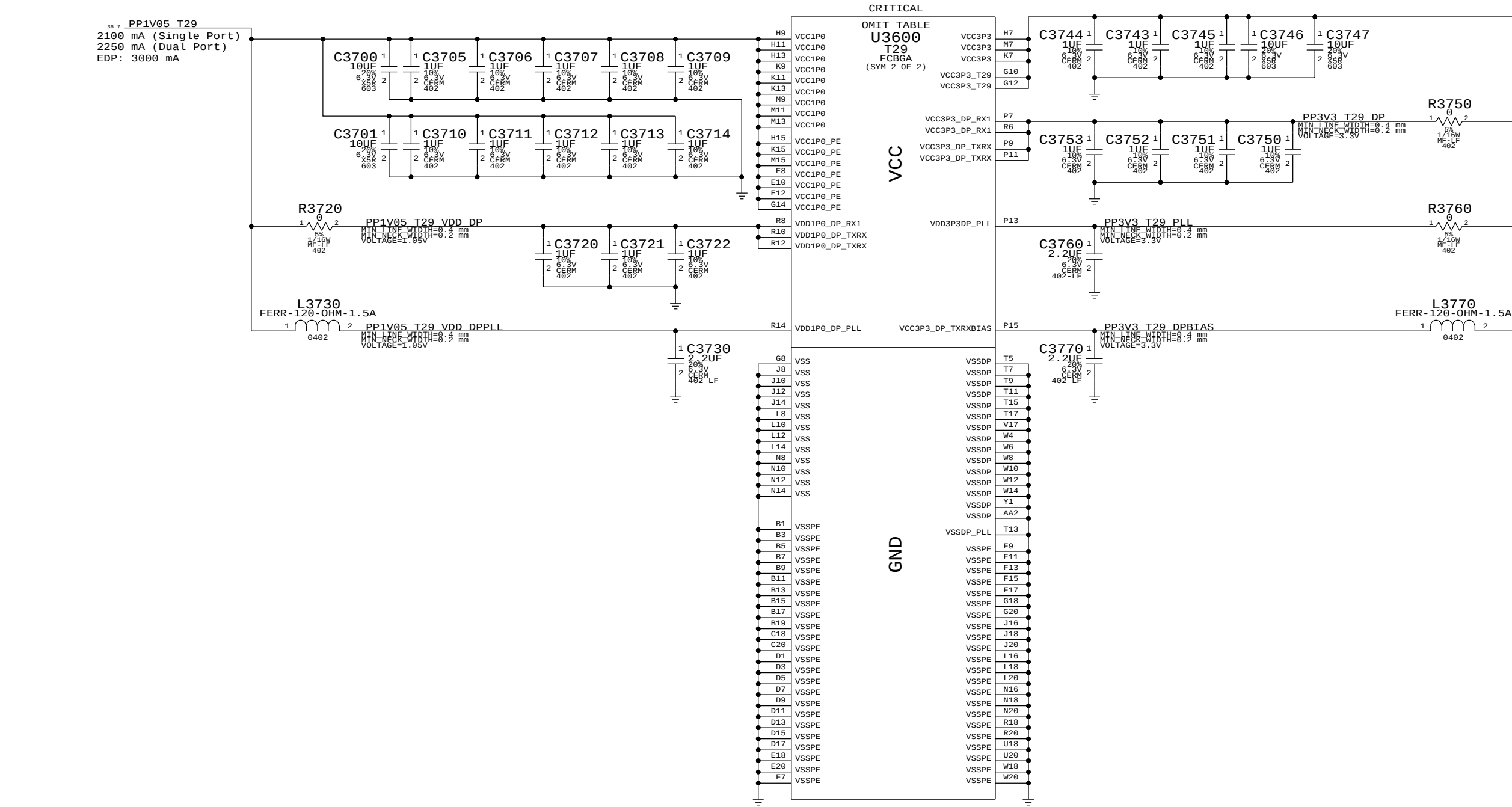
A

8 7 6 5 4 3 2 1

36 7 PP1V05 T29
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA

PP3V3 T29
135 mA (Single-Port)
152 mA (Dual-Port)
EDP: 200 mA

0-ohms are placeholders for now, replace with proper values after characterization.

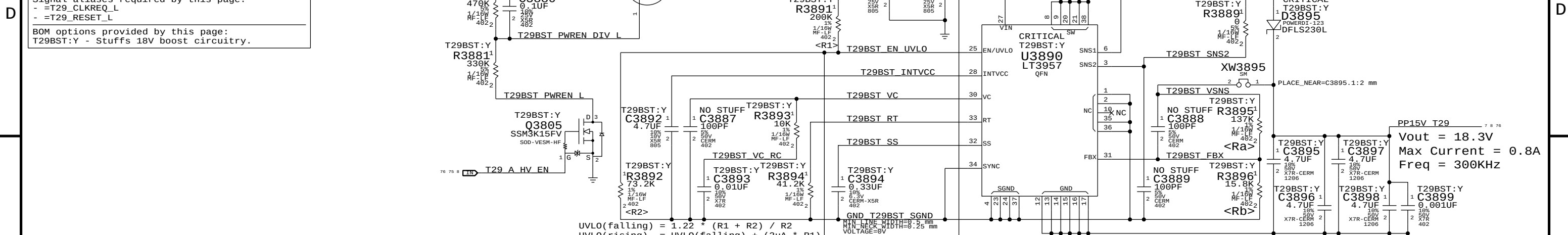


8 7 6 5 4 3 2 1

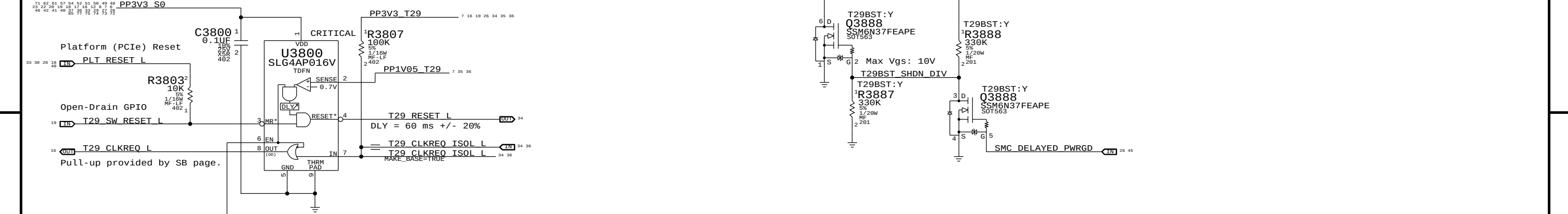
SYNC MASTER=T29		SYNC DATE=10/12/2010	
PAGE TITLE			
T29 Host (2 of 2)			
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
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8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

BOM options provided by this page:
T29BST:Y - Stuffs 18V boost circuitry.



C	Supervisor & CLKREQ# Isolation	$UVLO(\text{rising}) = UVLO(\text{falling}) + (20\mu A \times R1)$ $UVLO = 4.55V (\text{falling}), 4.95 (\text{rising})$	SGND shorted to GND inside package, no Xw necessary.	$V_{out} = 1.6V \times (1 + R_a / R_b)$	C
---	--------------------------------	--	--	---	---



B	3.3V T29 Switch	B
---	-----------------	---



	1.05V T29 Switch	
	U3815	



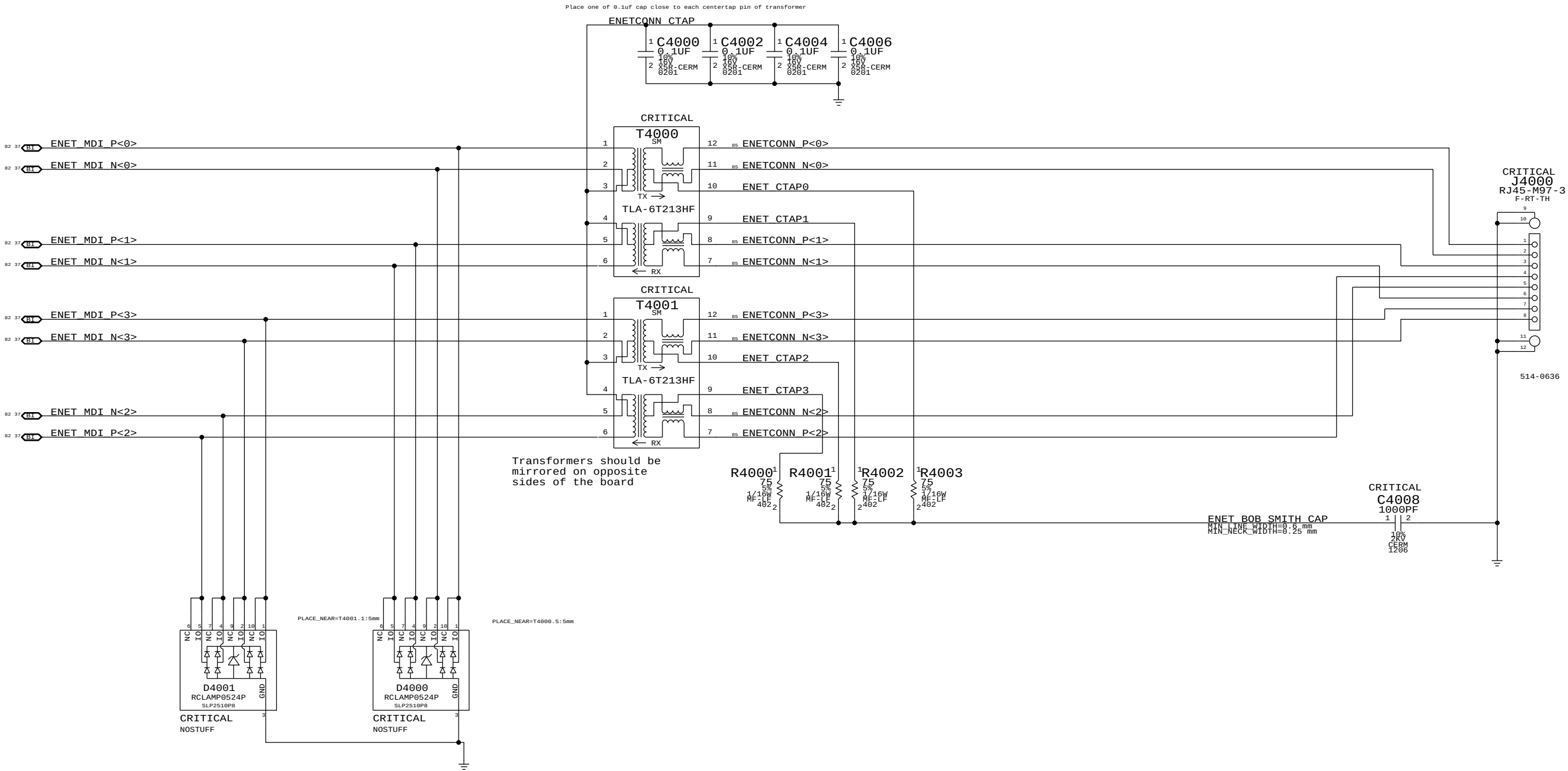
							IV ALL RIGHTS RESERVED	30 OF 80
8	7	6	5	4	3	2	1	

Page Notes

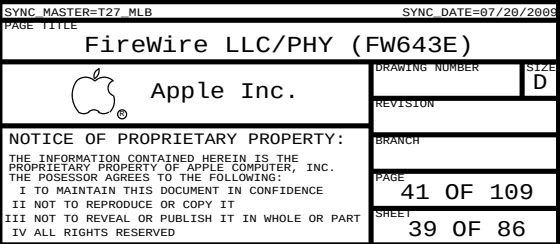
Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC_MASTER=K91_MLB		SYNC_DATE=05/26/2010	
PAGE TITLE		Ethernet Connector	
Apple Inc.		DRAWING NUMBER	SIZE D
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		PAGE	40 OF 109
		SHEET	38 OF 86



Page Notes

Power aliases required by this page:

- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

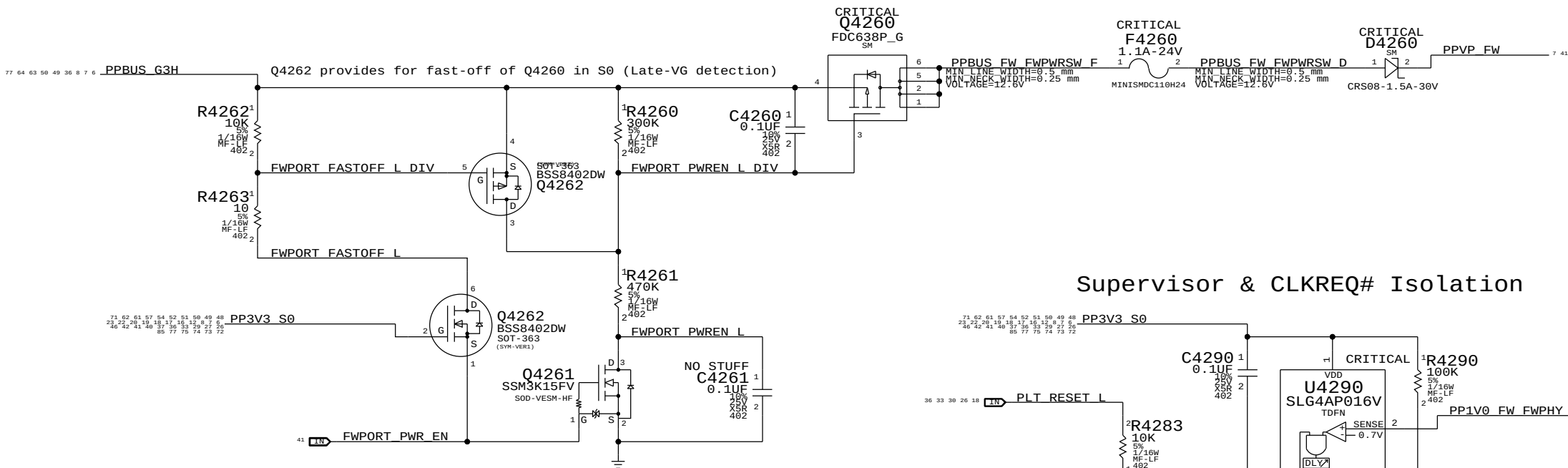
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

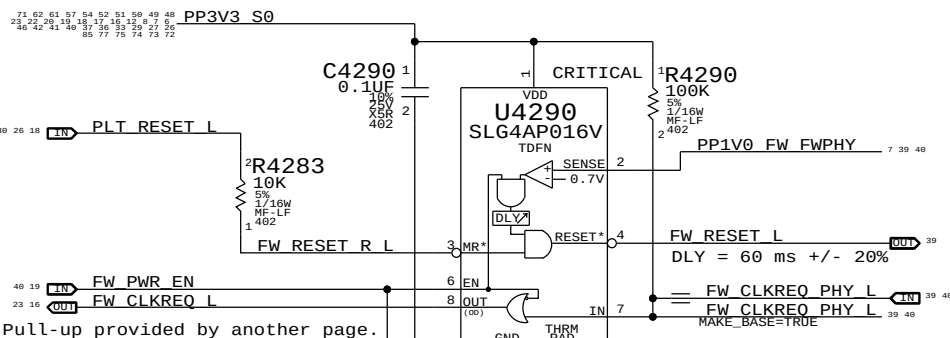
BOM options provided by this page:

(NONE)

FireWire Port Power Switch

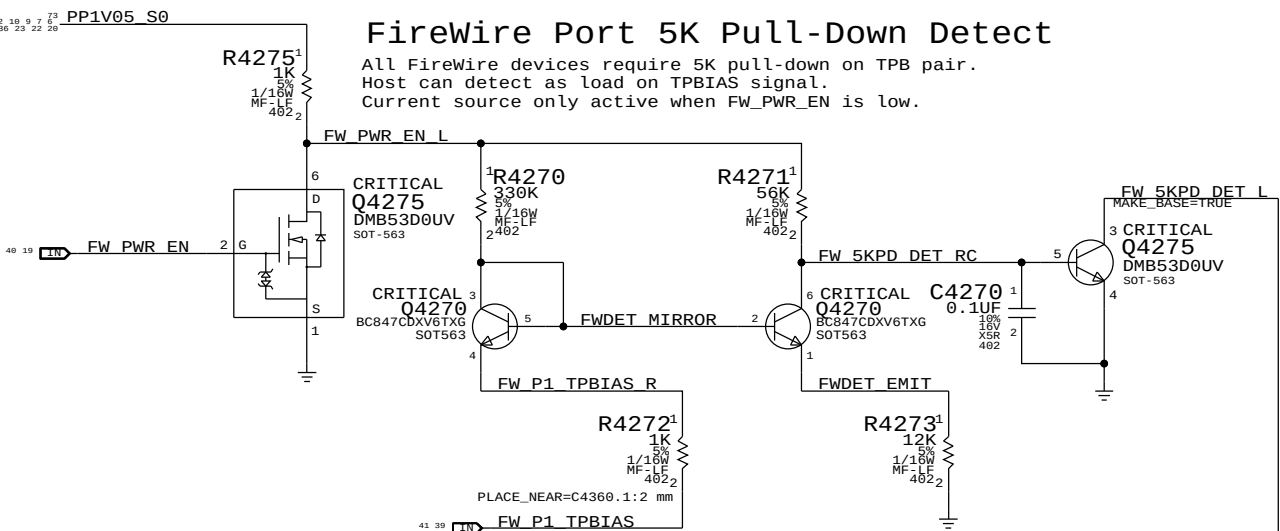


Supervisor & CLKREQ# Isolation



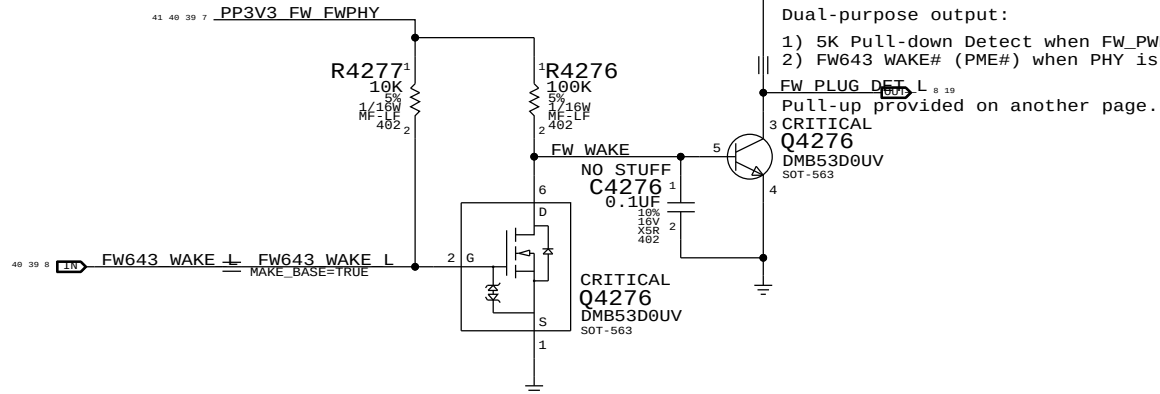
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

When PHY is powered, FW_5KPD_DET_L acts as legacy PME# signal.

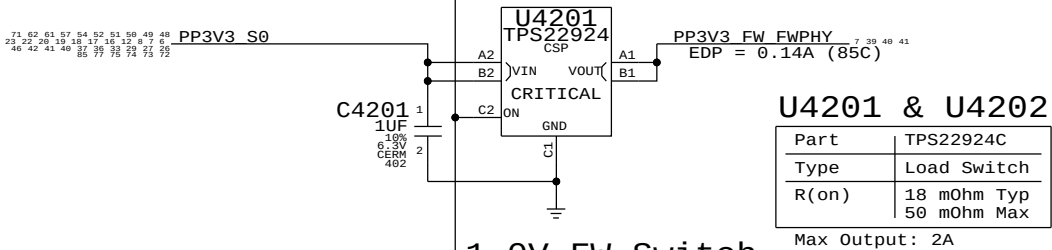


Dual-purpose output:

- 1) 5K Pull-down Detect when FW_PWR_EN is low.
- 2) FW643 WAKE# (PME#) when PHY is powered.

Pull-up provided on another page.

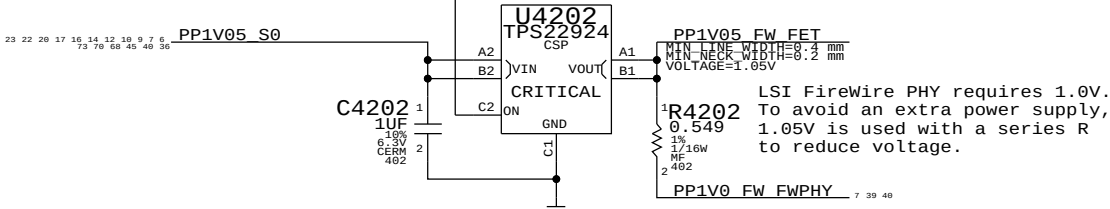
3.3V FW Switch



Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

1.0V FW Switch



LSI FireWire PHY requires 1.0V.
To avoid an extra power supply,
1.05V is used with a series R
to reduce voltage.

TEXT NOTE FOR 3.3V RAIL CURRENT CHANGED TO EDP NUMBER.

SYNC MASTER=T27_MLB		SYNC DATE=12/15/2009	
PAGE 12/12		DRAWING NUMBER	
FireWire Port & PHY Power		SIZE	
Apple Inc.		REVISION	
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Power aliases required by this page:

- =PPVP_FW_PORT1
- =PPVP_FW_PHY_CPS_FET (From Port)
- =PPVP_FW_PHY_CPS (To PHY)
- =PP3V3_FW_FWPHY
- =PP3V3_S0_FWLATEVG

Signal aliases required by this page:

- =FW_PHY_DS0
- =FW_PHY_DS1
- =FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

BOM options provided by this page:

(NONE)

1394b implementation based on Apple
FireWire Design Guide (FWDG 0.6, 5/14/03)

FW643 has internal leakage path through the TPCPS pin to VDD33.
FET blocks current to TPCPS until VDD33 is powered.

41 48 7 PPVP FW
From Port

41 41 PPVP FW CPS
MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=12.6V
MAKE_BASE=TRUE
To FW643

R4311
470K
5%
1/10W
MF-LF
482

Q4300
BSS8402DW
SOT-363
(SYM-VER1)

CPS EN L DIV

R4312
330K
5%
1/10W
MF-LF
482

CPS EN L

Q4301
BSS8402DW
SOT-363
(SYM-VER1)

41 40 39 7 PPVP3 FW FWPHY

Disabled per LSI instructions (All unused port signals TP/NC)									
41	39	6		NC FW0 TPBIAS	==	NC FW0 TPBIAS	MAKE_BASE=TRUE	NO_TEST=TRUE	39 41
82	41	39	6		NC FW0 TPAP	==	NC FW0 TPAP	MAKE_BASE=TRUE	NO_TEST=TRUE 6 39 41 82
82	41	39	6		NC FW0 TPAN	==	NC FW0 TPAN	MAKE_BASE=TRUE	NO_TEST=TRUE 39 41 82
82	41	39	6		NC FW0 TPBP	==	NC FW0 TPBP	MAKE_BASE=TRUE	NO_TEST=TRUE 6 39 41 82
82	41	39	6		NC FW0 TPBN	==	NC FW0 TPBN	MAKE_BASE=TRUE	NO_TEST=TRUE 6 39 41 82
41	39	6		NC FW2 TPBIAS	==	NC FW2 TPBIAS	MAKE_BASE=TRUE	NO_TEST=TRUE	6 39 41
41	39	6		NC FW2 TPAP	==	NC FW2 TPAP	MAKE_BASE=TRUE	NO_TEST=TRUE	6 39 41
41	39	6		NC FW2 TPAN	==	NC FW2 TPAN	MAKE_BASE=TRUE	NO_TEST=TRUE	6 39 41
41	39	6		NC FW2 TPBP	==	NC FW2 TPBP	MAKE_BASE=TRUE	NO_TEST=TRUE	6 39 41
41	39	6		NC FW2 TPBN	==	NC FW2 TPBN	MAKE_BASE=TRUE	NO_TEST=TRUE	6 39 41

Configures PHY for:
- Port "1" Bilingual (1394B)

41 48 39 7 PP3V3_FW_FWPHY

R4382¹
10K
1%
1/16W
MF-LF
402

R4380¹
10K
1%
1/16W
MF-LF
402

R4381¹
10K
1%
1/16W
MF-LF
402

FwPHY_DS0
MAKE_BASE=TRUE

FwPHY_DS1
MAKE_BASE=TRUE

FwPHY_DS2
MAKE_BASE=TRUE

FwPHY_DS0

FwPHY_DS1

FwPHY_DS2

41 48 39 7

Place close to FireWire P1

40 39 100 FW_P1_TPBIAS

1 0.33uF
10%
6.3V
CERM-XSR
492

2

SIGNAL_MODEL=EMPTY

1 R4360 56.2 1% 1/16W MF-LF 492

2

41 39 01 FW_PORT1_TPA_P

41 39 01 FW_PORT1_TPA_N

41 39 01 FW_PORT1_TPB_P

41 39 01 FW_PORT1_TPB_N

SIGNAL_MODEL=EMPTY

1 R4361 56.2 1% 1/16W MF-LF 492

2

FW_PORT1_TPB_C

1 R4364 4.99k 1% 1/10W MF-LF 492

2

1 C4364 220pF 5% 25V CERM 492

2

Cable Power

41 40 7 PPVP FW

CRITICAL
L4310
FERR-250-OHM

1 2

C4314
0.01uF
10%
50V
X7R
402

1 2

PPVP FW PORT1 F
MIN_LINE_WIDTH=0.5 mm
MIN_NECK_WIDTH=0.25 mm
VOLTAGE=32V

PORT 1
BILINGUAL

CRITICAL
J4310
1394B-M97
F-RT-TH

1 2 3 4 5 6 7 8

TPB- TPB(R)
TPB-CR
VP
NC
SC/NC
TPA- VG
TPA-CR
TPA+ TPA(R)

(FW_PORT1 TPB_N)
(FW_PORT1 BRFE)
(FW_PORT1 TPB_P)

(GND)
(FW_PORT1 TPA_N)
FW_PORT1 AREF
(FW_PORT1 TPA_P)

NC X

5/6 AND 7/8 ARE
ED FOR BETTER ROUTING)

C4319
0.1uF
10%
50V
X7R
603-1

1 2

R4319
1M
5%
1/10W
MF-LF
402

1 2

PLACE_NEAR=J4310.5:2 mm

10 11 12 13

CHASSIS
GND

514S0605

AREF needs to be isolated from all
local grounds per 1394b spec

When a bilingual device is connected to a
beta-only device, there is no DC path
between them (to avoid ground offset issue)

BREF should be hard-connected to logic
ground for speed signaling and connection

Diagram illustrating the connection of the J4310 M97 component (Critical J4310 1394B-M97 F-RT-TW) to various signals and grounds.

Connections:

- TPB (Pin 1) to TPB(R)
- TPB (Pin 9) to TPB(R)
- TPB (Pin 2) to TPB(R)
- TPB (Pin 8) to TPB(R)
- VP (Pin 7) to VP
- SC/NC (Pin 6) to SC/NC
- TPA (Pin 5) to TPA
- TPA (Pin 4) to TPA
- TPA* (Pin 3) to TPA*
- TPA* (Pin 4) to TPA*

Grounding and Power:

- (FW_PORT1 TPB_N) to GND
- (FW_PORT1 BRFF) to GND
- (FW_PORT1 TPA_P) to GND
- (FW_PORT1 TPA_N) to GND
- FW_PORT1 AREF to GND
- (FW_PORT1 TPA_P) to GND

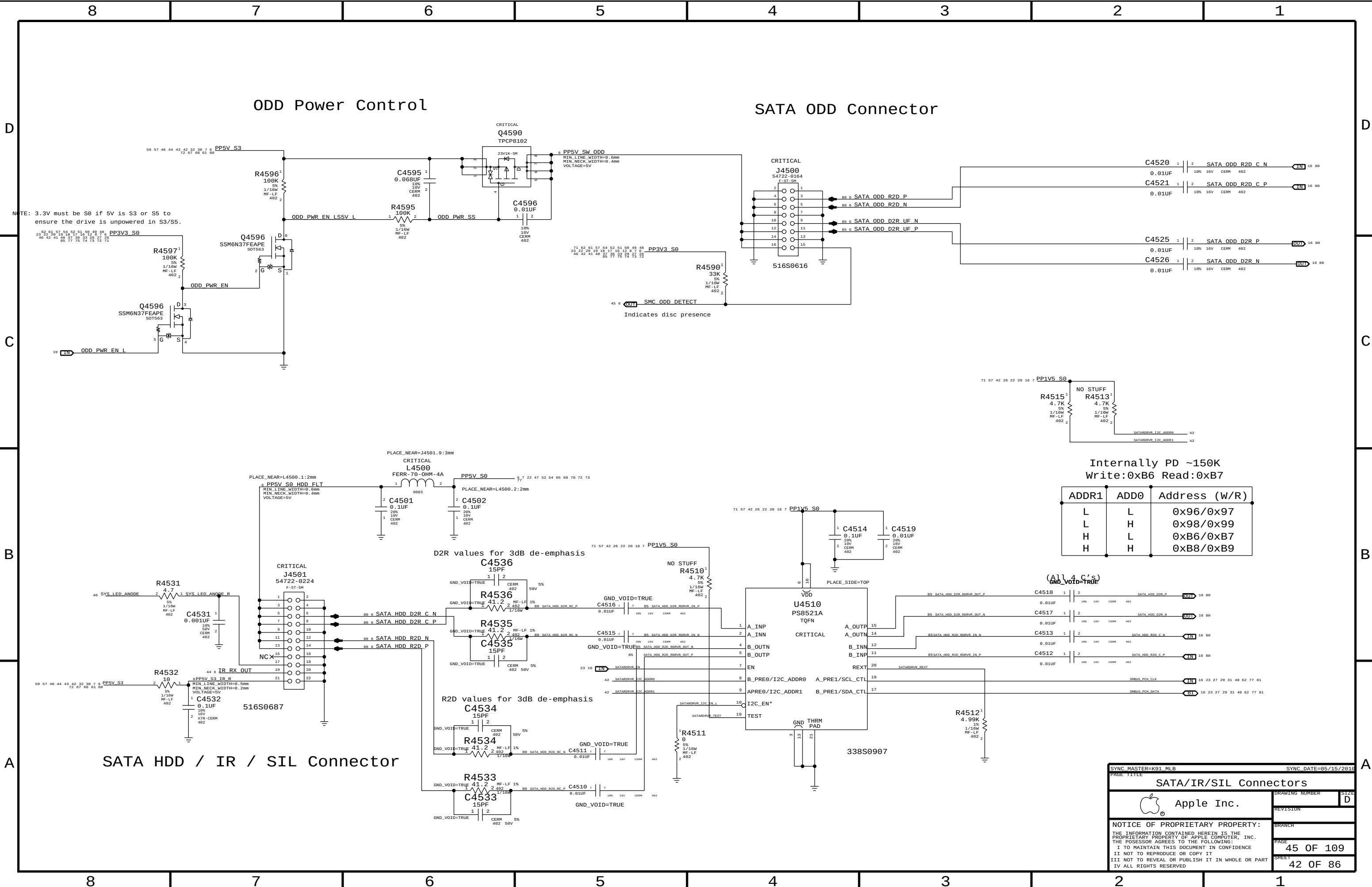
Component Details:

- C4319: 0.1uF, 10%, 50V, X7R, 603-1
- R4319: 1M, 5%, 1/4W, MF-LF, 402
- 514S0605: Chassis Ground

Notes:

- PLACE_NEAR=J4310, 5:2 mm
- TPB, TPB(R), VP, SC/NC, TPA, TPA*, TPA* are labeled as INPUT.
- TPB, TPB(R), VP, SC/NC, TPA, TPA*, TPA* are labeled as OUTPUT.

SYMC MASTER-T27 MLB		SYMC DATE=07/28/2005	
PAGE TITLE			
FireWire Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
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D

D

C

C

B

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A

A

ODD Power Control

SATA ODD Connector

NOTE: 3.3V must be S0 if 5V is S3 or S5 to ensure the drive is unpowered in S3/S5.

Internally PD ~150K
Write:0xB6 Read:0xB7

ADDR1	ADD0	Address (W/R)
L	L	0x96/0x97
L	H	0x98/0x99
H	L	0xB6/0xB7
H	H	0xB8/0xB9

SATA HDD / IR / SIL Connector

SYNC MASTER=K91 MLB

SYNC DATE=85/15/2810

PAGE TITLE

SATA/IR/SIL Connectors

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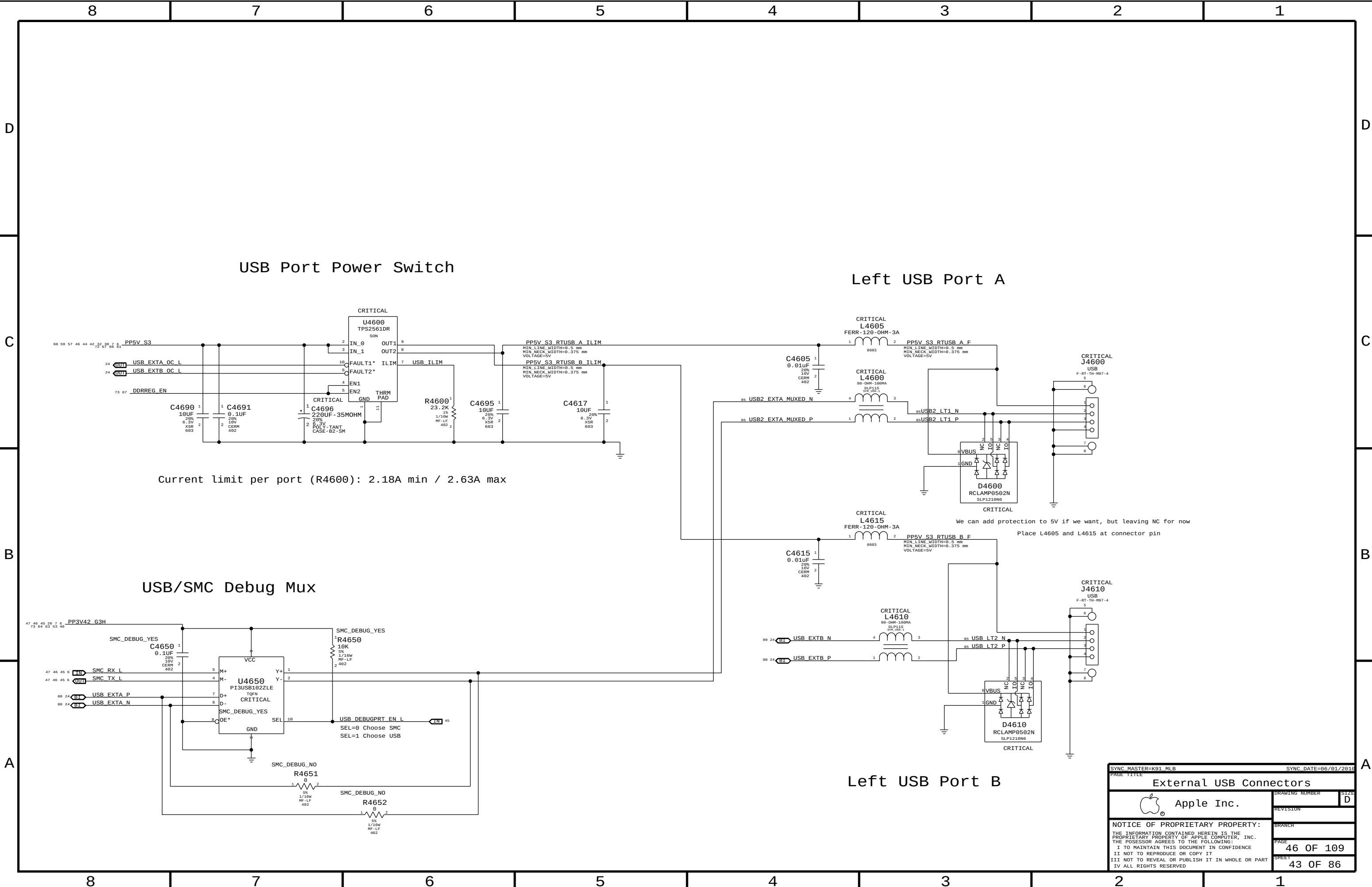
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SIZE

D

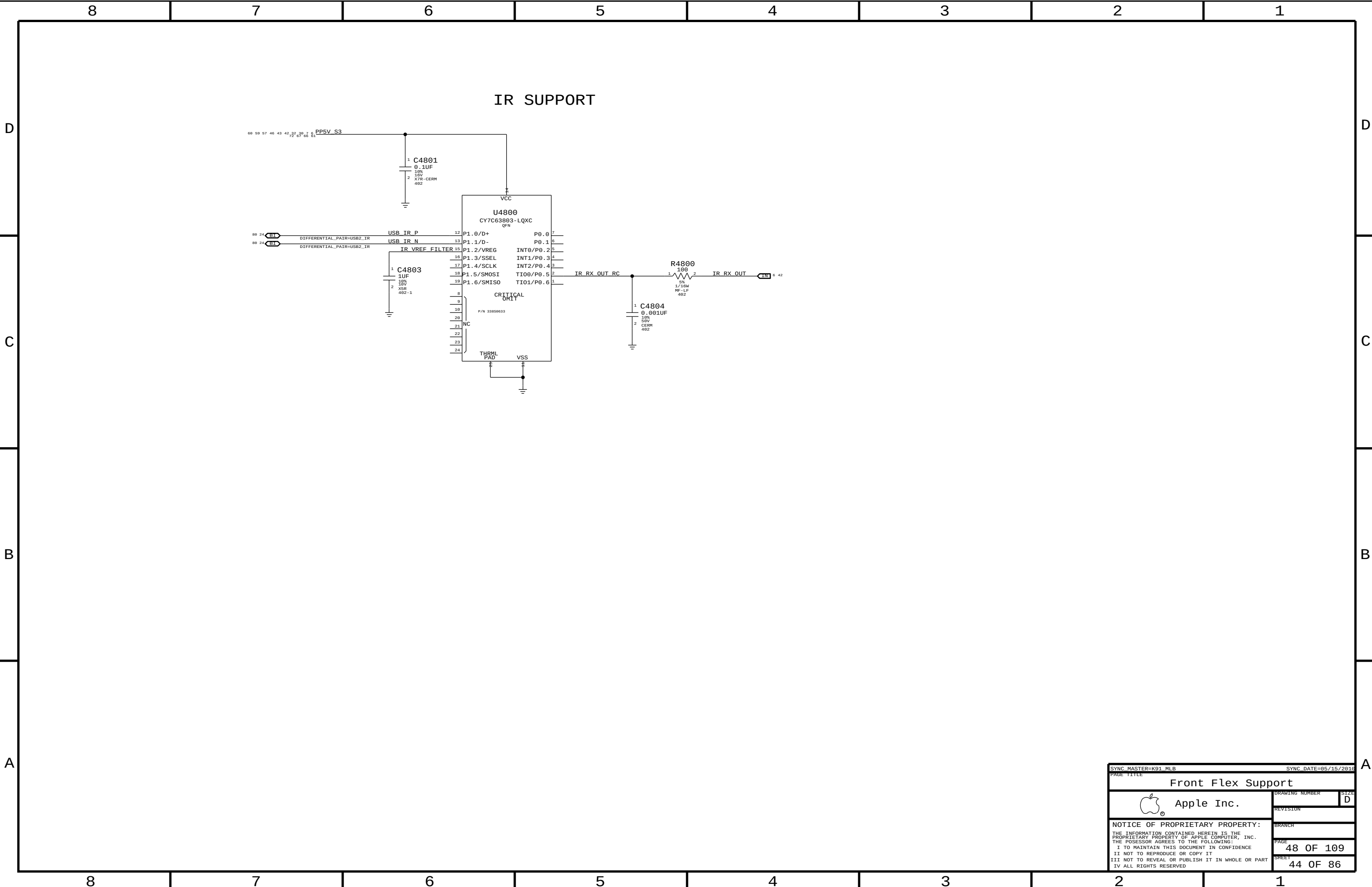
45 OF 109

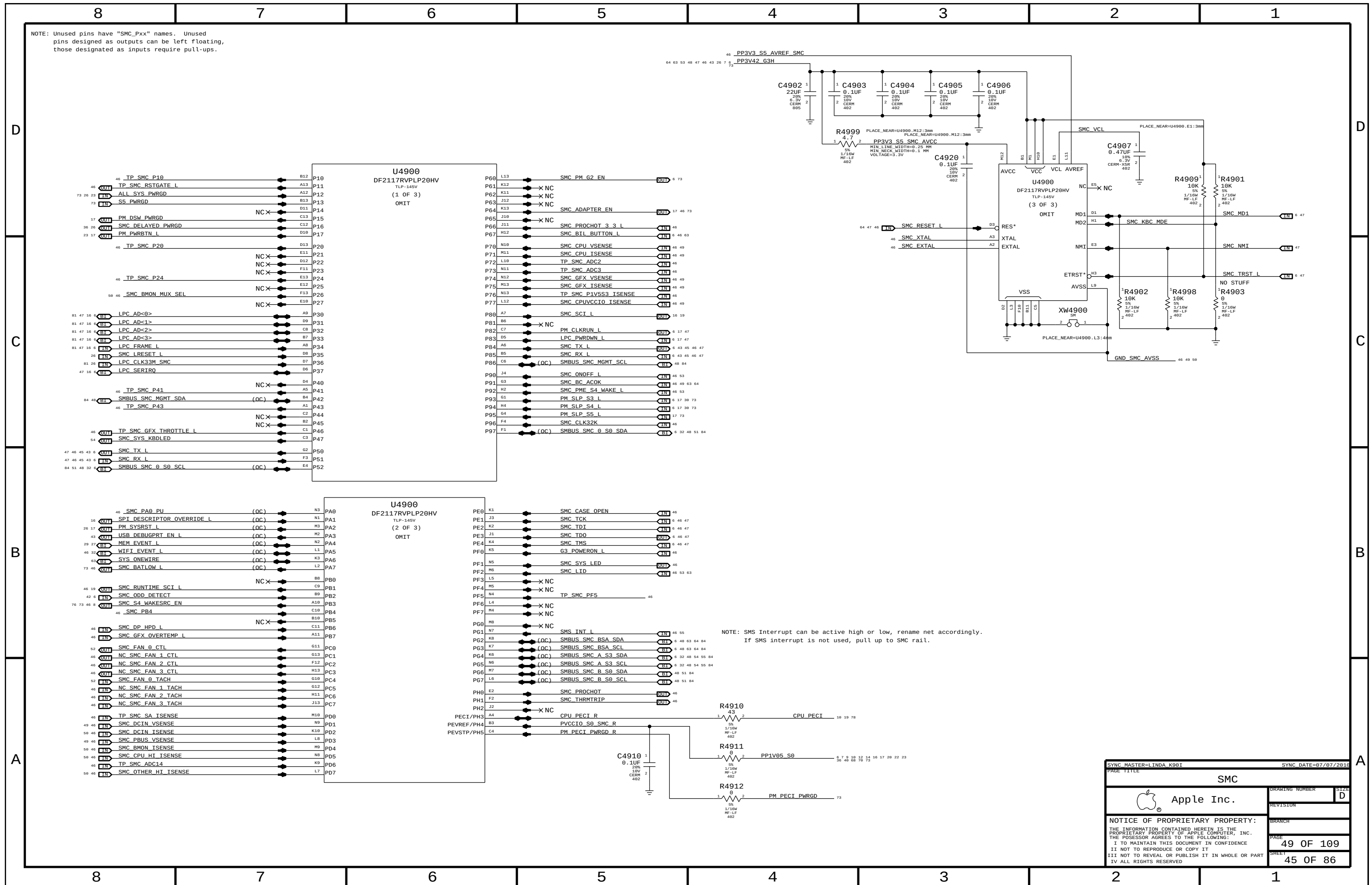
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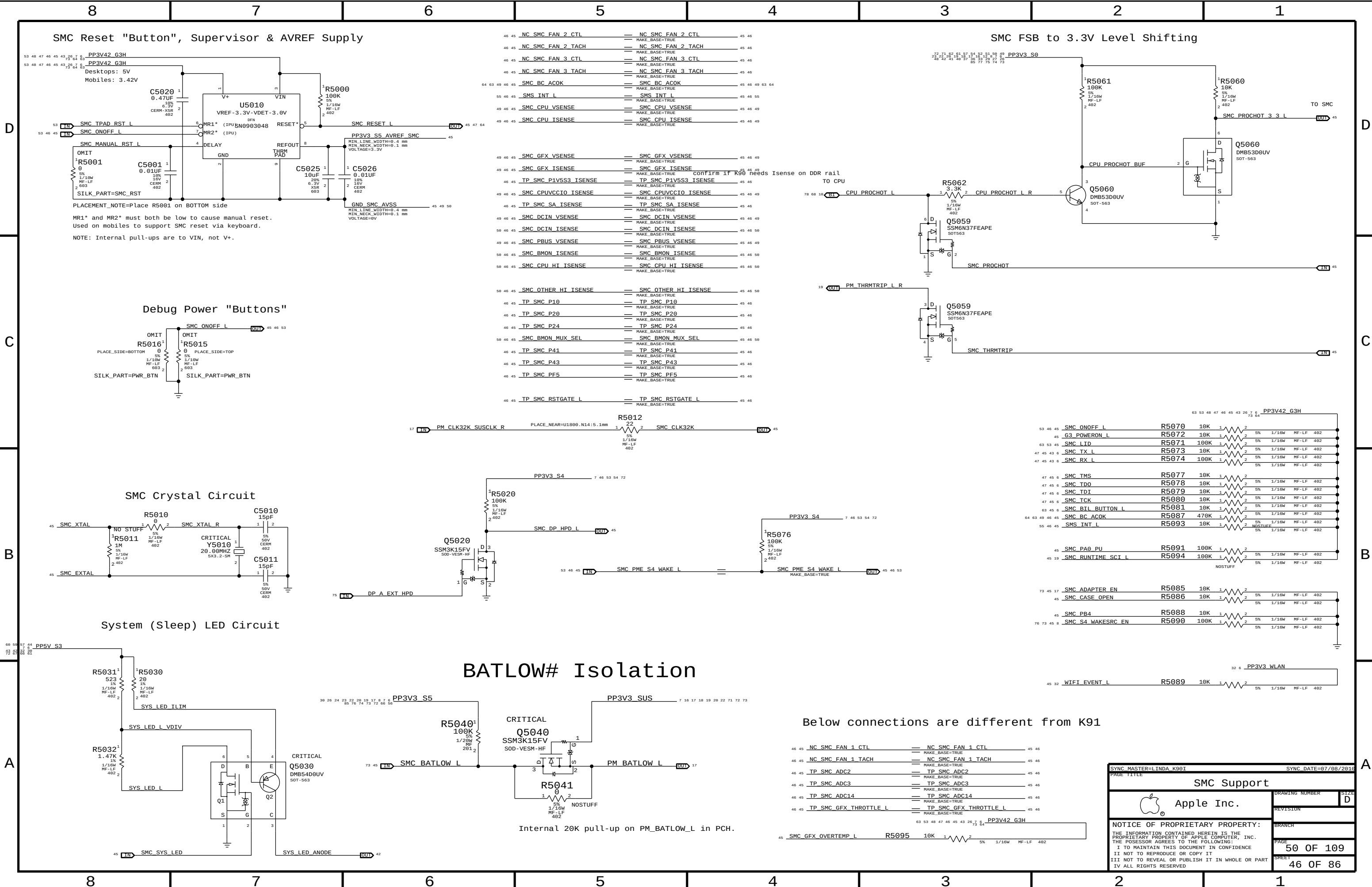


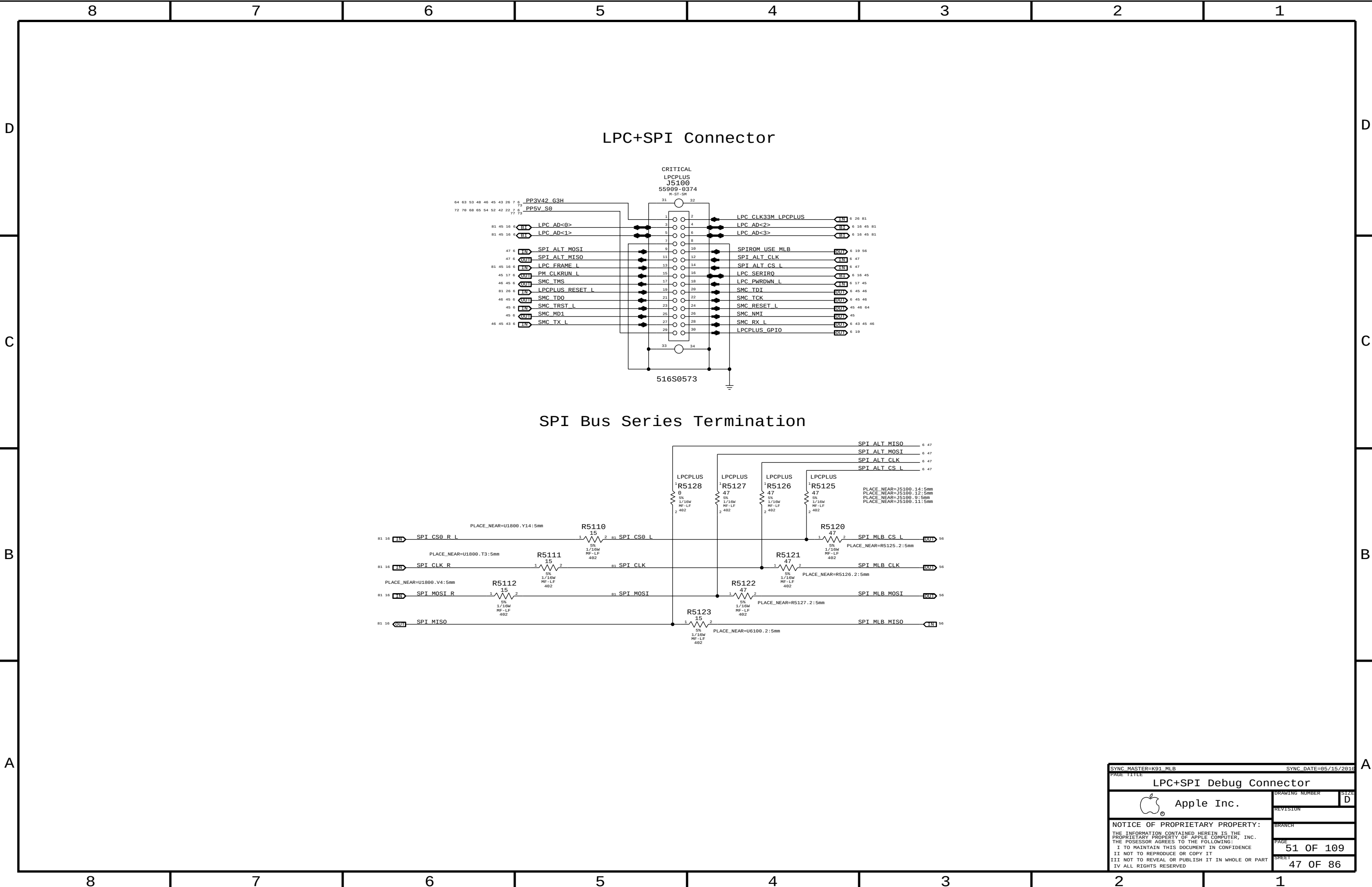
Current limit per port (R4600): 2.18A min / 2.63A max

We can add protection to 5V if we want, but leaving NC for now
Place L4605 and L4615 at connector pin

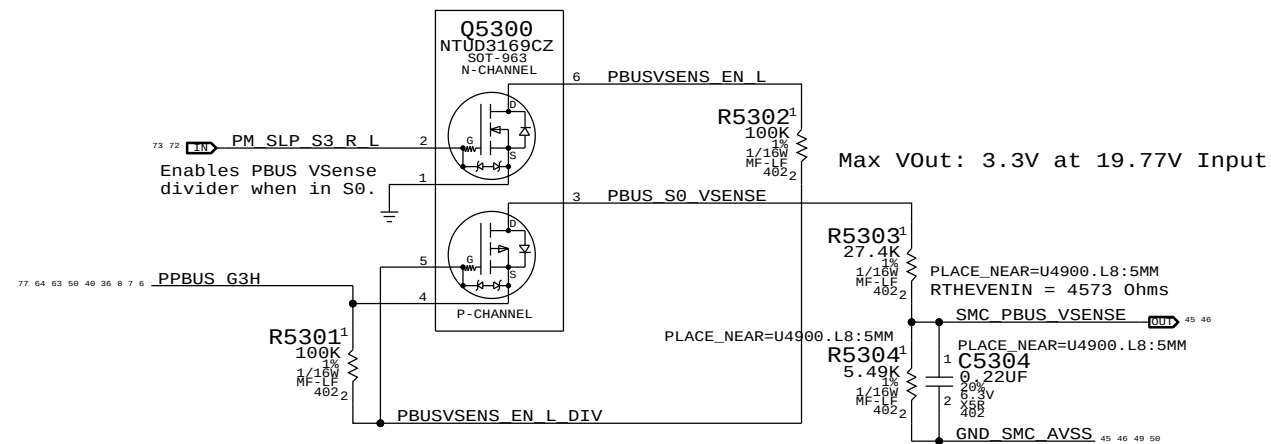




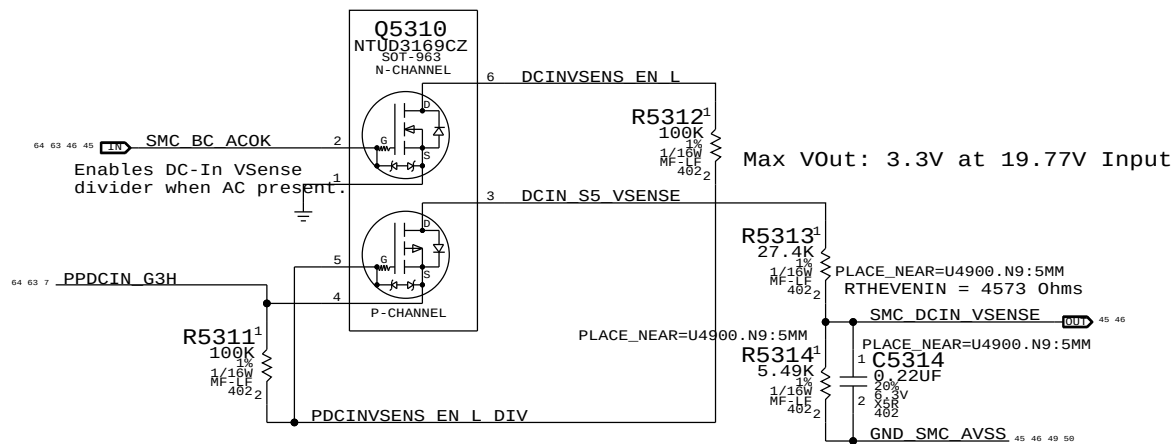




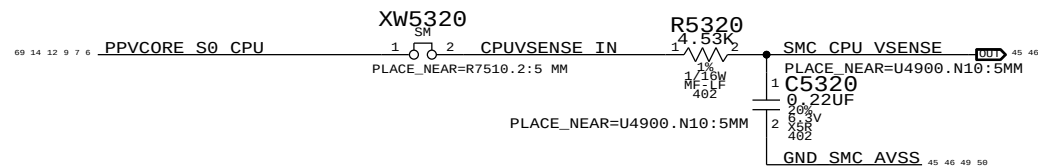
PBUS Voltage Sense Enable & Filter



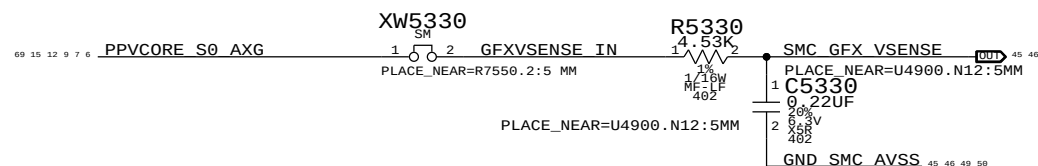
DC-In Voltage Sense Enable & Filter



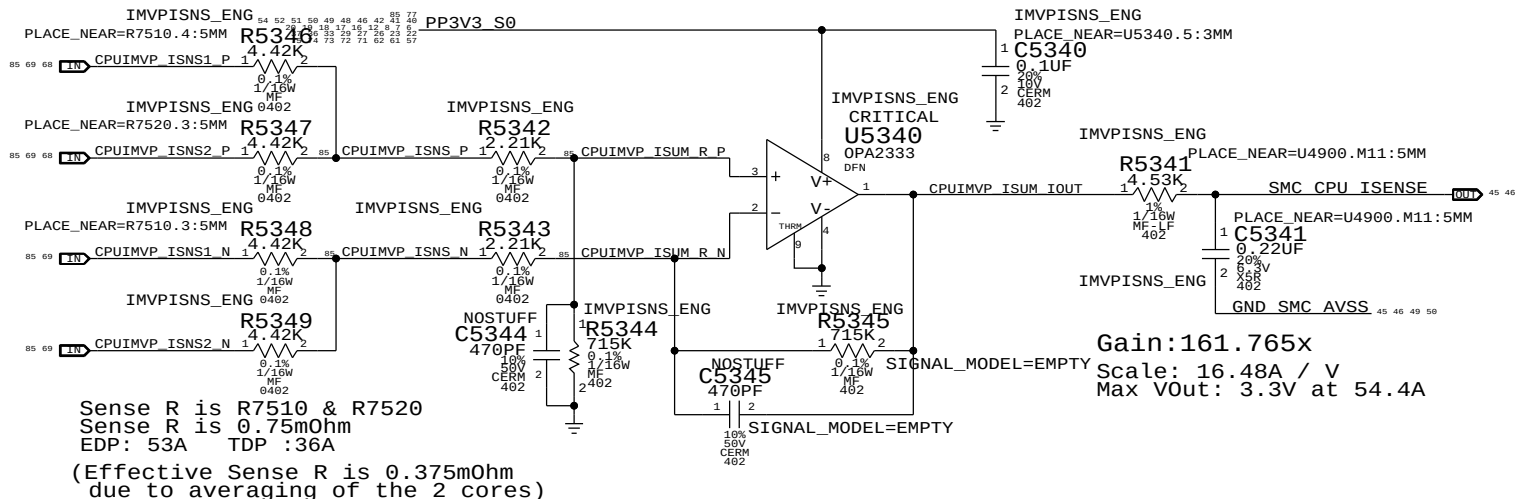
CPU Vcore Voltage Sense / Filter



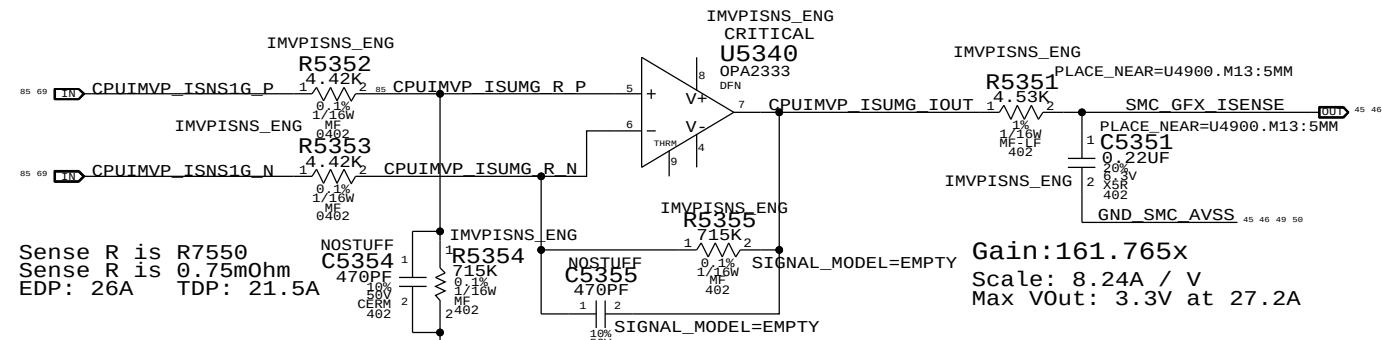
GFX/IG Vcore Voltage Sense / Filter



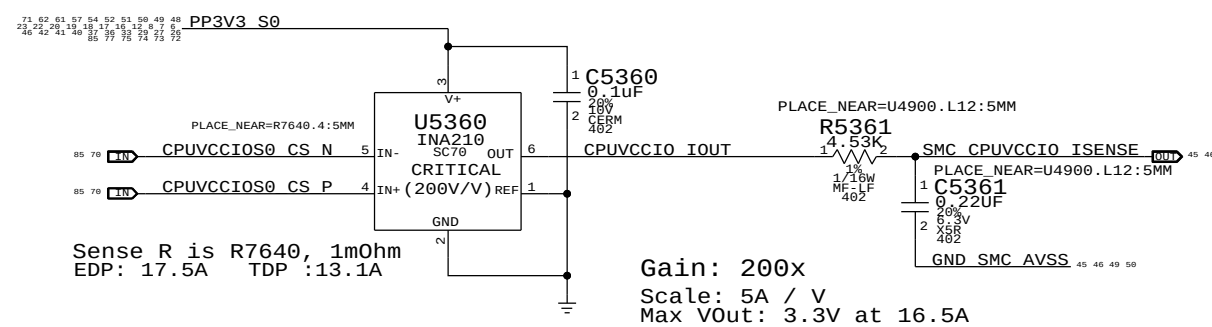
CPU VCore Load Side Current Sense / Filter



GFX/IG VCore Load Side Current Sense / Filter



CPU 1.05V VCCIO Current Sense / Filter



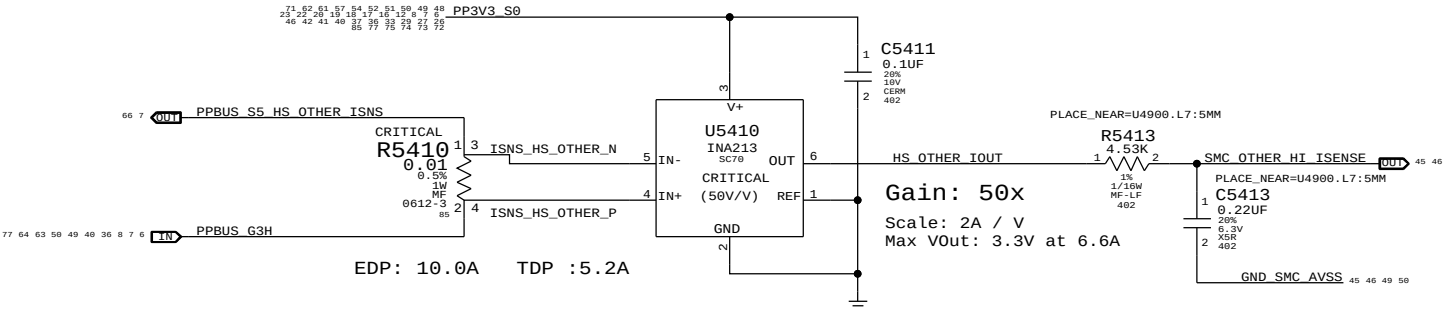
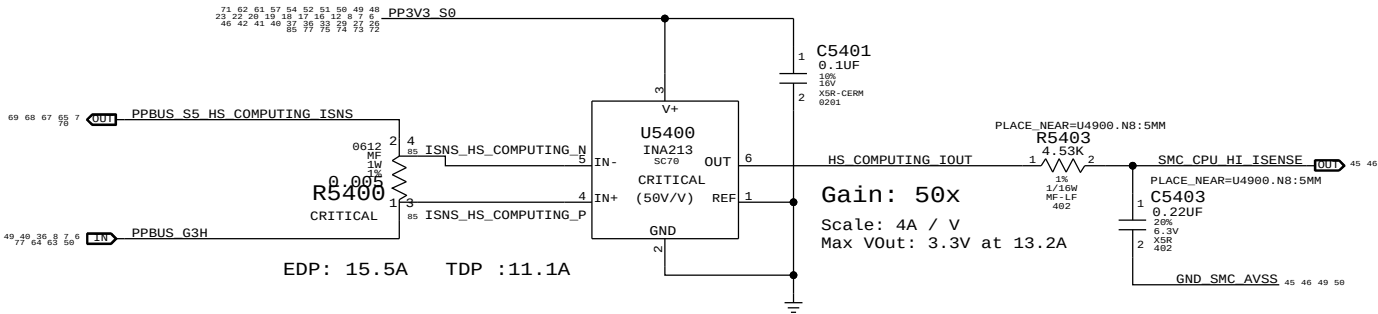
SYNC MASTER=LINDA K90I		SYNC DATE=10/22/2016	
PAGE TITLE			
Voltage & Load Side Current Sensing			
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D

D

COMPUTING High Side Current Sense / Filter

OTHER High Side Current Sense / Filter

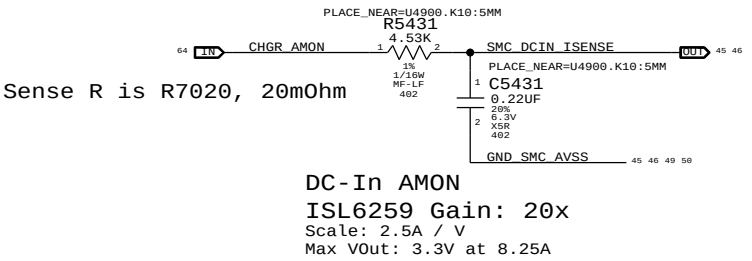
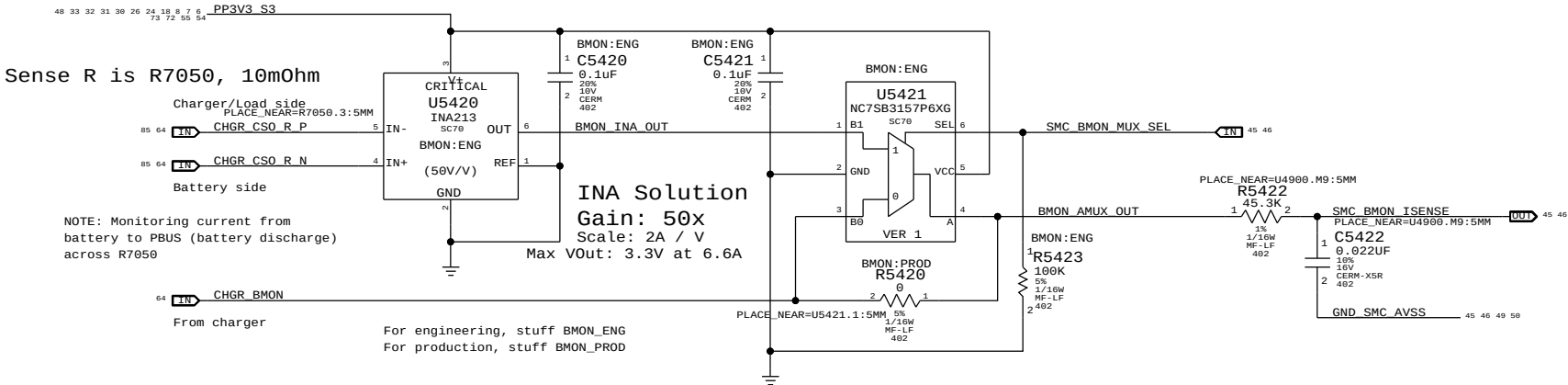


C

C

CHARGER BMON High Side (BATTERY DISCHARGE) Current Sense, MUX & Filter

DC-IN (AMON) Current Sense Filter



INA (Engineering) Solution
Gain: 50x
Scale: 2A / V
Max VOut: 3.3V at 6.6A

Charger BMON (Production) Solution
ISL6259 Gain: 36x
Scale: 2.78A / V
Max VOut: 3.3V at 9.167A

A

A

SYNC MASTER=LINDA K901		SYNC DATE=10/22/2010	
PAGE TITLE		High Side Current Sensing	
Apple Inc.		DRAWING NUMBER	SIZE
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D

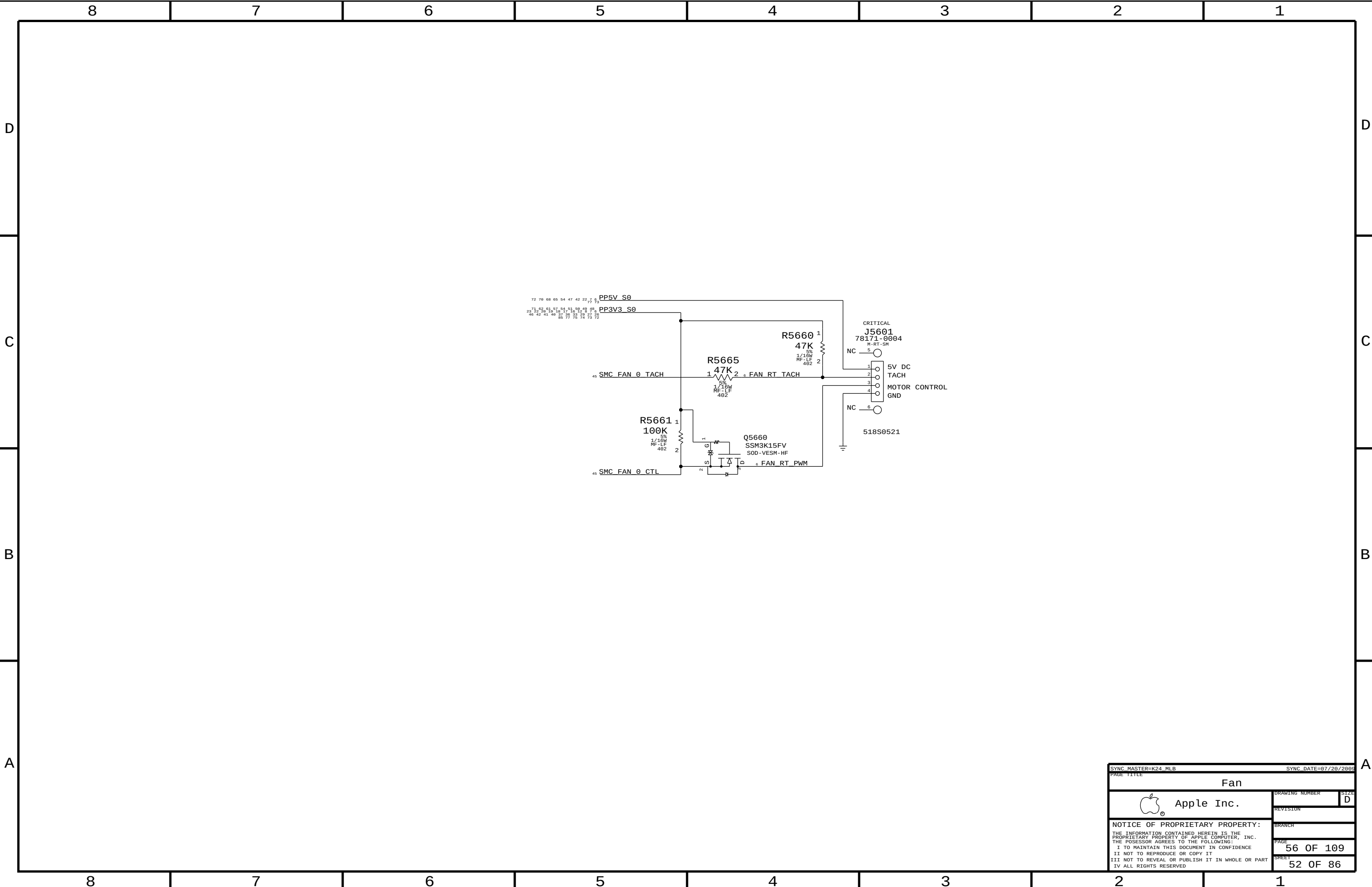


B



D





PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER

IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

Keyboard Connector

SMC Manual Reset & Isolation

Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with PSOC power to isolate when PSOC is not powered.

TPAD Buttons Disable

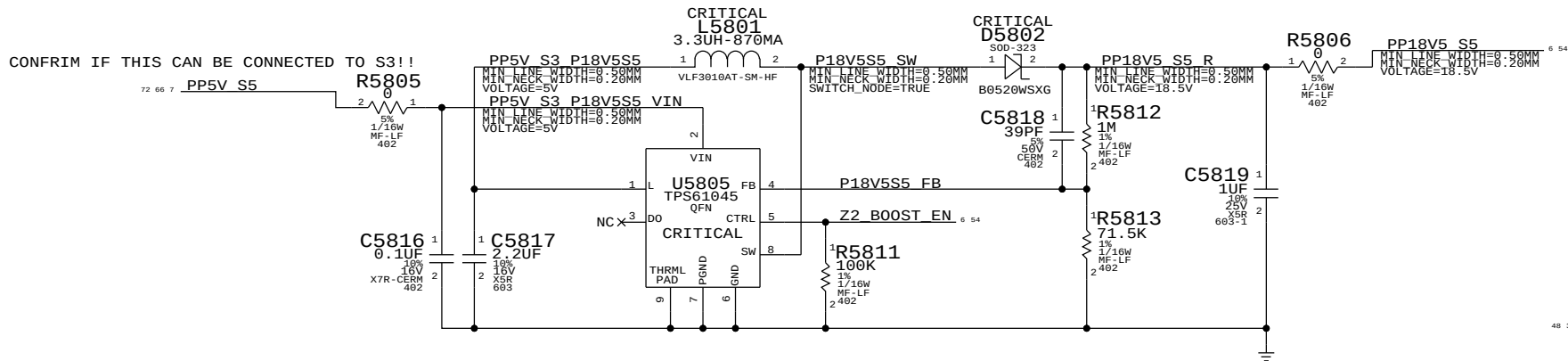
PLACE THESE COMPONENTS CLOSE TO J5800
THIS ASSUMES THERE'S A PP3V42_G3H PULL UP ON MLB

THE TPAD BUTTONS WILL BE DISABLE
WHEN THE LID IS CLOSED
LID OPEN => SMC_LID_LC ~ 3.42V
LID CLOSE => SMC_LID_LC < 0.50V

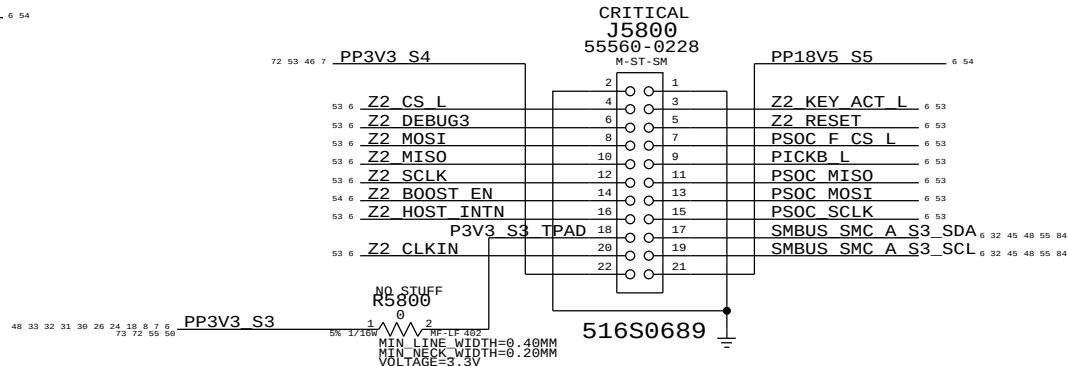
PAGE TITLE		SYNC DATE=07/12/2016	
WELLSPRING 1		SIZE D	
Apple Inc.		DRAWING NUMBER	
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PAGE 57 OF 109		SHEET 53 OF 86	

BOOSTER +18.5VDC FOR SENSORS

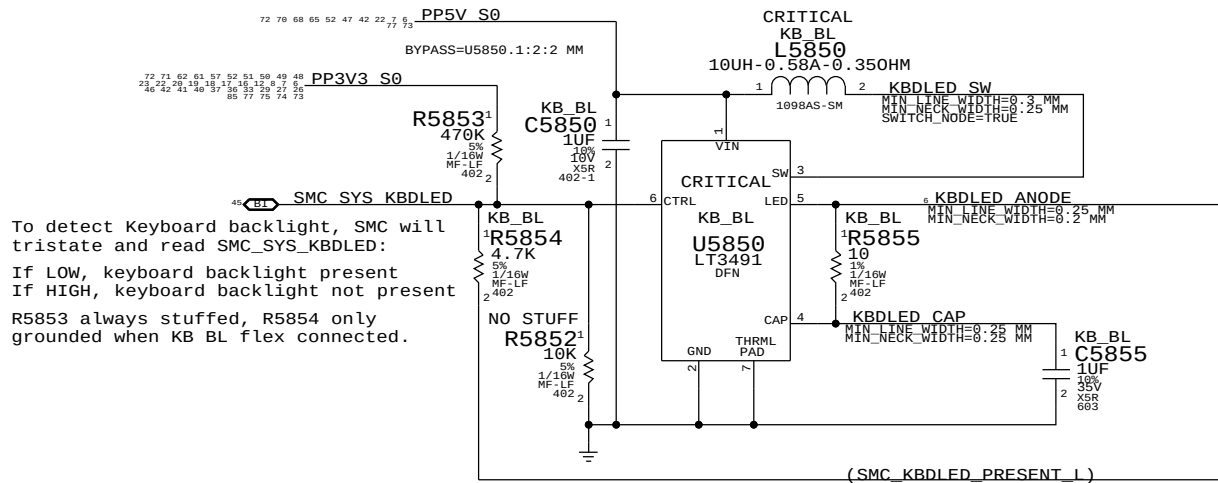
- BOOSTER DESIGN CONSIDERATION:
- POWER CONSUMPTION
 - DROOP LINE REGULATION
 - RIPPLE TO MEET ERS
 - 100-300 KHZ CLEAN SPECTRUM
 - STARTUP TIME LESS THAN 2MS
 - R5812,R5813,C5818 MODIFIED



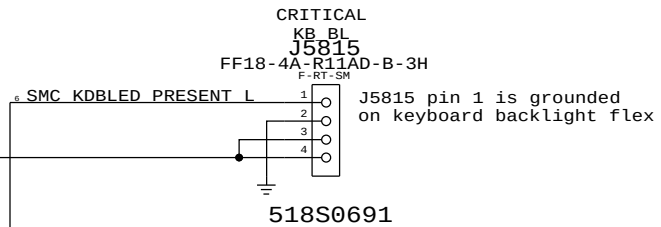
IPD Flex Connector



Keyboard Backlight Driver & Detection

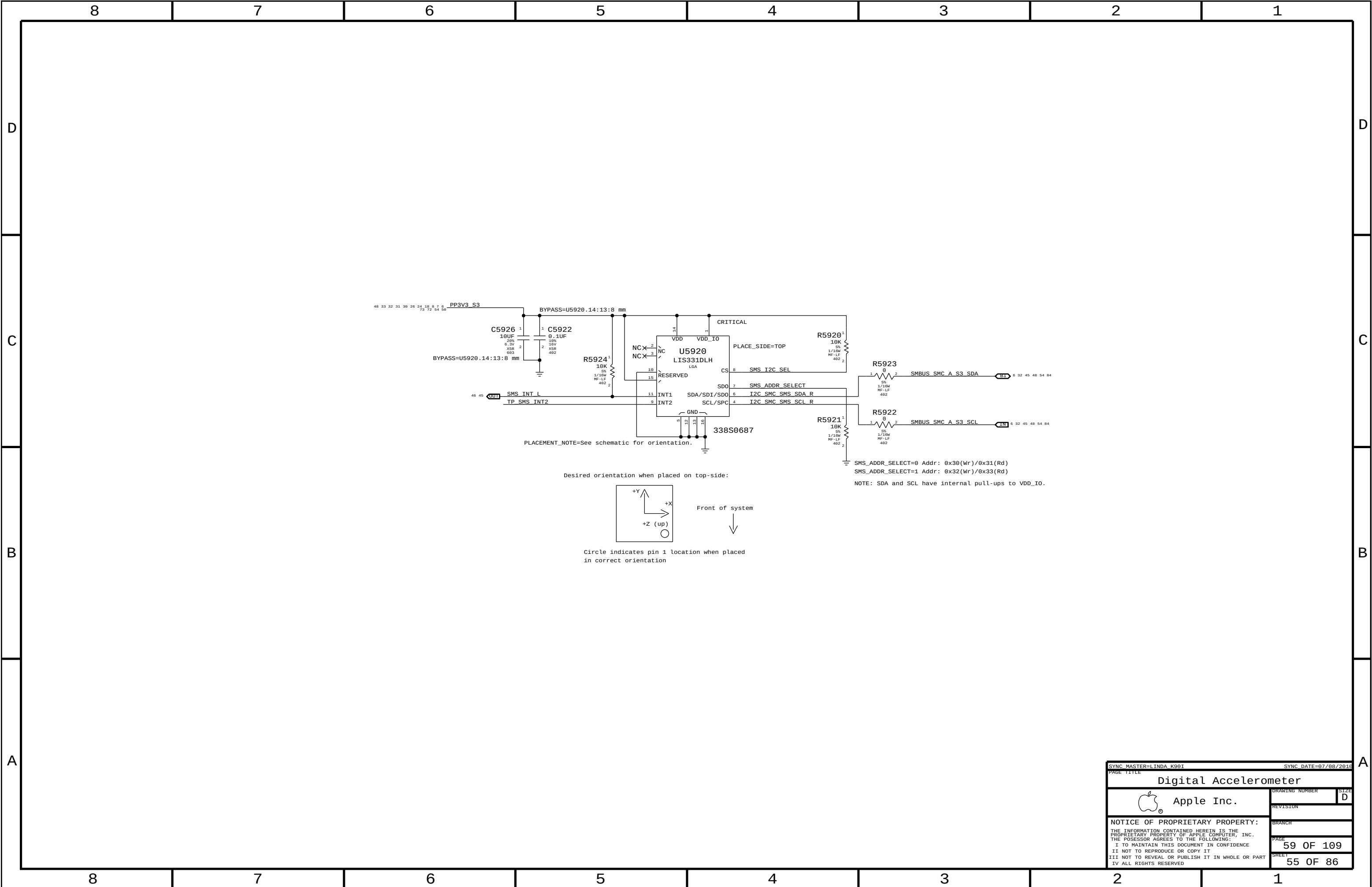


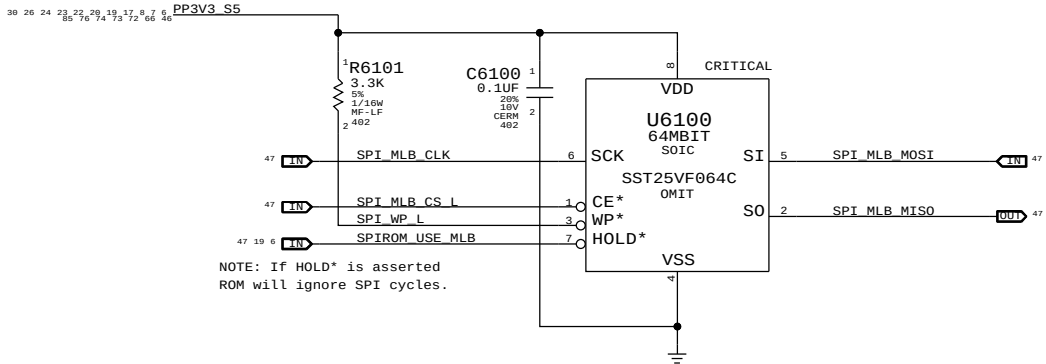
Keyboard Backlight Connector

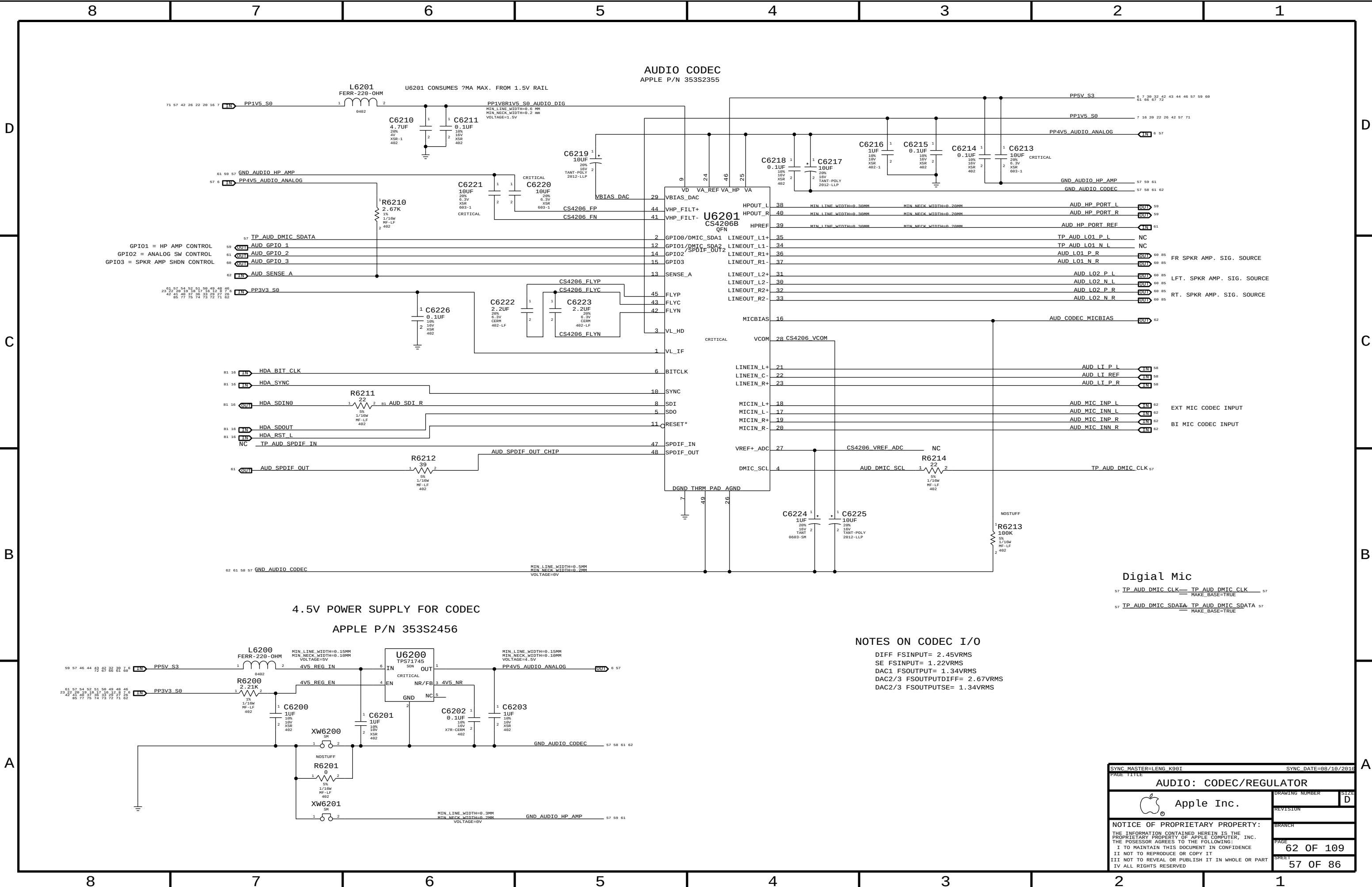


K6 NOTES : C5850 HAS BYPASS PROPERTY, SHOULD BE ADDED INCASE THIS PAGE IS SYNC'ED FROM T27

SYNC MASTER=LINDA K90I		SYNC DATE=07/12/2016	
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		PAGE	58 OF 109
		SHEET	54 OF 86







Digital Mic

57 TP_AUD_DMIC_CLK TP_AUD_DMIC_CLK 57
MAKE_BASE=TRUE

57 TP_AUD_DMIC_SDATA TP_AUD_DMIC_SDATA 57
MAKE_BASE=TRUE

NOTES ON CODEC I/O

DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC MASTER=LENG K901		SYNC DATE=08/10/2010	
PAGE TITLE			
AUDIO: CODEC/REGULATOR			
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
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8 7 6 5 4 3 2 1

D

C

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8 7 6 5 4 3 2 1

AUDIO: LINE INPUT FILTER

Apple Inc.

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8

LINE INPUT VOLTAGE DIVIDER

CODEC RIN = 20K OHMS
NET RIN = 10.36K OHMS (INCLUDING PULL-DOWNS AT ANALOG SWITCH COM PINS)
FC_HP = 3.6 HZ
FC_LP = 43KHZ
VIN = 2VRMS, CODEC VIN = 1.14 VRMS

CRITICAL
C6301
2.2UF
1 2
20%
10V
X5R-CERM
402

AUD LI L DIV
MIN LINE WIDTH=.1MM
MIN NECK WIDTH=.1MM

R6301
7.87K
1 2
1%
1/16W
MF-LF
402

AUD LI L
MIN LINE WIDTH=.1MM
MIN NECK WIDTH=.1MM

AUD LI REF
MIN LINE WIDTH=.1MM
MIN NECK WIDTH=.1MM

CRITICAL
C6302
2.2UF
1 2
20%
10V
X5R-CERM
402

AUD LI GND
MIN LINE WIDTH=.1MM

R6300
10
1 2
1%
1/16W
MF-LF
402

GND AUDIO CODEC

CRITICAL
C6312
2.2UF
1 2
20%
10V
X5R-CERM
402

CRITICAL
C6313
2.2UF
1 2
20%
10V
X5R-CERM
402

AUD LI R DIV
MIN LINE WIDTH=.1MM
MIN NECK WIDTH=.1MM

R6311
7.87K
1 2
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1/16W
MF-LF
402

AUD LI R
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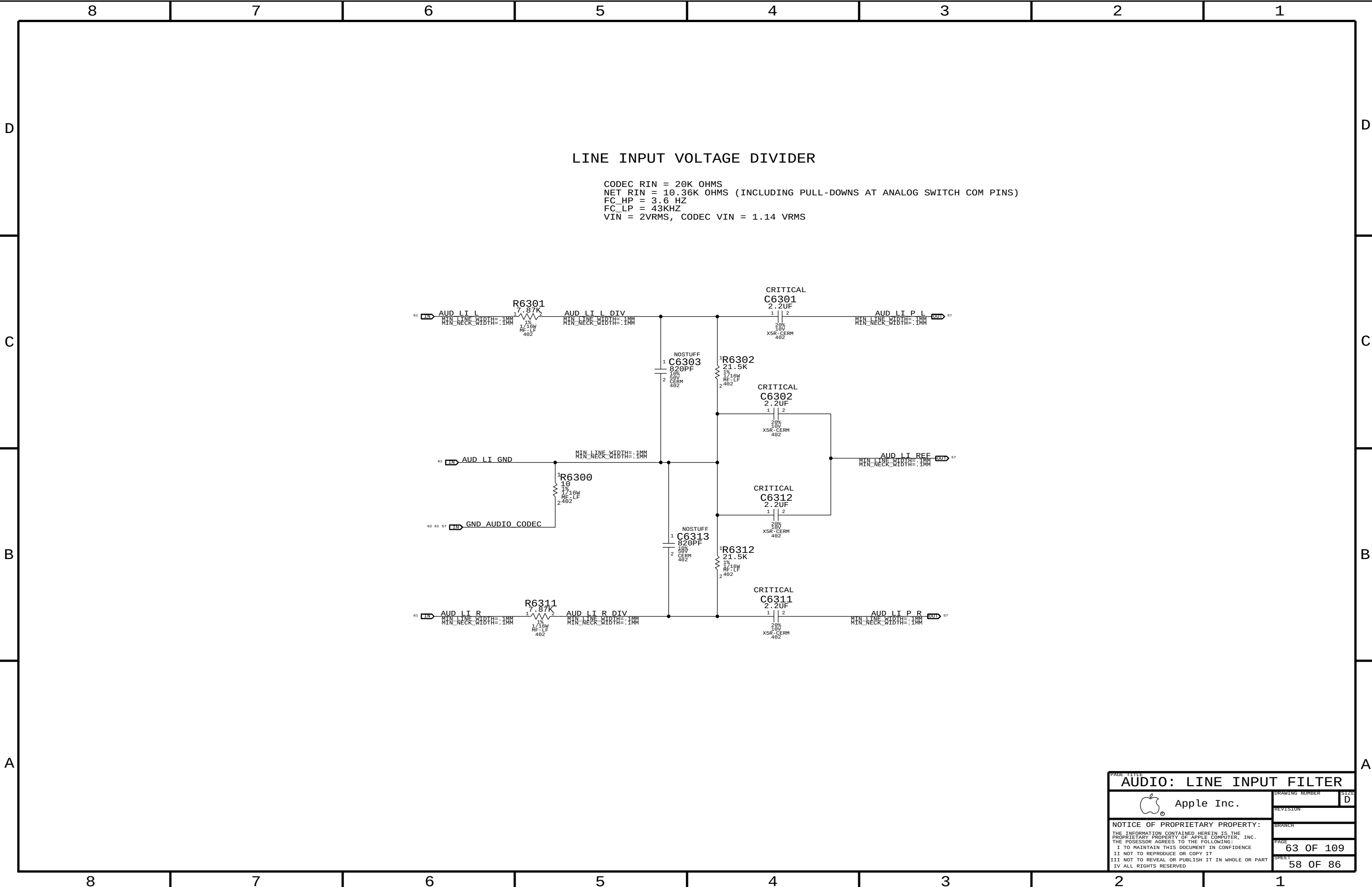
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X5R-CERM
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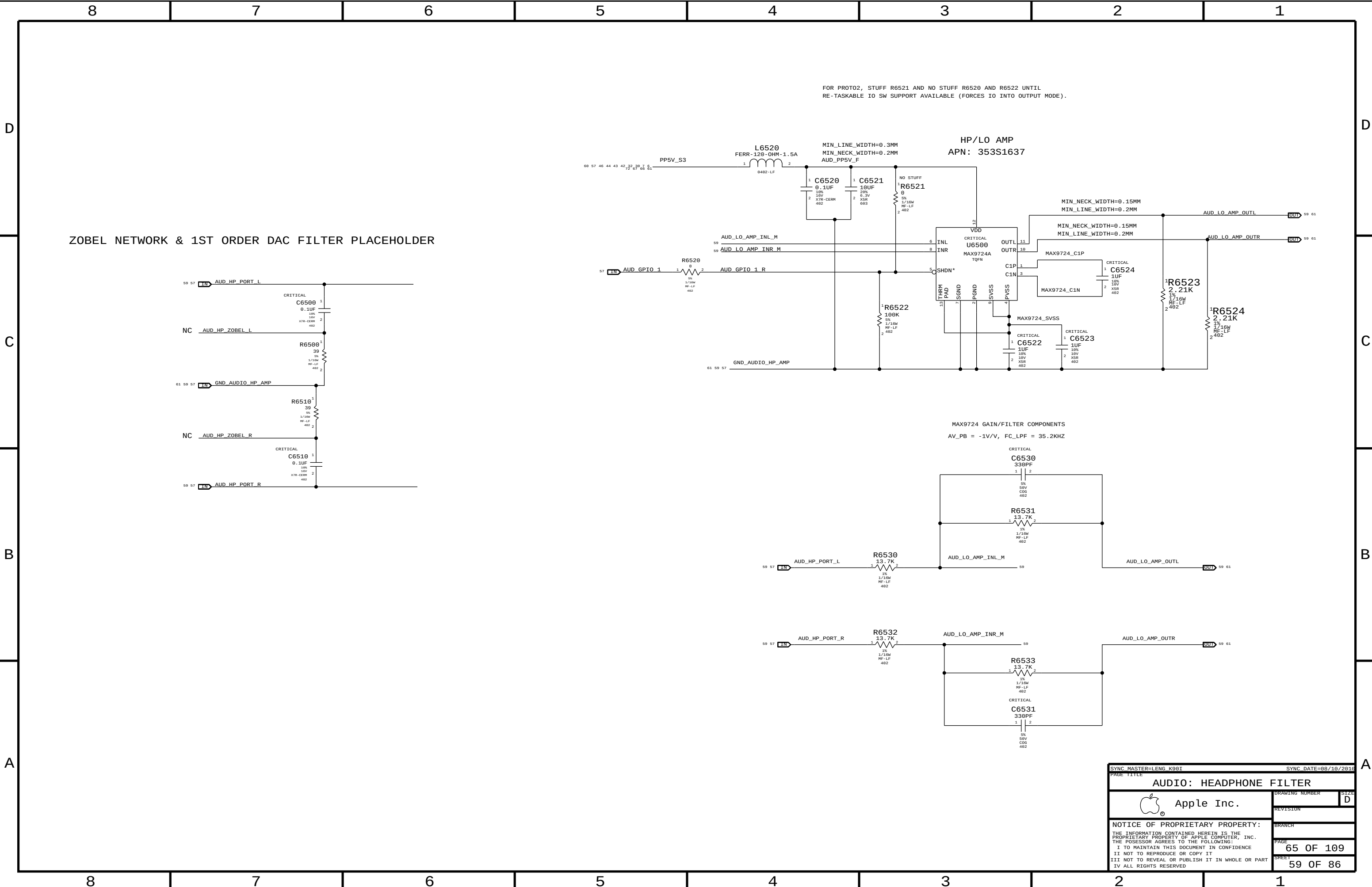
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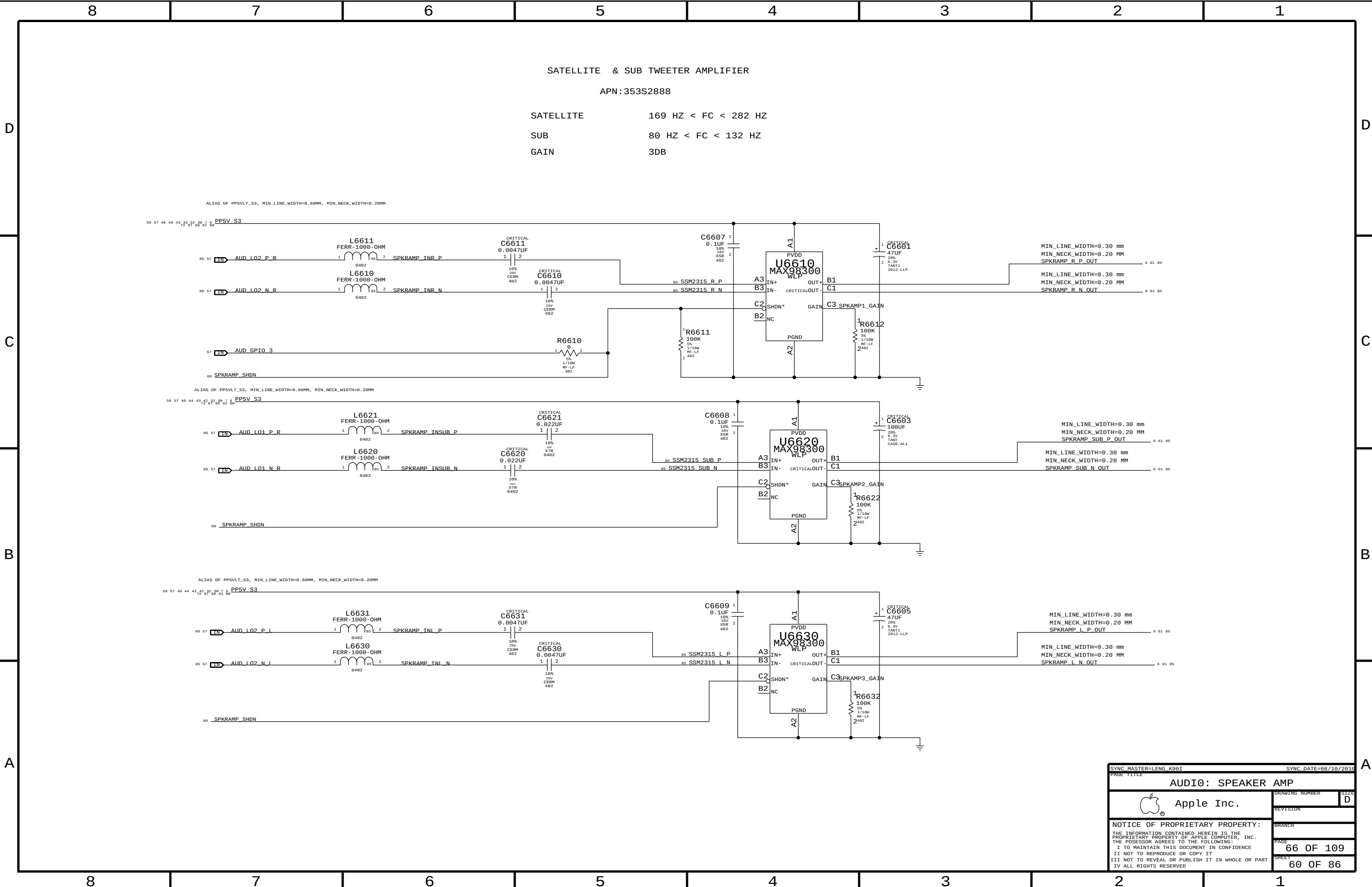


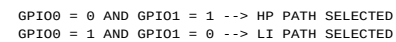
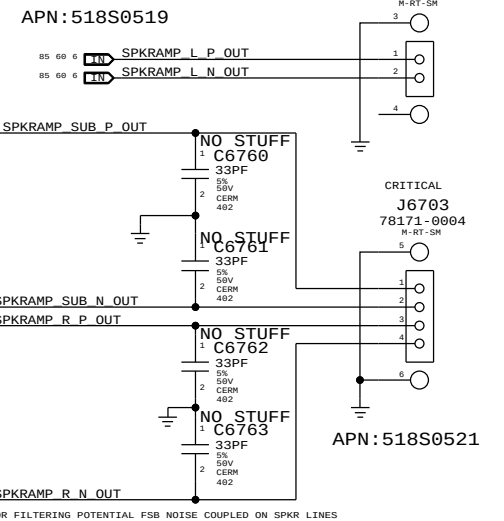
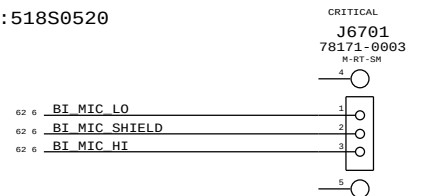
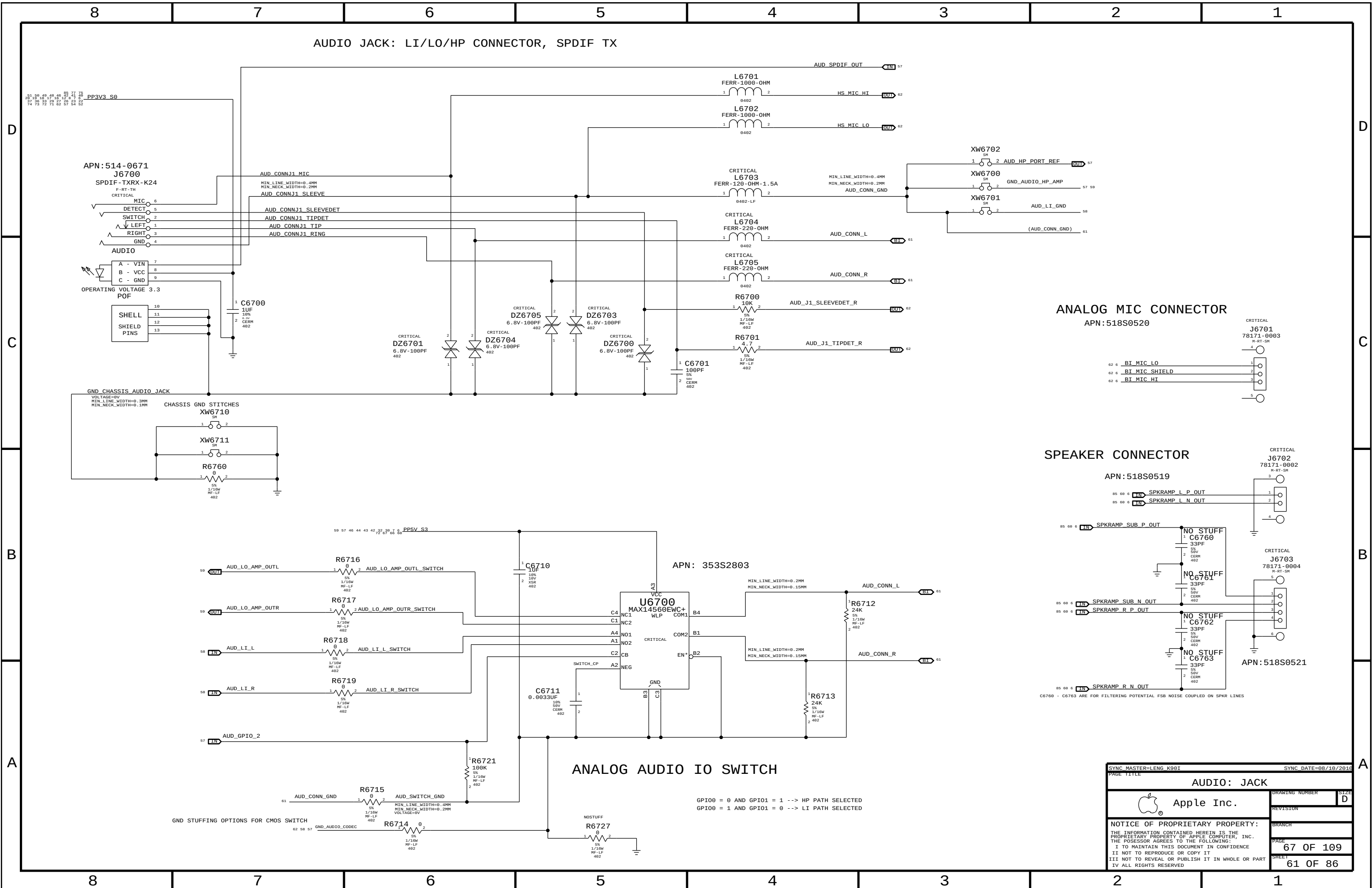
FOR PROT02, STUFF R6521 AND NO STUFF R6520 AND R6522 UNTIL
RE-TASKABLE IO SW SUPPORT AVAILABLE (FORCES IO INTO OUTPUT MODE).

HP/LO AMP
APN: 353S1637

MAX9724 GAIN/FILTER COMPONENTS
AV_PB = -1V/V, FC_LPF = 35.2KHZ

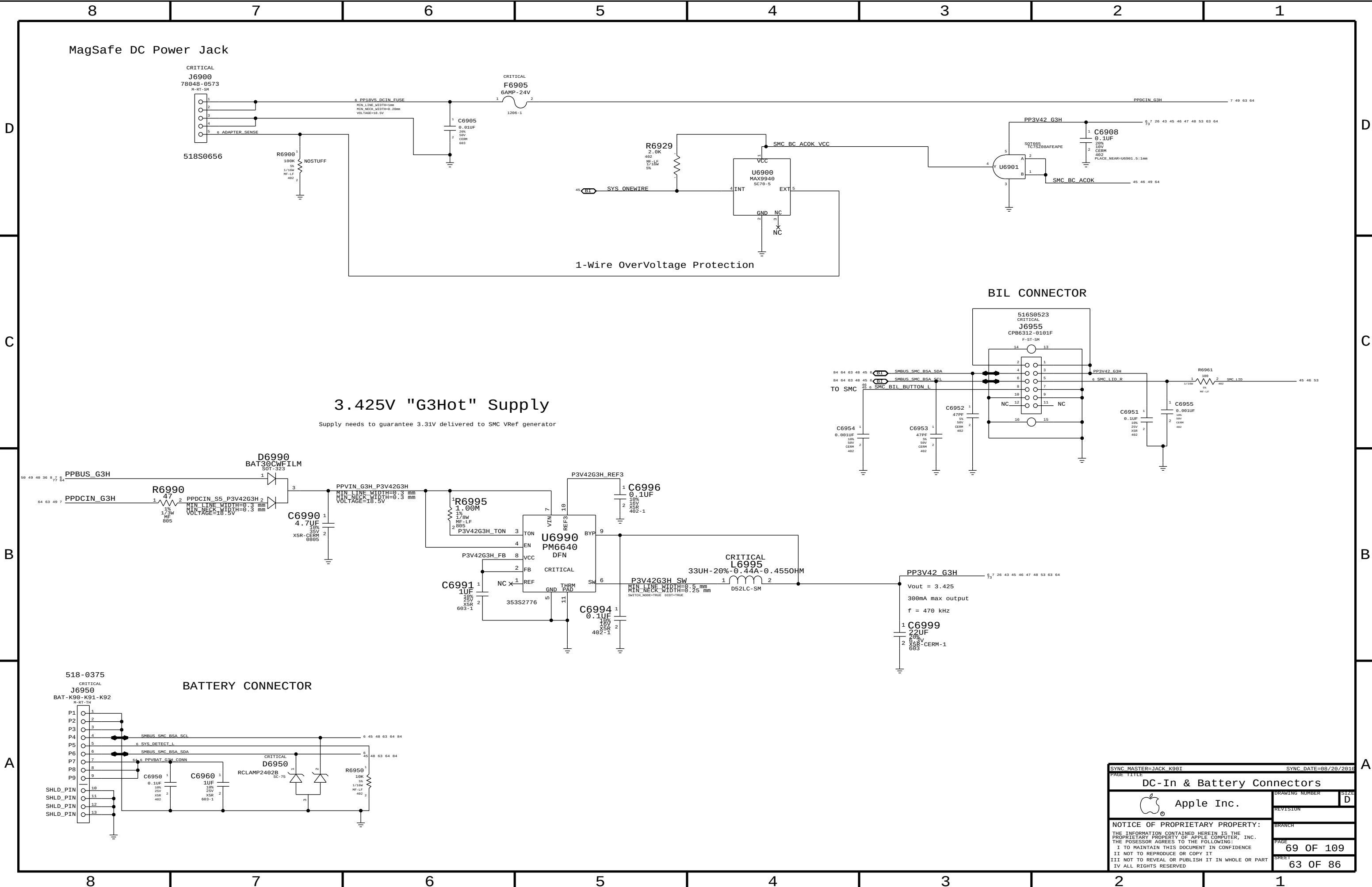
SYNC MASTER=LENG K901		SYNC DATE=08/10/2010	
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AUDIO: HEADPHONE FILTER			
 Apple Inc.		DRAWING NUMBER	SIZE
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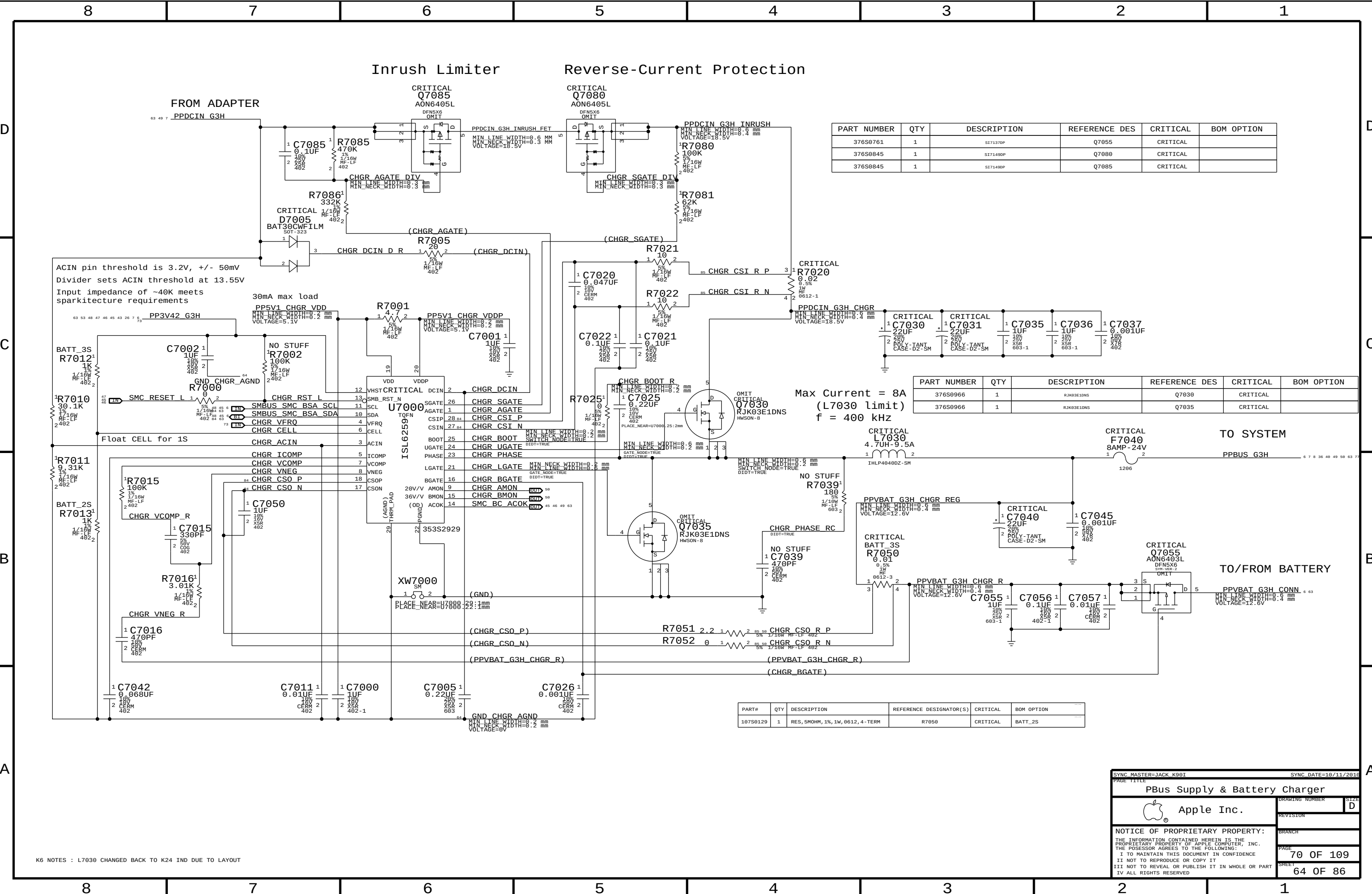




SYNC MASTER-LENG K901		SYNC DATE=08/10/2010	
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AUDIO: JACK			
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		REVISION	
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SYNCS MASTER-LENG K90I		SYNC DATE=08/10/2016	
PAGE TITLE			
AUDIO: JACK TRANSLATORS			
 Apple Inc.		DRAWING NUMBER	
		SIZE D	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0761	1	SL7137DP	Q7055	CRITICAL	
376S0845	1	SL7149DP	Q7080	CRITICAL	
376S0845	1	SL7149DP	Q7085	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0966	1	R3K03E1DNS	Q7030	CRITICAL	
376S0966	1	R3K03E1DNS	Q7035	CRITICAL	

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
107S0129	1	RES, 5MOHM, 1%, 1W, 0612, 4-TERM	R7050	CRITICAL	BATT_2S

K6 NOTES : L7030 CHANGED BACK TO K24 IND DUE TO LAYOUT

SYNC MASTER=JACK K901

SYNC DATE=10/11/2010

PAGE TITLE

PBus Supply & Battery Charger

Apple Inc.

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SIZE

REVISION

BRANCH

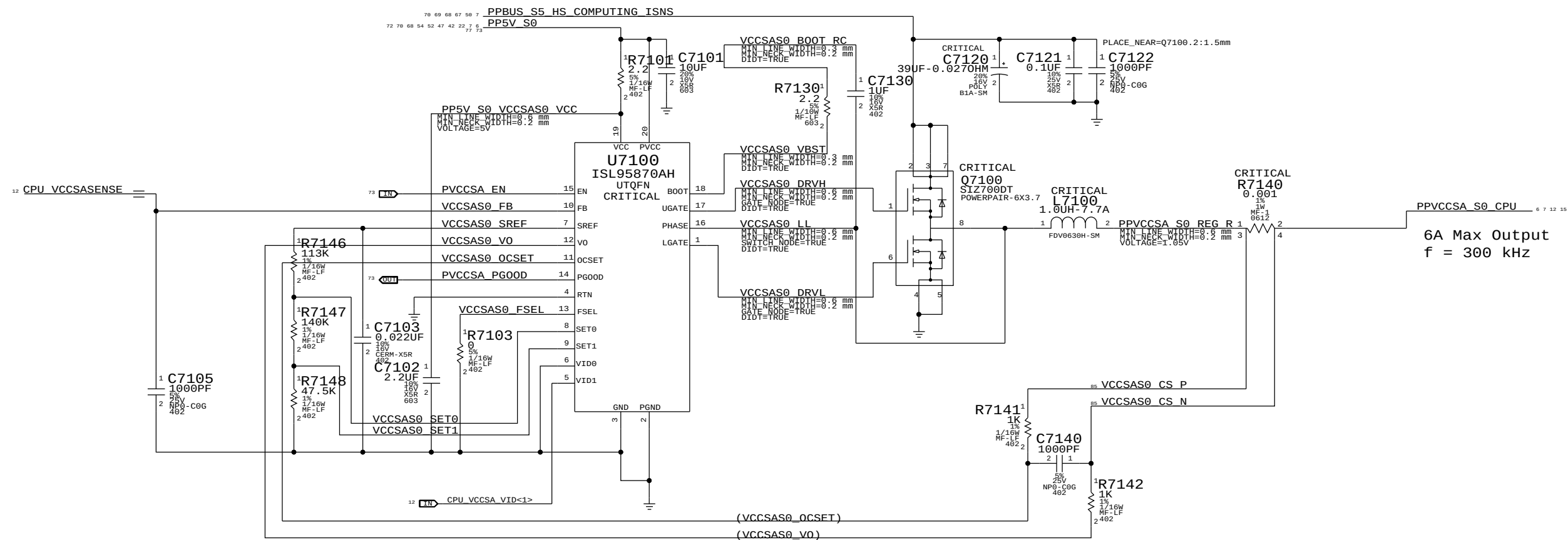
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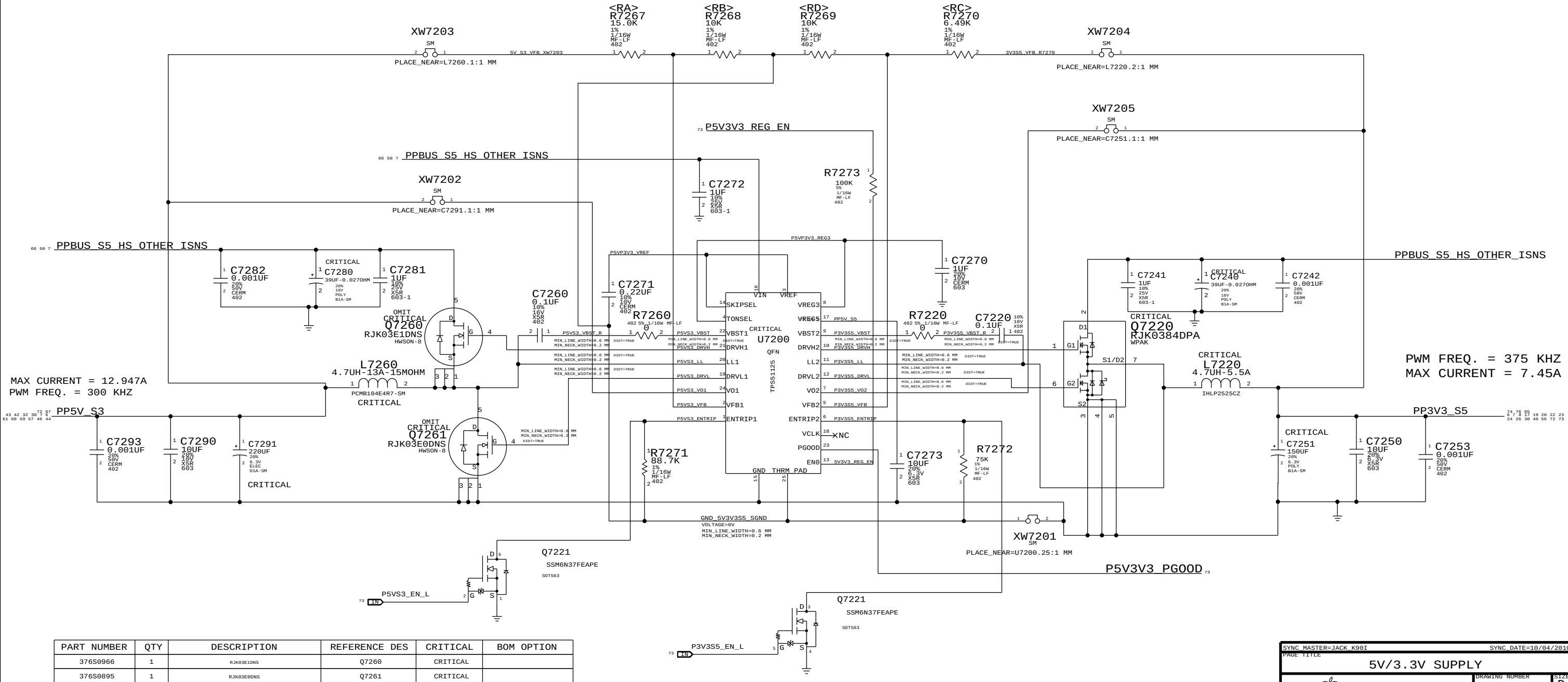
System Agent Power Supply



5V_S3/3.3V_S5 POWER SUPPLY

$V_{OUT} = (2 * R_A / R_B) + 2$

$V_{OUT} = (2 * R_C / R_D) + 2$



SEPERATED MASTER PG00D FOR BOTH 5V AND 3V3.

SYNC MASTER=JACK K901

SYNC DATE=10/04/2010

5V/3.3V SUPPLY

Apple Inc.

DRAWING NUMBER

REVISION

BRANCH

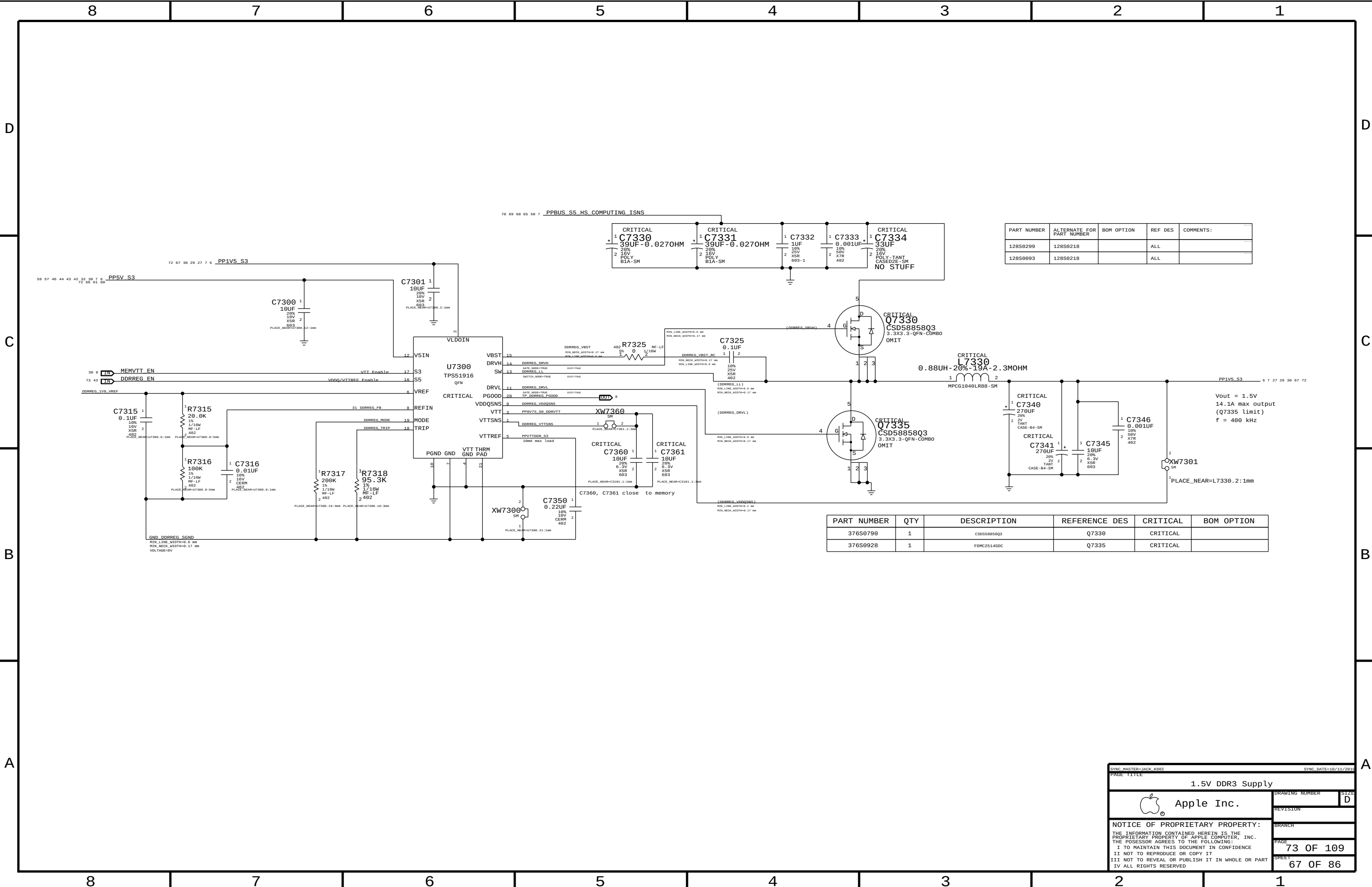
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0299	128S0218		ALL	
128S0093	128S0218		ALL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0790	1	CSD58858Q3	Q7330	CRITICAL	
376S0928	1	FMC25145DC	Q7335	CRITICAL	

Vout = 1.5V
14.1A max output
(Q7335 limit)
f = 400 kHz

SYNC MASTER: JACK_K881

SYNC DATE: 10/11/2011

1.5V DDR3 Supply

Apple Inc.

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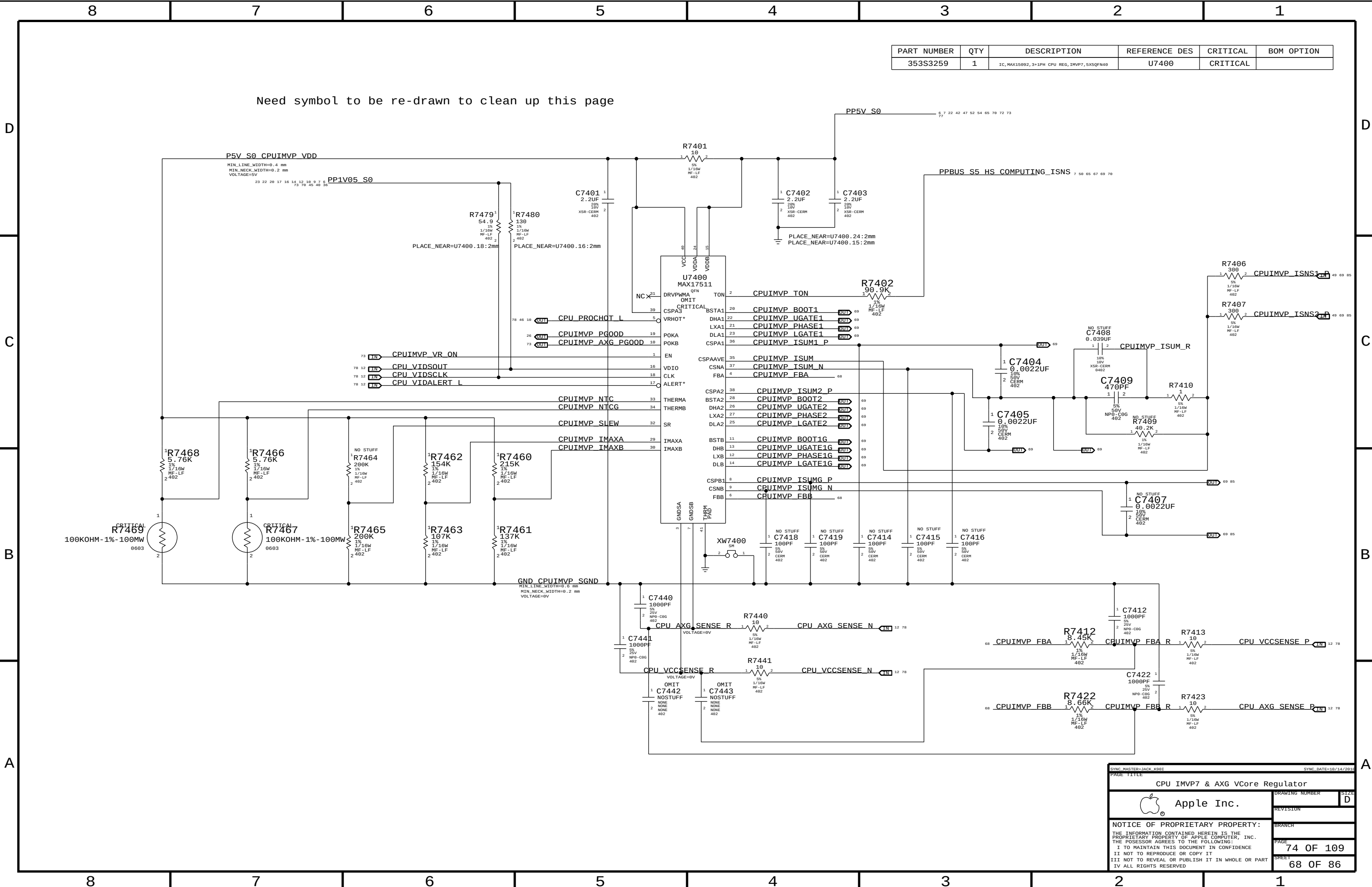
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SIZE D



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S3259	1	IC, MAX15092, 3+1PH CPU REG, 1MVP7, 5X5QFN40	U7400	CRITICAL	

SYNC MASTER=JACK_K981

SYNC DATE=10/14/2011

PAGE TITLE

CPU IMVP7 & AXG VCore Regulator

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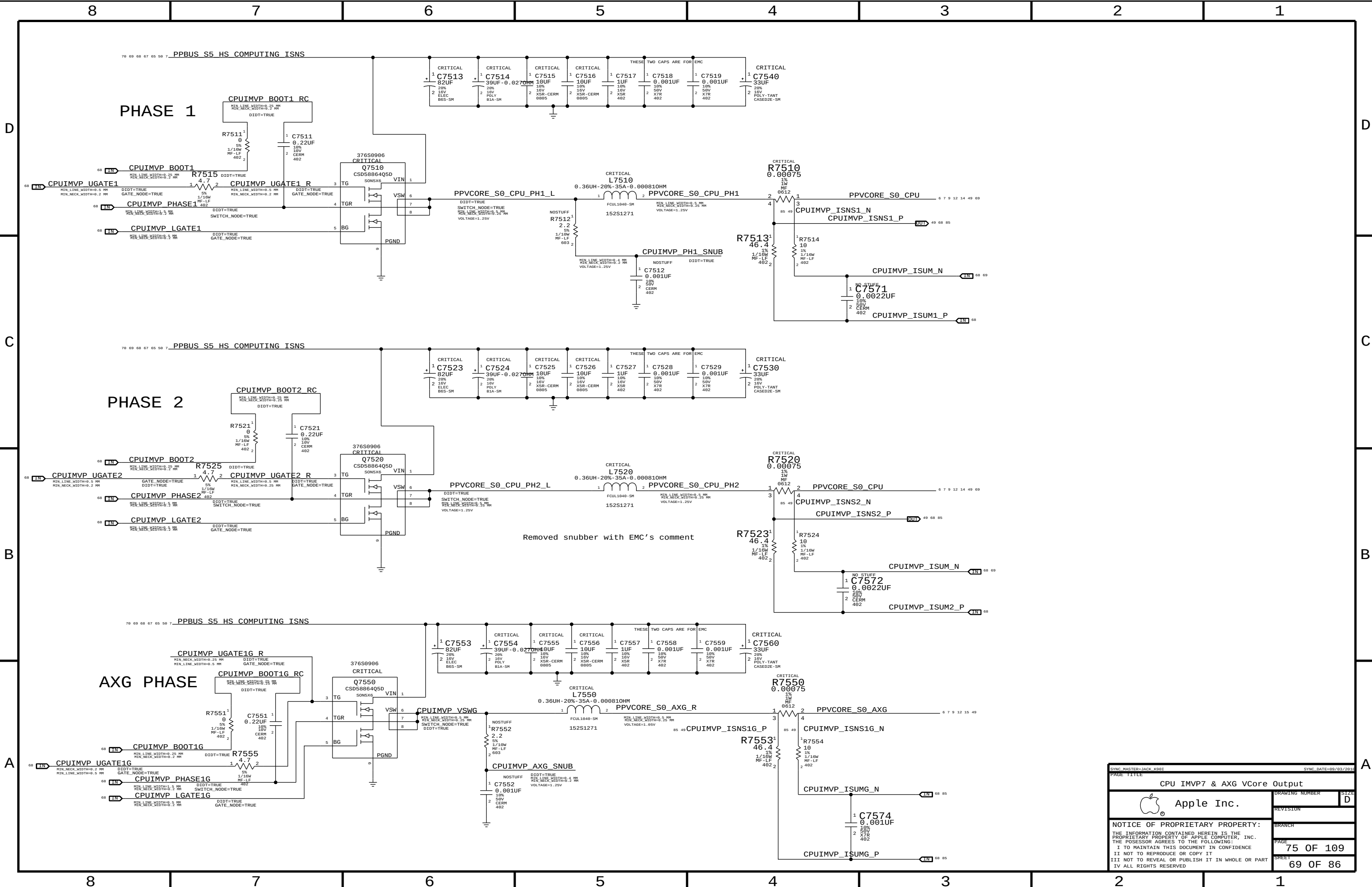
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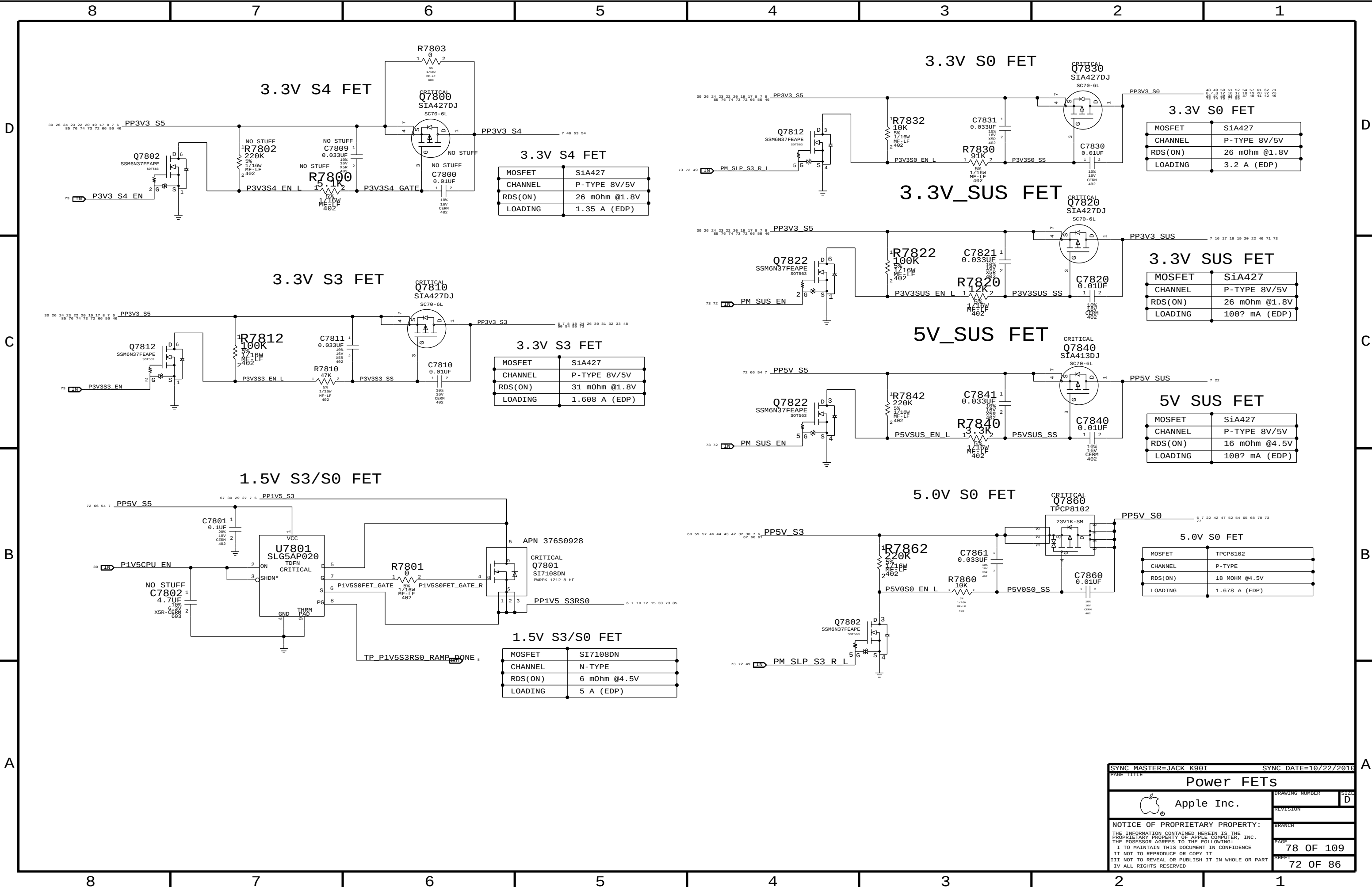
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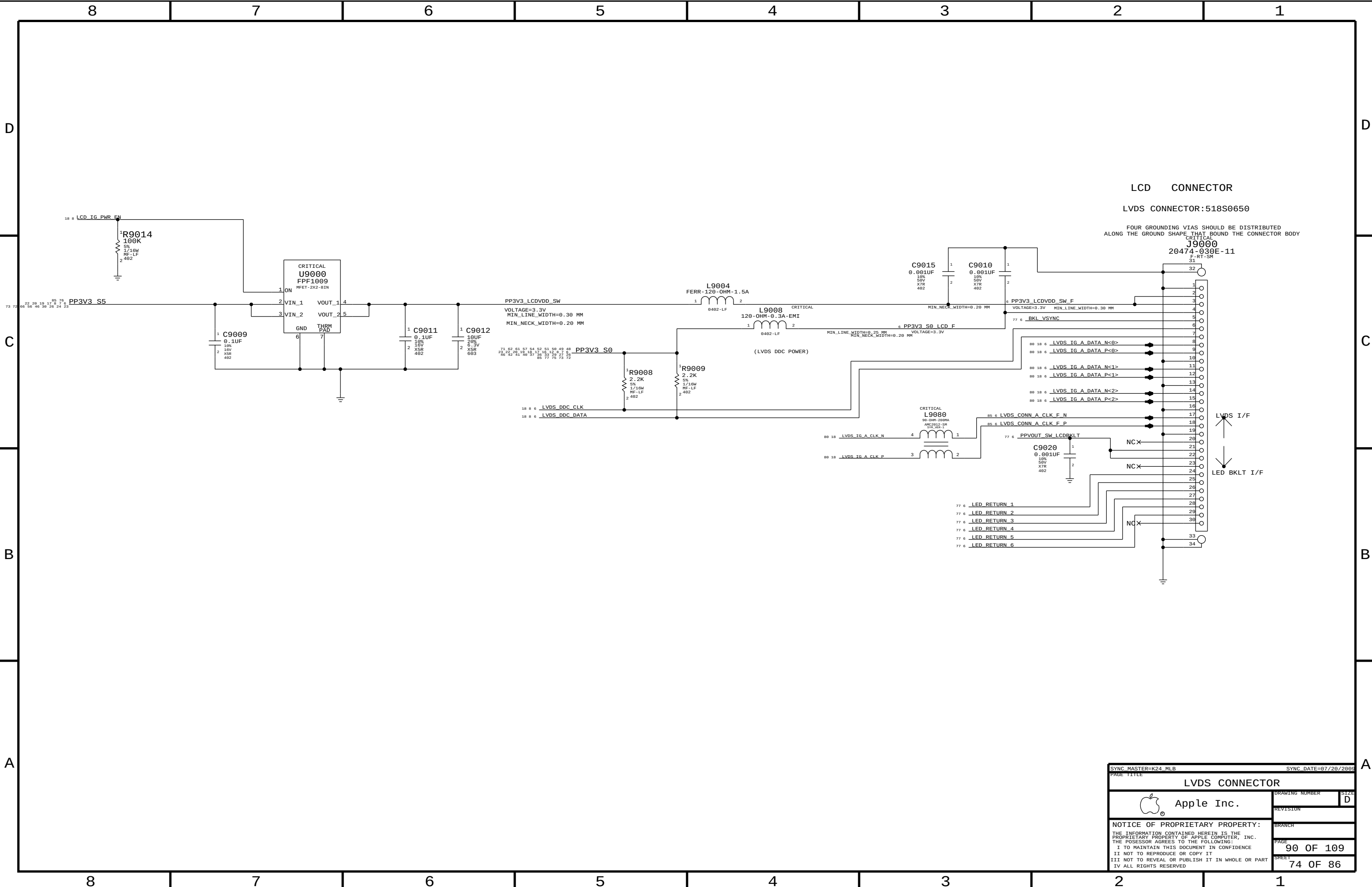
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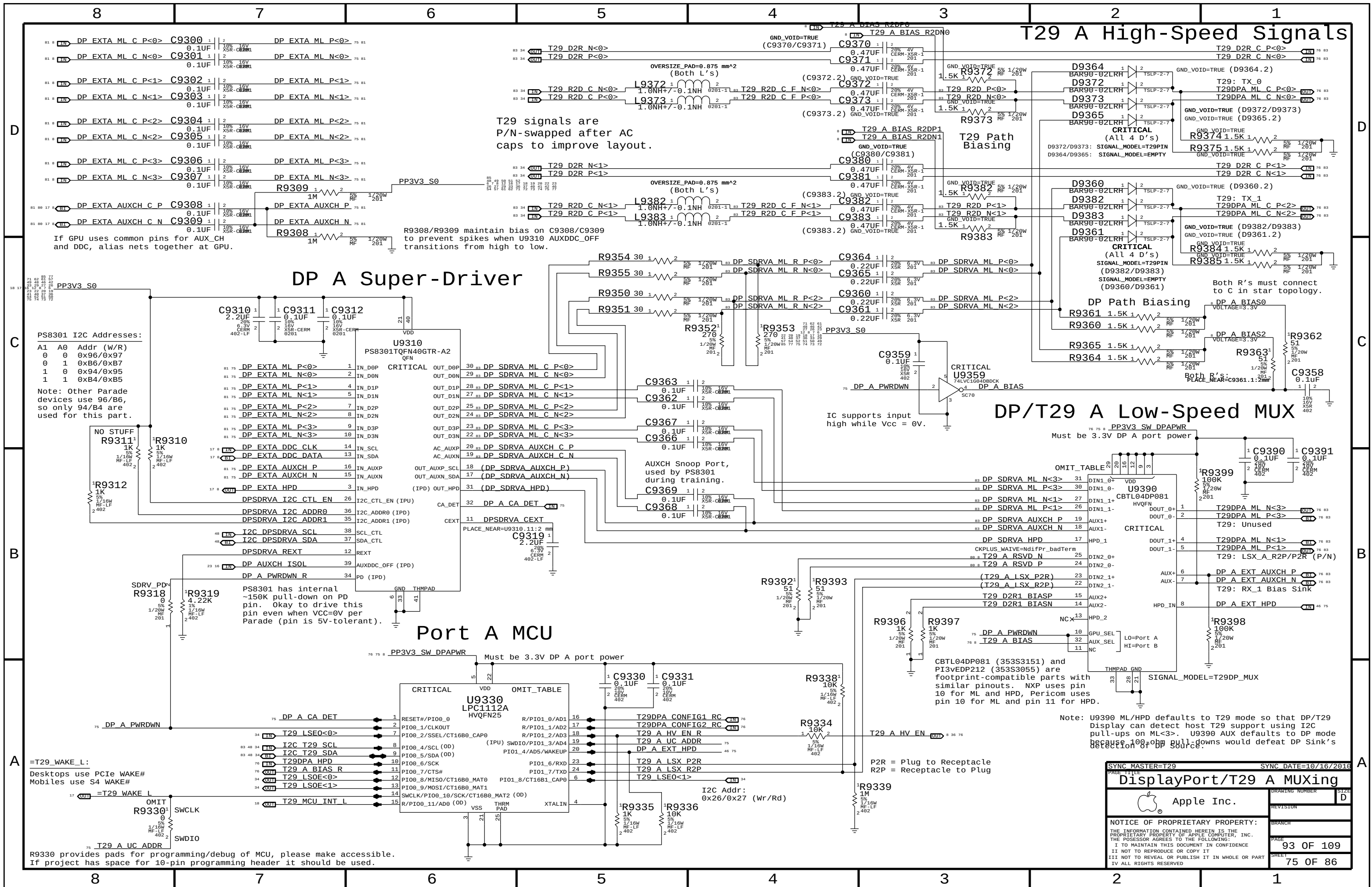
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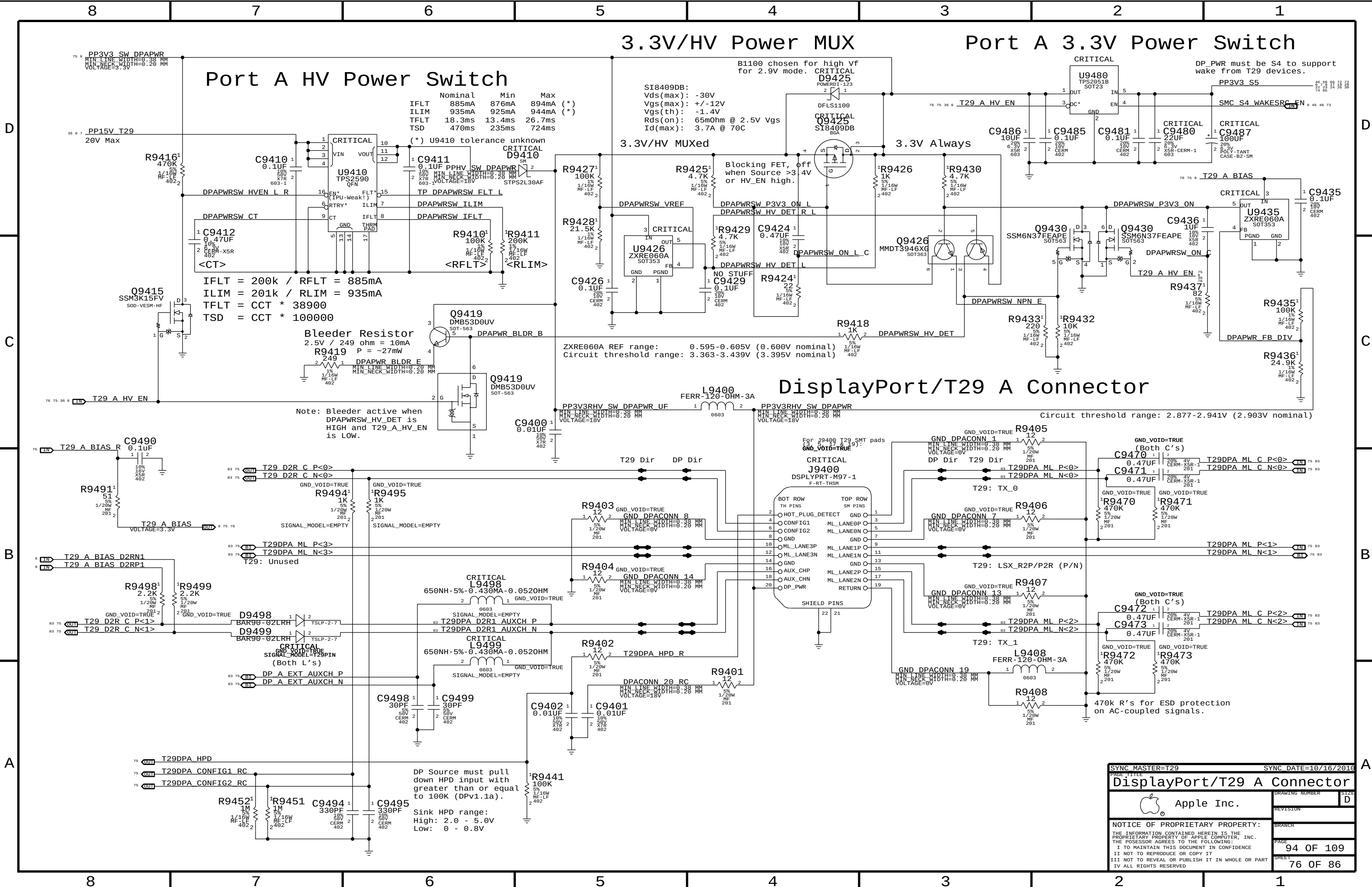
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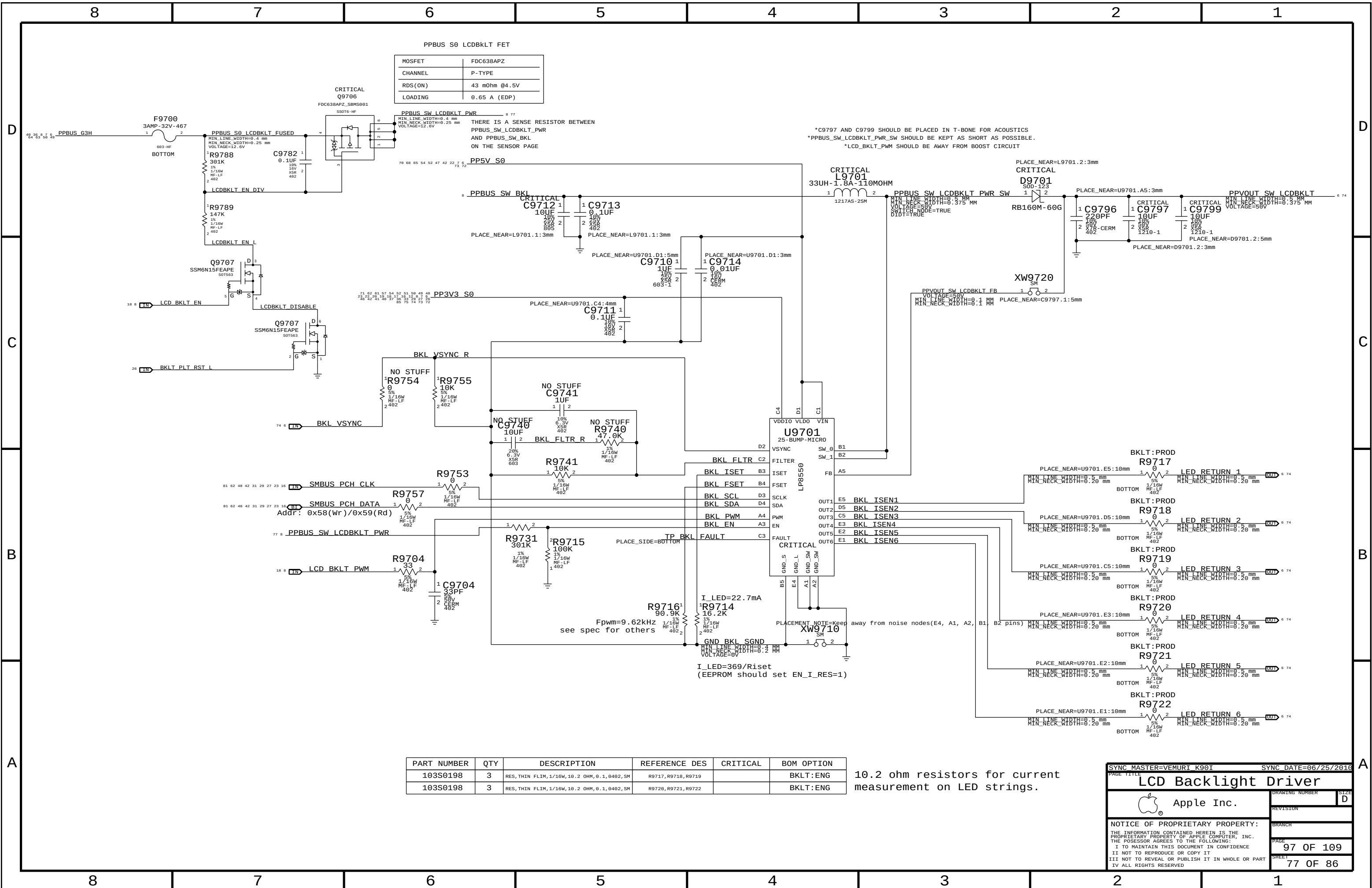
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

SYNC MASTER=VEMURI K901		SYNC DATE=06/25/2016	
PAGE TITLE		LCD Backlight Driver	
Apple Inc.		DRAWING NUMBER	SIZE D
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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

DisplayPort Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?

PCI-Express Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_98D	*	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	LPC_AD	LPC_50S	LPC	LPC_AD<3...0>
	LPC_FRAME_L	LPC_50S	LPC	LPC_FRAME_L
	LPC_RESET_L	LPC_50S	LPC	LPCPLUS_RESET_L
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS
	SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK
	SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS_PCH_DATA
	SMBUS_PCH_0_CLK	SMB_50S	SMB	SML_PCH_0_CLK
	SMBUS_PCH_0_DATA	SMB_50S	SMB	SML_PCH_0_DATA
	SMBUS_PCH_1_CLK	SMB_50S	SMB	SML_PCH_1_CLK
	SMBUS_PCH_1_DATA	SMB_50S	SMB	SML_PCH_1_DATA
	HDA_RTT_CLK	HDA_50S	HDA	HDA_BIT_CLK
	HDA_SYNC	HDA_50S	HDA	HDA_BIT_CLK_R
	HDA_RST_L	HDA_50S	HDA	HDA_SYNC
	HDA_SDIN0	HDA_50S	HDA	HDA_SYNC_R
	HDA_SDOUT	HDA_50S	HDA	HDA_RST_R_L
	PM_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	HDA_RST_L
	SPT_CLK	SPT_55S	SPT	HDA_SDIN0
	SPT_MOSI	SPT_55S	SPT	AUD_SDI_R
	SPT_MISO	SPT_55S	SPT	HDA_SDOUT
	SPT_CS0	SPT_55S	SPT	HDA_SDOUT_R
	PCIE_ENET_R2D	PCIE_85D	PCIE	PM_CLK32K_SUSCLK
	PCIE_ENET_D2R	PCIE_85D	PCIE	SPT_CLK_R
	PCIE_AP_R2D	PCIE_85D	PCIE	SPT_CLK
	PCIE_AP_D2R	PCIE_85D	PCIE	SPT_MOSI
	PCIE_FW_R2D	PCIE_85D	PCIE	SPT_MISO
	PCIE_FW_D2R	PCIE_85D	PCIE	SPT_CS0_R_L
	PCIE_AP_D2R	PCIE_85D	PCIE	SPT_CS0_L
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_ENET_R2D_C_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_ENET_R2D_C_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_ENET_D2R_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_ENET_D2R_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_C_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_C_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_AP_R2D_C_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_AP_R2D_C_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_D2R_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_AP_D2R_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_FW_R2D_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_FW_R2D_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_FW_R2D_C_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_FW_R2D_C_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_FW_D2R_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_FW_D2R_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_FW_D2R_C_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_FW_D2R_C_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_AP_D2R_PI_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_AP_D2R_PI_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_PI_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_PI_N
	PCIE_FW_D2R	PCIE_85D	PCIE	NC_PEG_CLK100MP
	PCIE_AP_D2R	PCIE_85D	PCIE	NC_PEG_CLK100MN
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_CLK100M_ENET_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_CLK100M_ENET_N
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_CLK100M_AP_P
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_CLK100M_AP_N
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_CLK100M_FW_P
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_CLK100M_FW_N
	PCIE_FW_D2R	PCIE_85D	PCIE	NC_PEG_CLK100M_EXCARDP
	PCIE_AP_D2R	PCIE_85D	PCIE	NC_PEG_CLK100M_EXCARDN
	PCIE_AP_R2D	PCIE_85D	PCIE	
	PCIE_FW_R2D	PCIE_85D	PCIE	
	PCIE_FW_D2R	PCIE_85D	PCIE	
	PCIE_AP_D2R	PCIE_85D	PCIE	
	PCIE_AP_R2D	PCIE_85D	PCIE	
	PCIE_FW_R2D	PCIE_85D	PCIE	
	PCIE_FW_D2R	PCIE_85D	PCIE	
	PCIE_AP_D2R	PCIE_85D	PCIE	

Chipset Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
10000	DP EXTA ML	DP 850	DISPLAYPORT	DP EXTA ML C P<3...0>	8 75
10010	DP EXTA ML	DP 850	DISPLAYPORT	DP EXTA ML C N<3...0>	8 75
10020	DP EXTA ML	DP 850	DISPLAYPORT	DP EXTA ML P<3...0>	75
10030	DP EXTA ML	DP 850	DISPLAYPORT	DP EXTA ML N<3...0>	75
10040	DP EXTA AUXCH	DP 850	DISPLAYPORT	DP EXTA AUXCH C P	8 17 75
10050	DP EXTA AUXCH	DP 850	DISPLAYPORT	DP EXTA AUXCH C N	8 17 75
10060	DP EXTA AUXCH	DP 850	DISPLAYPORT	DP EXTA AUXCH P	75
10070	DP EXTA AUXCH	DP 850	DISPLAYPORT	DP EXTA AUXCH N	75
10080	DP INT ML	DP 850	DISPLAYPORT	DP INT ML C P<3...0>	
10090	DP INT ML	DP 850	DISPLAYPORT	DP INT ML C N<3...0>	
10000	DP INT AUXCH	DP 850	DISPLAYPORT	DP INT AUXCH C P	
10010	DP INT AUXCH	DP 850	DISPLAYPORT	DP INT AUXCH C N	
10020	PCIe T29_R2D	PCIe 850	PCIe	PCIe T29 R2D C P<3...0>	8 34
10030	PCIe T29_R2D	PCIe 850	PCIe	PCIe T29 R2D C N<3...0>	8 34
10040	PCIe T29_R2D	PCIe 850	PCIe	PCIe T29 R2D P<3...0>	34
10050	PCIe T29_R2D	PCIe 850	PCIe	PCIe T29 R2D N<3...0>	34
10060	PCIe T29_D2R	PCIe 850	PCIe	PCIe T29 D2R P<3...0>	8 34
10070	PCIe T29_D2R	PCIe 850	PCIe	PCIe T29 D2R N<3...0>	8 34
10080	PCIe T29_D2R	PCIe 850	PCIe	PCIe T29 D2R C P<3...0>	34
10090	PCIe T29_D2R	PCIe 850	PCIe	PCIe T29 D2R C N<3...0>	34
10000	PCIe CLK100M_T29	CLK PCIe 900	CLK PCIe	PCIe CLK100M_T29 P	16 34
10010	PCIe CLK100M_T29	CLK PCIe 900	CLK PCIe	PCIe CLK100M_T29 N	16 34

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
[P23]	SYSCLK CLK32K_RTC	CLK SLOW 55S	CLK SLOW	SYSCLK CLK32K_RTC	16 26
[P25]	SYSCLK CLK25M_SB	CLK 25M 55S	CLK 25M	SYSCLK CLK25M_SB	16 26
[P26]		CLK 25M 55S	CLK 25M	SYSCLK CLK25M_SB R	16
[P27]		CLK 25M 55S	CLK 25M	SYSCLK CLK25M_ENET	26 37
[P28]		CLK 25M 55S	CLK 25M	SYSCLK CLK25M_ENET R	
[P29]	SYSCLK CLK25M_T29	CLK 25M 55S	CLK 25M	SYSCLK CLK25M_T29	26 34
[P30]		CLK 25M 55S	CLK 25M	SYSCLK CLK25M_T29 R	34

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AD

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DD

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18	19
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18	19
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18	19
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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1T01_DIFFPAIR	*		=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

		NET_TYPE		
ELECTRICAL_CONSTRAINT_SET		PHYSICAL	SPACING	
	SMBUS_SMC_A_S3_SCL	SMB_50S	SMB	SMBUS_SMC_A_S3_SCL 6 32 45 48 54 55
	SMBUS_SMC_A_S3_SDA	SMB_50S	SMB	SMBUS_SMC_A_S3_SDA 6 32 45 48 54 55
	SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_B_S0_SCL 45 48 51
	SMBUS_SMC_B_S0_SDA	SMB_50S	SMB	SMBUS_SMC_B_S0_SDA 45 48 51
	SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_B_S0_SCL 6 32 45 48 51
	SMBUS_SMC_B_S0_SDA	SMB_50S	SMB	SMBUS_SMC_B_S0_SDA 6 32 45 48 51
	SMBUS_SMC_BSA_SCL	SMB_50S	SMB	SMBUS_SMC_BSA_SCL 6 45 48 63 64
	SMBUS_SMC_BSA_SDA	SMB_50S	SMB	SMBUS_SMC_BSA_SDA 6 45 48 63 64
	SMBUS_SMC_MGMT_SCL	SMB_50S	SMB	SMBUS_SMC_MGMT_SCL 45 48
	SMBUS_SMC_MGMT_SDA	SMB_50S	SMB	SMBUS_SMC_MGMT_SDA 45 48

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	CHGR_CS1	1T01_DIEFPATR		CHGR_CS1 P	64
		1T01_DIEFPATR		CHGR_CS1 N	64
	CHGR_CS0	1T01_DIEFPATR		CHGR_CS0 P	64
		1T01_DIEFPATR		CHGR_CS0 N	64

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_I701_55S	"	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_I701_55S	"	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	"	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets
PCIE	GND	*	GND_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets
SATA	GND	*	GND_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets
USB	GND	*	GND_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets
CLK_PCIE	SB_POWER	*	PWR_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets
SATA	SB_POWER	*	PWR_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets
USB	SB_POWER	*	PWR_P23H1 https://www.cadence.com/US/en/products/pcb-design/pcb-design-automation/pcb-design-automation-features.html#rule-sets

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MH
MEM_CMD	GND	*	GND_P2MH
MEM_CTRL	GND	*	GND_P2MH
MEM_DATA	GND	*	GND_P2MH
MEM_DQS	GND	*	GND_P2MH

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.076 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
CLK_PCIE_90D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEP_72D	BOTTOM			0.127 MM	6.35 MM		
MEP_85D	TOP			0.1 MM	6.35 MM		

K90i Specific Net Properties

ELECTRICAL CONSTRAINT SET	PHYSICAL	NET TYPE	SPACING		
	ENET_1000	ENETCONN	ENETCONN_P<3...0>	38	
	ENET_1000	ENETCONN	ENETCONN_N<3...0>	38	
	SATA_900	SATA	SATA_ODD_D2R_UF_P	6 42	
	SATA_900	SATA	SATA_ODD_D2R_UE_N	6 42	
	SATA_900	SATA	SATA_HDD_D2R_RDRVR_OUT_P	42	
	SATA_900	SATA	SATA_HDD_D2R_RDRVR_OUT_N	42	
	SATA_900	SATA	SATA_HDD_R2D_RDRVR_IN_P	42	
	SATA_900	SATA	SATA_HDD_R2D_RDRVR_IN_N	42	
FE20	SATA_900	SATA	SATA_HDD_D2R_RDRVR_IN_P	42	
FE20	SATA_900	SATA	SATA_HDD_D2R_RDRVR_IN_N	42	
FE20	SATA_900	SATA	SATA_HDD_R2D_RDRVR_OUT_P	42	
FE20	SATA_900	SATA	SATA_HDD_R2D_RDRVR_OUT_N	42	
	SENSE_DIFFPAIR	THERM_1701_500	THERM	CPUTHMSNS_D2_P	51
		THERM_1701_500	THERM	CPUTHMSNS_D2_N	51
SENSE_DIFFPAIR		THERM_1701_500	THERM	CPU_THERMD_P	9 51
SENSE		THERM_1701_500	THERM	CPU_THERMD_N	9 51
	SENSE_DIFFPAIR	THERM_1701_500	THERM	T29_THERMD_P	34 51
		THERM_1701_500	THERM	T29_THERMD_N	51
SENSE_DIFFPAIR		THERM_1701_500	THERM	T29THMSNS_D2_P	51
		THERM_1701_500	THERM	T29THMSNS_D2_N	51
	SENSE_DIFFPAIR	SENSE_1701_500	SENSE	ISNS_HS_COMPUTING_N	50
		SENSE_1701_500	SENSE	ISNS_HS_COMPUTING_P	50
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_HS_OTHER_N	50
		SENSE_1701_500	SENSE	ISNS_HS_OTHER_P	50
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUVCCIO50_CS_N	49 70
		SENSE_1701_500	SENSE	CPUVCCIO50_CS_P	49 70
	SENSE_DIFFPAIR	SENSE_1701_500	SENSE	CPUIMVP_ISNS1_P	49 60
		SENSE_1701_500	SENSE	CPUIMVP_ISNS1_N	49 60
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUIMVP_ISNS2_P	49 60
		SENSE_1701_500	SENSE	CPUIMVP_ISNS2_N	49 60
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUIMVP_ISNS1G_P	49 60
		SENSE_1701_500	SENSE	CPUIMVP_ISNS1G_N	49 60
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUIMVP_ISUM_R_P	49
		SENSE_1701_500	SENSE	CPUIMVP_ISUM_R_N	49
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUIMVP_ISUMG_R_P	49
		SENSE_1701_500	SENSE	CPUIMVP_ISUMG_R_N	49
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUIMVP_ISNS_P	49
		SENSE_1701_500	SENSE	CPUIMVP_ISNS_N	49
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	VCCSAS0_CS_P	65
SENSE		SENSE_1701_500	SENSE	VCCSAS0_CS_N	65
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	CPUIMVP_ISUMG_P	60 60
SENSE		SENSE_1701_500	SENSE	CPUIMVP_ISUMG_N	60 60
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_CPU_N	
SENSE		SENSE_1701_500	SENSE	ISNS_CPU_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_HDD_N	
SENSE		SENSE_1701_500	SENSE	ISNS_HDD_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_HDD_R_N	
SENSE		SENSE_1701_500	SENSE	ISNS_HDD_R_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_LCDBKLT_N	
SENSE		SENSE_1701_500	SENSE	ISNS_LCDBKLT_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_ODD_N	
SENSE		SENSE_1701_500	SENSE	ISNS_ODD_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_ODD_R_N	
		SENSE_1701_500	SENSE	ISNS_ODD_R_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_P1VBGPU_N	
		SENSE_1701_500	SENSE	ISNS_P1VBGPU_P	
SENSE_DIFFPAIR		SENSE_1701_500	SENSE	ISNS_P1VBGPU_R_N	
SENSE		SENSE_1701_500	SENSE	ISNS_P1VBGPU_R_P	
FE20	LVDS_000	LVDS	LVDS_CONN_A_CLK_F_N	6 74	
FE20	LVDS_000	LVDS	LVDS_CONN_A_CLK_F_P	6 74	

K90i Specific Net Properties

[illegible]

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

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Project Specific Constraints			
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